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Improving Clinical Information Extraction with Public-Data Pretraining

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Abstract

Contents

1 Introduction

> "The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin. [...] We should build in only the meta-methods that can find and capture this arbitrary complexity. [...] The eventual success is tinged with bitterness, and often incompletely digested."
>
> - Richard S. Sutton, *The Bitter Lesson*

Healthcare has been one of the earliest and most prominent application domains for artificial intelligence. Use cases range from diagnostic support to medical record management, cohort selection, report generation, and clinical information extraction.

As early as the 1970s, MYCIN, the iconic expert system developed at Stanford, diagnosed bacterial infections and recommended antibiotics. It already achieved performance comparable to human specialists in its narrow domain. INTERNIST-1, later known as CADUCEUS, tackled internal medicine diagnosis across hundreds of diseases. However, these systems relied on explicit rules and static knowledge bases, which limited their ability to handle the complexity and variability of real-world medical data. They were rigid and required constant maintenance to keep up with medical advances. Building such systems demanded expensive medical experts to annotate clinical data and write rules by hand.

With the rise of machine learning in the 1990s and 2000s, data-driven approaches began to dominate healthcare AI. Supervised and unsupervised learning algorithms trained models directly from data. The interpretability of rule-based systems was traded for the performance of deep neural networks capable of capturing complex patterns. R2 ImageChecker (1998) became the first FDA-approved computer-aided detection system for mammography, using neural networks to spot suspicious micro-calcifications. PAPNET applied neural networks to cytological screening for cervical cancer.

Then came the scaling era. The recipe became simple: pretrain a language model on billions of words of raw text, then fine-tune on the target task. This approach routinely outperformed systems built with years of domain expertise. The need for expensive annotation, hand-crafted rules, and curated knowledge bases faded, beaten by data and compute.

Despite healthcare offering many structured and curated knowledge bases such as UMLS, SNOMED-CT, and RxNorm, experience has shown that general-purpose language models pretrained on massive unstructured data outperform approaches based on domain-specific rules and knowledge bases. It seems our models cannot effectively integrate these structured resources, or perhaps these knowledge bases do not cover the full complexity and variability of medical knowledge as it is used in practice.

However, healthcare faces a limitation that other domains do not: the confidentiality and scarcity of annotated clinical data. Medical data is sensitive and difficult to obtain in large quantities for model training. Works adapting language models to the medical domain often rely on public data such as PubMed or patient forums, but these do not always reflect the real clinical language used in electronic health records. The clinical domain has unique jargon, abbreviations, and writing styles that differ from general medical texts. Moreover, while MIMIC exists in the United States, public clinical datasets remain scarce in other countries such as France.

This leads to a fundamental problem: the paradigm that has won is pretraining at scale, but in healthcare we lack massive public clinical data to pretrain language models. Can we find new ways to exploit public data for pretraining language models adapted to the clinical domain, without using sensitive clinical data?

Part I

What is Biomedical Text?

2 Clinical Text

FRANÇAIS

CENTRE HOSPITALIER DE SAINT-AUBIN
Dossier n^{o} 4471902
Cardiologie, Soins Intensifs
Séjour : 03/11–08/11/2024

Patient : CARON Julien, né le 14/03/1962 (62 ans). **Dest. :** D^{r} S. Lemaire.

Motif. Douleur thoracique constrictive rétrosternale, irradiation bras G, depuis 2h.

Antécédents. HTA. DT2 sous metformine. Tabac 30 PA. Dyslipidémie.

Examen. TA 148/92, FC 98, SpO_2 96% AA, EVA 7/10. Pas de signe d'IVG.

Paraclinique. ECG : sus-décalage ST V1–V4. Tropo T 4520 ng/L (N <14). Coro : occlusion IVA proximale, angioplastie + stent actif, TIMI 3. ETT : FEVG 40%.

Au total. SCA ST+ antérieur (I21.0).

Ttt sortie. Kardégic 75, ticagrélor 90x2, atorvastatine 80, bisoprolol 2,5, ramipril 5.

CAT. Réadaptation cardiaque. Consultation cardio à 1 mois. D^{r} C. Després

ENGLISH · TRANSLATION

SAINT-AUBIN HOSPITAL
Record no. 4471902
Cardiology, Intensive Care
Stay: 03/11–08/11/2024

Patient: CARON Julien, b. 14/03/1962 (62 y). **To:** Dr S. Lemaire.

Reason. Constrictive retrosternal chest pain, radiating to L arm, for 2h.

History. HTN. T2DM on metformin. Smoking 30 PY. Dyslipidaemia.

Exam. BP 148/92, HR 98, SpO_2 96% RA, pain 7/10. No sign of LVF.

Workup. ECG: ST elevation V1–V4. Trop T 4520 ng/L (N <14). Angiography: proximal LAD occlusion, angioplasty + drug-eluting stent, TIMI 3. Echo: LVEF 40%.

Summary. Anterior STEMI (I21.0).

Discharge tx. Aspirin 75, ticagrelor 90x2, atorvastatin 80, bisoprolol 2.5, ramipril 5.

Plan. Cardiac rehabilitation. Cardiology review at 1 month. Dr C. Després

Figure 2.1: A discharge summary for an acute myocardial infarction; French original, English translation.

Figure 2.1 is a letter from one cardiologist to a colleague. For an ordinary French reader, it is a text that is very hard to read and to understand. It has very few verbs, few adverbs, and sometimes neither. The sentences are telegraphic, the information very dense and simply separated by commas, like a table. One might think this is just carelessness, that doctors have very little time and rush their letters without paying them much attention. There is some truth to that. In reality, though, it is a style of communication tuned for the community it is written for. The writer and the reader share enough knowledge that the writer can give just the precise information he knows the other needs and go straight to what matters clinically.

This phenomenon has a precise name. Studying how specialised communities write, the linguist Zellig Harris observed in the 1960s that a technical field has a language of its own, with its own grammar, and not just everyday language with fewer words (Harris, 1968). He called it a *sublanguage*: the text a community produces about a specialised domain, following its own rules for how words combine. Some of these rules are stricter than those of everyday language, and some ignore everyday rules entirely. Harris and his colleagues later made this concrete, reconstructing the grammar of a small immunology literature from its articles alone, and showing that the sublanguage itself held the field's knowledge (Harris et al., 1989).

The clinical note is the best-documented case of this. One of the earliest attempts to make computers read medical records, Naomi Sager's Linguistic String Project at New York University, set out to turn free-text clinical reports into structured data, and in the process found that the notes already followed a specific grammar (Sager, 1981; Sager et al., 1987). To read them, the program needed rules for the short, verbless phrases that notes use but everyday language does not, together with categories taken from medicine itself, such as patient, finding, and treatment, all visible in Figure 2.1. A line like *afebrile, chest clear* is not broken language; it is written this way for a reason. The style emerged in the community of clinicians, for efficiency, and newcomers learn to write the same way (Kittredge, 1982). A grammar like this can live inside a hospital, kept up and extended by the people who use it.

Let us come back to the idea of carelessness. If the note were merely careless, its missing words would be scattered at random through the document, with no logic behind them. Instead, the omissions follow a pattern. Sager and colleagues showed that clinical shorthand obeys a small set of deletion rules: the writer drops a subject here, a verb there, a tense elsewhere, but only when the reader can put it back (Anderson et al., 1975). *Chest films normal* stands for *the chest films were normal*; *no bone pain* stands for *the patient has no bone pain*. What disappears is the part the reader already knows. The note is a form of compression: it leaves out the information the writer is sure the reader already has.

This works because the reader is an expert. Writing for a colleague who shares the same training, the clinician can leave out whatever the other already knows, and say only what is needed (Grice, 1975; Sperber and Wilson, 1995). The same words behave differently depending on who reads them. *Afebrile, vitals stable, continue antibiotics* is complete for a physician and almost empty for anyone else.

That shared background is what linguists call *common ground*. The note records only the update to a large body of assumptions the writer and reader already hold, their medical training, the patient's history, the routines of the ward (Stalnaker, 2002). A clinical note is also performative. Written by a doctor, it does not only describe; it can also declare or set off an action (Austin, 1962; Searle, 1969). A prescription is not just information, it is what lets the patient obtain the medication, and an order to transfer a patient sets the transfer in motion. The note is a document with real authority, signed and legally binding.

The layout of the note also has roots. Each finding is tied to a place in the body, so the patient appears as a list of lesions. This way of looking at the body grew out of the clinical gaze that took shape in the modern hospital (Foucault, 1963).

The note stays inside the hospital for a plainer reason: it is full of personal data. One report can hold the patient's name, age, and dates, a home address, a family situation, and a diagnosis rare enough to point to one person. Stripping out the obvious identifiers does not solve this. A pseudonymised note is still not anonymous, and a model trained on such text can memorise what it read and later give it back (Tannier et al., 2026). Confidentiality is not a rule laid on top of the text. It is part of what the text is.

For French the problem is sharper. There is no shareable corpus of real French hospital records to set beside the large English ones, and there has been none for a decade (Névéol et al., 2018). Public text takes its place: published case reports, drug leaflets, article abstracts. But a case report is written to teach. It reads like a short story, in full sentences and the past tense, as in this opening from a French case corpus: *"B.A., âgé de 36 ans, sans antécédents notables, a été admis en février 1994 pour des douleurs lombaires droites"* (Grabar et al., 2020). Set against the clipped lines of the note in Figure 2.1, the gap is easy to see.

Whether these substitutes really match hospital notes is an open argument. The team behind the case-report corpus is hopeful: the clinical part of a published case, they say, can be close enough to a real note that tools built on it still work on hospital text (Grabar et al., 2020). They also name the limits. Real notes are full of misspellings, carry administrative boilerplate, and present examinations and lab values in tables, while a published case rewrites all of this as clean sentences. A more recent project sees it differently. In building PARHAF, a corpus of 7,394 fictitious reports hand-written by 104 medical residents, its authors say flatly that "the style of case reports [...] is quite different from that of electronic health records" (Tannier et al., 2026). They wrote fake reports from scratch because the real ones, even pseudonymised, cannot be shared. The field still cannot agree on whether the text it may share even resembles the text it cannot, and the text we most want to model remains the text we can least obtain.

Put together, these traits explain why the clinical note is so hard to get. It is rare because it is private, and private because of what it is: a closed sublanguage, written in compressed form for an expert in-group, packed with personal data, and carrying out acts with legal weight. The note is not scarce by accident. It is scarce because of what it is.

Yet medicine produces text in huge amounts. Most of it is not the clinical note at all, but the published literature, which is plentiful and free to read. That sets up the question for the next chapter: can this abundance stand in for the language of the clinic?

3 Scientific Text

FRANÇAIS

L'insuffisance mitrale (IM) ischémique est une valvulopathie fonctionnelle, restrictive, liée au remodelage ventriculaire gauche post-infarctus et qui complique environ 20% des cardiopathies ischémiques. Sa présence grève lourdement le pronostic de la maladie coronarienne, tant en termes de morbidité que de mortalité. Encore mal connue, elle suscite depuis quelques années de nombreux travaux visant à préciser ses mécanismes physiopathologiques, ses facteurs pronostiques et sa prise en charge thérapeutique.

ENGLISH · TRANSLATION

Ischaemic mitral regurgitation is a functional, restrictive valve disease, linked to post-infarction left-ventricular remodelling, which complicates about 20% of ischaemic heart diseases. Its presence weighs heavily on the prognosis of coronary disease, in terms of both morbidity and mortality. Still poorly understood, it has prompted in recent years a large body of work seeking to clarify its pathophysiological mechanisms, its prognostic factors and its therapeutic management.

Figure 3.1: A research article about myocardial infarction; French original, English translation.

Figure 3.1 also concerns the heart attack of the previous chapter, here one of its long-term complications, yet nothing else about it resembles the note. The sentences are whole, each clause is tied to the next, and the passage even stops to point out what is not yet known, then promises to look into it. There is, moreover, an enormous quantity of writing like this, and unlike the clinical note it is free to read. It is tempting to think we have found the clinical language after all, just written more carefully and easy to find. We have not. The research article is abundant and public, but it is a language of its own.

The two texts describe the same disease, so the article can look like a better-written version of the note. But what sets the two apart is their purpose. Asking why texts take the forms they do, the rhetorician Carolyn Miller argued that a genre is a recurring social action, defined by what it is for rather than by how it looks on the page (Miller, 1984). The clinical note exists to record a patient and guide their care. The research article exists to make a contestable claim public and hold it up for scrutiny. That difference in purpose shapes everything about how each one is written.

The article's purpose leaves a visible mark on its shape. Studying how researchers open their papers, the linguist John Swales described a pattern so regular that he gave it a name: move from a broad field, to a specific gap in what is known, to the work that fills it (Swales, 1990). A typical opening establishes that a topic matters, observes that something about it is still unknown, and announces that the paper will settle it. In short, the article argues that it needs to exist. A clinical note never does this: it has no gap to open and nothing to justify.

The body of the article follows the same logic. Its four parts, introduction, methods, results, and discussion, retrace the steps of a controlled study: a question, a procedure, an observation, an interpretation. The arrangement feels timeless, yet it is surprisingly recent. In a survey of 1,297 articles from four leading medical journals between 1935 and 1985, not one used the structure in 1935; by the 1970s more than four in five did, and by the 1980s it was the only accepted form (Sollaci and Pereira, 2004). A format only a few decades old now carries the shape of scientific reasoning in its headings alone.

The clinic has a structure of its own, and it encodes a different kind of reasoning. The problem-oriented record, built to organise care around a patient's problems, and its familiar subjective-objective-assessment-plan note, follow the loop of diagnosis and management rather than experiment (Weed, 1968). The article asks whether a hypothesis holds; the note asks what is wrong with the patient and what to do next. Both are about the same medicine, but they reason in different ways, and one is not a worse version of the other.

The difference reaches down to the sentence. An article does more than arrange its argument. It constantly signals how sure it is. Ken Hyland's studies of academic writing describe this layer as metadiscourse, the words an author uses to adjust how strongly they commit to a claim (Hyland, 2005; Hyland and Tse, 2004). Hedges such as *may* or *suggests* hold a statement back, while boosters such as *clearly* or *shows* push it forward. *These results suggest that X may contribute to Y* is a claim offered for debate, whereas the bare *X causes Y* would be a much bolder claim. The note hedges too, but sparingly and for another reason: its *rule out myocardial infarction* is a plan for action, not a position in an argument.

These impressions can be counted. Measured across many texts, scientific prose is about as far from everyday conversation as written English gets: dense with nouns, long words, and prepositions (Biber, 1988). And it is not uniform within a single article: the methods read flat and impersonal, the discussion hedged and personal, so the same shift in certainty shows up as a measurable difference between sections (Biber and Finegan, 1994).

The pattern reaches even the smallest scale. The medical article's rhetorical moves have been catalogued in detail, eleven of them across the usual sections (Nwogu, 1997), and its set phrases are just as fixed: frames such as *the role of* or *has been shown to* recur with a narrow, discipline-specific set of fillers (Luzón Marco, 2000). From its overall plan down to its smallest phrases, the article is built for a purpose the clinical note does not share.

This hedging does real work. A finding hardens into a fact as its qualifiers drop away from one citation to the next, so the hedges that fill the literature are the visible trace of claims still under negotiation (Latour and Woolgar, 1979).

There is a more formal way to put all of this. Comparing patient reports with biology articles, Carol Friedman and colleagues found two distinct sublanguages, each needing its own grammar to be processed by a computer (Friedman et al., 2002). The lesson for what follows is direct: a model trained on the abundant public literature learns to argue about medicine rather than to record a patient. More articles, by themselves, do not bring us closer to the clinic.

So the scientific article is very different from the clinical note. It is public where the note is private, spells things out where the note leaves them unsaid, and weighs each claim where the note simply acts. It is abundant and free to read, but it is its own language. There is, however, a third kind of medical text, more abundant still and written for the widest audience of all, the patients themselves. Whether that brings us any nearer to the clinic is where we turn next.

4 Lay and Web Text

FRANÇAIS

Bonjour, mon père a fait une crise cardiaque la semaine dernière. Il a eu une grosse douleur dans la poitrine qui descendait dans le bras gauche, il était tout pâle et en sueur, on a appelé le 15 tout de suite. Est-ce que quelqu'un sait combien de temps dure la convalescence après un infarctus ? Merci d'avance.

ENGLISH · TRANSLATION

Hello, my father had a heart attack last week. He had a strong pain in the chest that went down into the left arm, he was all pale and sweating, we called emergency services right away. Does anyone know how long recovery takes after a heart attack? Thanks in advance.

Figure 4.1: A patient forum about myocardial infarction; French original, English translation.

Figure 4.1 is the same heart attack once more, now in the words of the people who live through it. There is no technical vocabulary at all: a worried relative writes *crise cardiaque*, not *infarctus du myocarde*, and describes the symptoms in plain, everyday terms. Text like this is the most abundant medical writing of all: there is far more health information written for patients than there are journal articles, let alone clinical notes. It is plainly not a clinical record. But it is worth a look, because it shows what happens to medical language when it reaches the public, and in particular what happens to names.

That a fact changes as it travels is an old observation. In 1935 the physician Ludwik Fleck argued that knowledge is produced inside a community of specialists and is transformed as it moves outward to a wider audience (Fleck, 1979). The cautious *it appears that* of the laboratory becomes the public's *doctors have shown*; the careful technical term becomes an everyday word. Writing for patients is its own layer of the same knowledge, with its own form.

Not all of this lay writing is unofficial. The most systematic patient-facing text of all is the humble drug leaflet, the *notice* folded inside every box of medicine. In the European Union it is not optional: by law every authorised drug carries one, in the official language of each market, written "in clear and understandable terms for the users" (European Parliament and Council of the European Union, 2001). A common template fixes the same six headings everywhere, from *what the medicine is and what it is used for* to *possible side effects*, and a draft must be tested on real patients for readability before it can be approved (European Medicines Agency, 2025). Here the move from specialist to public that Fleck described is not left to chance but codified and quality-checked. The leaflet also stages the variation in names directly, since regulation requires it to print the brand name beside the common one, *Doliprane* next to *paracétamol*, so that a single substance wears its several names side by side on one page.

What changes most, for our purpose, is the naming. The founding ideal of terminology, set out by the engineer Eugen Wüster, was one concept and one term, each thing with a single correct name (Wüster, 1979). Real usage does not behave this way. The linguist María Teresa Cabré showed that a single concept is named differently depending on who is speaking and why, and that this variation is normal rather than a defect (Cabré, 2003); in French, Gaudin's socioterminology makes the same point about how terms spread from specialists to the public (Gaudin, 2003). The heart attack of the clinical note, *infarctus du myocarde*, coded *I21*, is the same event the patient calls a *crise cardiaque*. One concept, several names, and most of them come from the public.

The trouble runs deeper than names. Rita Temmerman showed that many medical categories have no sharp edges: they cluster around typical cases, shift over time, and are often understood through metaphor, the heart as a pump that can fail (Temmerman, 2000). Normalisation therefore meets a double difficulty: a single concept wears many names, and the concept itself has soft boundaries.

Medicine has not solved this; it has only catalogued it. The Unified Medical Language System, described by Bodenreider, gathers more than sixty vocabularies into a single map that links names to concepts (Bodenreider, 2004). The map is many-to-many: in its 2004 release it held about 2.5 million names for some 900,000 concepts. It does not choose one name per concept; it records them all. The dream of one concept, one term did not come true, and our tools simply store every name instead.

This is why a formal ontology does not simply replace the text. An ontology is built to state what exists, in clean and consistent terms (Smith and Ceusters, 2010); a clinical note states what a clinician believes at one moment, with its hedges, its timeline, and everything left unsaid. The two are complementary, not interchangeable: the UMLS can list every name for a heart attack, but it cannot record that this patient's was *probable*, *three days ago*, and now *resolved*. Linking the running text to the codes means translating between the two, which is harder than looking a word up in a table.

Naming in medicine is not neutral. As Ian Hacking observed, classifying people can change the people classified, so a label can reshape the very thing it names (Hacking, 1999). And the registers sit measurably apart: the language of patients leans on pronouns and everyday words, where clinical and scientific writing pile up nouns and exact terms (Biber, 1988).

The three texts complete the picture. The same medicine is written in three ways. They differ in how much there is, from the rare clinical note to the abundant article to the even more abundant page for patients. They also differ in form, in reasoning, and most of all in how they name things. The text we can gather most easily is the furthest from the text we actually need. So using public data for the clinic is not just a question of quantity: it means crossing from one register to another, and turning the many names of a thing into the one the clinic uses. That is the problem the rest of this thesis takes up.

Part II

Related Works

5 Language Models

5.1 Introduction

Long before computers, people already had the idea that you can guess the next symbol in a text from the ones that came before. In 1913 the mathematician Andrei Markov tried it out by hand. He took twenty thousand letters from Pushkin's *Eugene Onegin*, marked each one as a vowel or a consonant, and counted: a vowel was followed by another vowel only 13% of the time, but by a consonant 66% of the time (Markov, 2006). Letters do not stand alone. What comes before shapes what comes next, and you can measure it.

Claude Shannon turned this counting into a theory. He treated a text as a random source and defined its *entropy*, the average information each symbol carries, as

$$H = -\sum_i p_i \log_2 p_i,$$

measured in bits (Shannon, 1948). This number is at its highest when every symbol is equally likely, and it gets smaller as some symbols become easier to guess.

Shannon showed this with a simple demonstration. Generate letters independently and the result is noise, *xfoml rxkhrjffjuj zlpwcfwkcyj*. Keep three-letter statistics and the output is already pronounceable and nearly English, *in no ist lat whey cratict froure birs*. Stitch whole words together by pairs and the fragments start to read like real sentences, *the head and in frontal attack on an english writer that the character of this point*. Each step uses a little more context, and each step looks more like the language.

How predictable is English, then? Shannon answered with an experiment in which a person guessed the next letter of a text one symbol at a time (Shannon, 1951). Shown the preceding text, his reader guessed correctly about two times in three, usually on the first try; from how the guesses fell, Shannon derived bounds on the entropy. An alphabet of 27 symbols, the 26 letters and a space, could in principle carry $\log_2 27 \approx 4.76$ bits per character. Yet given enough preceding text, Shannon estimated the true entropy of English at about one bit per character, somewhere between roughly 0.6 and 1.3 (Figure 5.1). Nearly four of every five possible bits are redundant, already fixed by what came before. That redundancy is the whole opportunity: a model that predicts the next symbol well has learned how the language works.

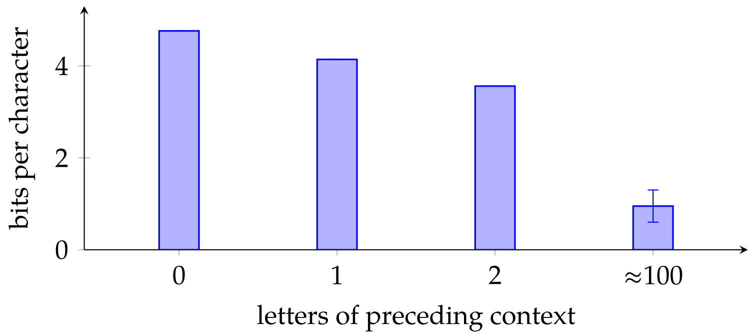

Figure 5.1: The entropy of printed English falls as more preceding context is used, from $\log_2 27 \approx 4.76$ bits per character with no context to about one bit (between 0.6 and 1.3) once roughly a hundred letters are taken into account (Shannon, 1951).

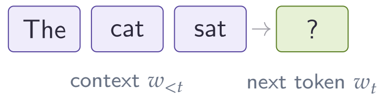

Figure 5.2: A language model predicts each token from the ones before it. The probability of a whole text is the product of these next-token predictions.

This is what a language model is. By the chain rule, the probability of a whole text factorises exactly into a product of next-symbol predictions,

$$P(w_1, \dots, w_T) = \prod_{t=1}^{T} P(w_t \mid w_{<t}),$$

so modeling language reduces to estimating a single conditional distribution: the probability of the next token given those before it (Figure 5.2). Every model in this chapter is an estimator of that one quantity, and the history of the field is the history of estimating it better.

Estimating that conditional is harder than it sounds. The simplest idea, the one Markov and the early n-gram models used, is to count: see how often each short sequence is followed by each next word. But counting runs out almost at once. Most word sequences never appear twice, even in a large corpus, so the counts are zero exactly where we need them. Much of the history of language modeling is the search for estimators that get around this problem, and the rest of this chapter follows that search, from n-gram counts to neural networks to the Transformer.

For this thesis the point is practical. Clinical information extraction rests entirely on these models: they are pretrained on large corpora and then adapted to a task. Adaptation is the hard part, because the target combines three scarcities at once, French, biomedical, and clinical, and, as the previous part showed, the text we can gather most easily is the least like the text we need. This chapter reviews the methods that make this adaptation possible: the architectures and pretraining objectives, how they scale, how they are specialised to a domain through continual pretraining, and the tokenization and long-context choices that matter for clinical documents. Continual pretraining, the mechanism at the heart of this thesis, is treated last and in the most detail.

5.2 From Statistical to Neural Language Models

5.2.1 N-gram models

The first estimators keep only a short history. Markov's assumption is that the next token depends only on the last few, so the full conditional $P(w_t \mid w_{<t})$ is approximated by $P(w_t \mid w_{t-n+1}, \dots, w_{t-1})$, the probability of a token given the $n-1$ before it. These probabilities are read straight from counts in a training corpus. The trouble is the one raised above, only sharper: the number of possible n-grams grows as $|\mathcal{V}|^n$, so for any real vocabulary most n-grams are never seen and their counts are zero. Smoothing methods, of which Kneser-Ney smoothing (Ney et al., 1994) is the best known, move some probability onto unseen sequences and fall back to shorter contexts when a longer one is missing. They made n-gram models good enough to be the state of the art for years. But they cannot share what they learn between similar words: *heart attack* and *cardiac arrest* are two separate entries in the table, with nothing in common.

5.2.2 Neural language models

The neural language model fixes exactly that. Bengio et al. (2003) replace the count table with a small neural network. Each word is first mapped to a vector, its *embedding*; the network reads the embeddings of the context and predicts the next word with a softmax over the vocabulary. Because it learns the embeddings at the same time as the prediction, words used in similar ways end up with similar vectors and can pool their evidence. A sequence never seen in training is no longer a dead end, because its words resemble others the model has seen. This alone cut the perplexity of the best n-gram models by about a quarter. The embeddings turned out to be useful on their own: word2vec (Mikolov et al., 2013) showed that simple arithmetic on them captures regularities of meaning; the textbook example is that $king - man + woman$ lands near *queen*. These early vectors were still *static*: one vector per word, whatever the sentence. ELMo (Peters et al., 2018) made them contextual, giving a word a different vector in each sentence, and showed that representations pretrained on plain text and then reused improve a wide range of tasks. This is an early form of the pretrain-then-adapt recipe the rest of this thesis depends on.

5.2.3 Recurrent networks

All of the models so far read a fixed window of context. A *recurrent neural network* removes that limit by reading the sequence one token at a time and carrying a running summary forward. The idea goes back to Rumelhart and McClelland (1987) and was first made to work as a language model by Mikolov et al. (2010). At each step t the network takes the current token x_t and the previous hidden state h_{t-1}, and combines them into a new hidden state,

$$h_t = f(x_t, h_{t-1}),$$

where f is a small learned function and h_t is a fixed-size vector that summarises everything read up to position t. The next token is predicted from h_t with a softmax over the vocabulary, just as in the neural model above. Because the hidden state is passed from one step to the next (Figure 5.3), a prediction can in principle depend on the whole history rather than the last $n-1$ tokens.

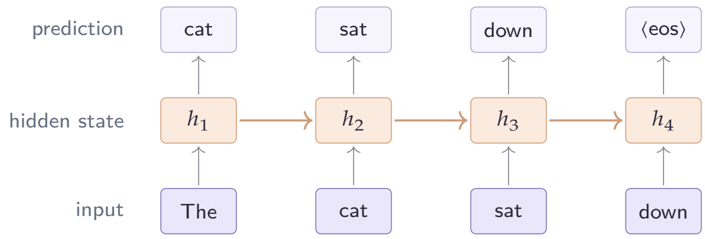

Figure 5.3: A recurrent language model, drawn one step per token. At each position the network combines the current input with the hidden state carried over from the previous step (the horizontal arrows) and predicts the next token, so the running state lets a prediction depend on the whole past rather than a fixed window.

In practice this memory is short. The network is trained by propagating the error back through every step, and Pascanu et al. (2013) showed that this backward signal is scaled by a similar factor at each step, so it shrinks or grows geometrically as it travels into the past (Figure 5.4). When the factor is below one the gradient vanishes and the model cannot learn links between distant tokens; when it is above one the gradient explodes and training becomes unstable. Bengio et al. (1994) had already identified this *vanishing* and *exploding gradient* problem as inherent to plain recurrence.

The *long short-term memory* network, or LSTM, was built to ease it (Hochreiter and Schmidhuber, 1997). Alongside the hidden state it keeps a separate memory cell that runs along the sequence, controlled by a set of *gates* that decide at each step what to erase from the cell, what to write into it, and what to read out. Because information can travel along the cell almost unchanged, useful facts survive over longer spans, and so does the gradient that carries them. A lighter variant, the gated recurrent

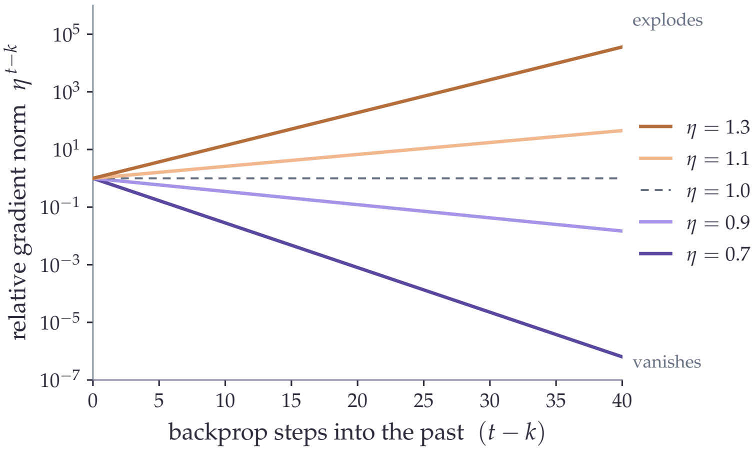

Figure 5.4: How the gradient of a recurrent network scales as it is propagated $t-k$ steps back in time, following the geometric bound η^{t-k} of Pascanu et al. (2013): below one it vanishes and above one it explodes, so links across long spans are hard to learn. The curves are this analytical bound, not measured gradients.

unit, drops one of the gates and performs comparably (Chung et al., 2014). LSTMs became the standard language model of the early 2010s.

Two limits remained. Even with gates, very long dependencies are hard to capture, and the step-by-step processing cannot be parallelised, which makes training on large corpora slow. A related weakness showed up in translation, where an encoder had to compress a whole input sentence into its final hidden state before a decoder could read it. Bahdanau et al. (2015) relieved this bottleneck by letting the decoder look back at all of the encoder's states through an early form of attention. Long-range dependencies, parallelism, and this idea of direct access between positions are what the next architecture was built around.

5.3 Transformer

5.3.1 Self-attention

The Transformer (Vaswani et al., 2017) keeps the goal, predicting the next token, and changes how context is read. Its core operation is *self-attention*, which lets every token look directly at every other. From the input representations X, the model computes three projections, a set of queries, keys, and values:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V.$$

Each token's query is compared with every key through a dot product, which measures how relevant the other tokens are to it. These scores are turned into weights with a softmax, and the token's new representation is the weighted average of the values:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V.$$

The division by $\sqrt{d_k}$, the size of the key vectors, keeps the dot products from growing too large, which would push the softmax to put almost all the weight on a single token.

Two things follow. Any two tokens now interact in a single step, however far apart they are, whereas a recurrent network had to pass information through every position in between. And since the scores for all positions are computed together as matrix products, the whole sequence is processed at once, which is what made training on very large corpora practical. The price is cost: every token is compared with every other, so time and memory grow with the square of the sequence length, $O(L^2)$. In practice the operation runs in several *heads* side by side, each with its own projections, so that different heads can follow different things, such as syntax, meaning, or position.

5.3.2 Positional encodings

Self-attention has one blind spot: it treats its input as a set, with no sense of order. *The cat sat* and *sat the cat* look the same to it, so position has to be supplied separately. The original Transformer added fixed sinusoidal signals to the embeddings (Vaswani et al., 2017), and BERT instead learned a vector for each position (Devlin et al., 2019); both encode an *absolute* position and cope badly with sequences longer than those seen in training. Later methods encode *relative* position. ALiBi adds a small penalty that grows with the distance between two tokens (Press et al., 2022), and RoPE rotates the queries and keys by an angle that depends on their position, so that their dot product ends up depending only on how far apart they are (Su et al., 2024). RoPE handles long and variable lengths well and is now the default in recent models such as LLaMA, Mistral, and ModernBERT.

5.3.3 Encoders and decoders

The same Transformer block can be wired in two ways, and the only difference is which tokens each one is allowed to see (Figure 5.5). In an *encoder*, every token attends to the whole sequence, left and right. This two-sided view suits tasks that read a text and label it, such as classification or entity recognition, and it underlies BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019). In a *decoder*, a token may attend only to the tokens before it, a restriction imposed by a causal mask. This is what makes generation possible, one token at a time, as in GPT (Radford and Narasimhan, 2018) and LLaMA. A third design keeps both: an encoder to read an

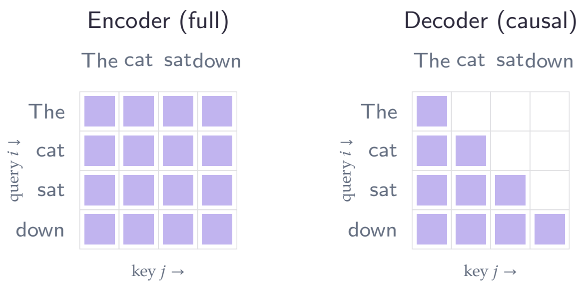

Figure 5.5: Attention masks as matrices. Row i is a query token and column j a key token; a filled cell means token i may attend to token j. The encoder allows every pair, so the matrix is full; the decoder allows only $j \leq i$, the lower triangle, so each token sees itself and the tokens before it but none of those that follow.

input and a decoder to write an output, as in T5 (Raffel et al., 2020), which is natural for translation and other input-to-output tasks but less common today. The two are built the same way. What really separates them is the training objective, the subject of the next section.

5.4 Pretraining Objectives

The architecture fixes how a model reads its context; the pretraining objective fixes what it is asked to predict, and that choice shapes the representations it learns. Two objectives dominate, and they differ in a way that matters for the rest of this thesis (Figure 5.6).

5.4.1 Causal language modeling

The most direct objective follows the chain rule: predict the next token. The model reads the sequence once and, at each position t, produces a probability distribution over the vocabulary for the token that should come next. Training lowers the negative log-probability it assigns to the token that actually follows, summed over the whole sequence,

$$\mathcal{L}_{\text{CLM}} = -\sum_{t=1}^{L} \log P(w_t \mid w_{<t}),$$

where $w_{<t}$ is the tokens before position t. A causal mask is what lets this happen in a single pass: each position may look only to its left, so the model can be scored at every position at once without seeing the very token it must predict. Every position therefore yields a prediction and a gradient, so the supervision is dense. This is the objective behind GPT (Radford and Narasimhan, 2018) and the modern decoder models, and it fits generation naturally: at inference the model does exactly what it was trained to do, continue the text.

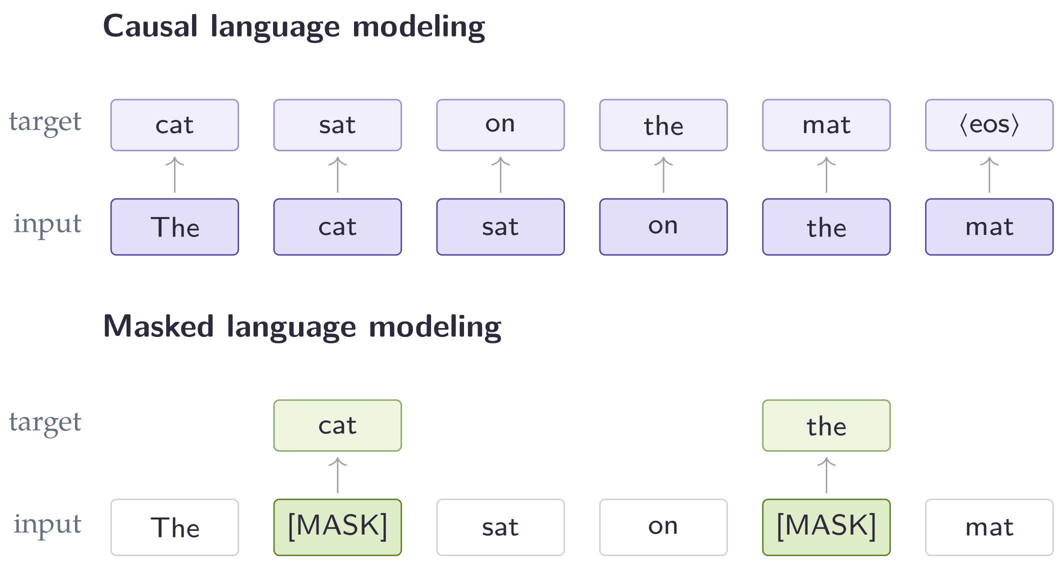

Figure 5.6: The same sentence under the two objectives. The bottom row is the input and the boxes above are the targets the model must predict, one arrow per predicted position. Causal language modeling predicts the next token at every position, so every token gives a training signal (dense). Masked language modeling hides about 15% of the tokens, here two, and predicts only those from the context on both sides, so most positions give no signal (sparse). This gap in how much signal each token carries is what the encoder pretraining of Chapter 12 exploits.

5.4.2 Masked language modeling

Understanding a text, rather than continuing it, calls for a different objective. BERT (Devlin et al., 2019) corrupts the input by replacing some tokens with a special `[MASK]` symbol, then trains the model to put the originals back. At each masked position the model predicts the missing token from the context on both sides, and the loss counts only those positions,

$$\mathcal{L}_{\text{MLM}} = -\sum_{t \in \mathcal{M}} \log P(w_t \mid \mathbf{w}_{\setminus t}),$$

where $\mathcal{M}$ is the set of masked positions and $\mathbf{w}_{\setminus t}$ is the rest of the (corrupted) sequence. About 15% of the tokens are chosen; of those, 80% become `[MASK]`, 10% are swapped for a random word, and 10% are left as they are, a mix that stops the model from only ever reacting to the `[MASK]` symbol. Seeing the whole sequence at once makes the representations bidirectional, which is what makes encoders strong at classification and entity recognition. The price is that only the masked positions produce a gradient, about one token in seven, against every token under the causal objective.

5.4.3 Mixing the two

The two objectives trade the same quantity against itself: causal training gives a signal at every token but only looks left, while masked training looks both ways but learns from few tokens. Recent work tries to get both at once. Some encoders are pretrained with a causal phase followed by a short masked phase, which beats masked pretraining alone (Gisserot-Boukhlef et al., 2025); others alternate the two objectives during training (Team, 2024). This line is central to the thesis: Chapter 12 shows that a temporary detour through causal pretraining leaves a lasting and useful mark on an encoder. An earlier idea, ELECTRA's replaced-token detection (Clark et al., 2020), also sought a denser signal, by training the model to spot tokens a small generator had swapped in, though recent encoders such as ModernBERT (Warner et al., 2024) show that a good architecture with plain masking is already enough.

With both the architecture and the objective settled, the next question is how far this recipe scales.

5.5 Scaling Laws & LLMs

5.5.1 Scaling

The Transformer settled how a model reads its context, and the pretraining objective settled what it learns from raw text. The remaining lever is size. The natural question is how far performance keeps improving as we add parameters and data, and whether anything new appears along the way. The answer, found around 2020, changed both how the field builds models and how it uses them.

Until then, a pretrained model was put to work by *fine-tuning*. After pretraining, each new task needed a further round of supervised training on labelled examples, producing a separate specialised copy of the model. Brown et al. (2020) showed a different route. Their model, GPT-3, a decoder with 175 billion parameters, can carry out a task from nothing more than a description and a few examples written into its input, with no change to its weights. They call this *in-context learning*, and the abstract states the mechanism plainly: the model is "applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text" (Brown et al., 2020). The practical shift is large. Instead of collecting labelled data and training a classifier for every task, one writes a prompt.

How much help the prompt carries defines three settings. In the *zero-shot* setting the model is given only the task, at most a short instruction; in *one-shot* it also sees one worked example; in *few-shot* it sees several, typically ten to a hundred, as many as fit in the 2048-token context window (the most text the model can read at once). Figure 5.7 places the three side by side on a simple arithmetic task. The examples are not training data, they are text the model reads before answering, and the same idea works across very different tasks: unscrambling letters (`skicts` → `sticks`),

Task	Zero-shot	One-shot	Few-shot
Two-digit addition	76.9	99.6	100.0
Two-digit subtraction	58.0	86.4	98.9
TriviaQA (open QA)	64.3	68.0	71.2

Table 5.1: GPT-3 (175B) accuracy as the number of in-context examples grows, with no weight updates. Numbers are verbatim from Brown et al. (2020).

word analogies (*lull is to trust as cajole is to compliance*), or translating a sentence into French.

zero-shot

```
Q: What is 98 plus 45?
A:
```

one-shot

```
Q: What is 12 plus 7?
A: 19
Q: What is 98 plus 45?
A:
```

few-shot

```
Q: What is 12 plus 7?
A: 19
Q: What is 31 plus 6?
A: 37
Q: What is 98 plus 45?
A: 143
```

Figure 5.7: The same question put to GPT-3 in three ways. With no example (zero-shot) the model sees only the question; with one or a few solved examples in the prompt (one- and few-shot) it picks up the pattern and answers (in green). No weights are updated; the examples are just text. The format and the final example are from Brown et al. (2020).

The striking part is that more demonstrations help, and they help more as the model grows. On two-digit addition GPT-3 goes from 76.9% with no example to 99.6% with one and 100% with a few; on the TriviaQA question-answering benchmark it goes from 64.3% to 68.0% to 71.2% (Table 5.1). The same trend holds across many tasks, and Brown et al. (2020) note that "the gap between zero-, one-, and few-shot performance often grows with model capacity", which they read as a sign that larger models are better at learning from their context. A capability that is weak or absent in small models and strengthens with scale is what later work called *emergent*.

These gains are not a fluke, and they rest on something the model is trained to do directly. Below the downstream abilities sits the loss, the model's average surprise on held-out text, and the loss falls with scale in a strikingly regular way. Kaplan et al. (2020) showed that it follows a smooth *power law* in two quantities, the number of parameters N and the number of training tokens D, which is a straight line when plotted on logarithmic axes. The loss drops predictably as either grows, so model design becomes a forecast: pick a budget of computation and read off the loss to expect. Their estimate, though, placed the optimum at large models trained on relatively little data.

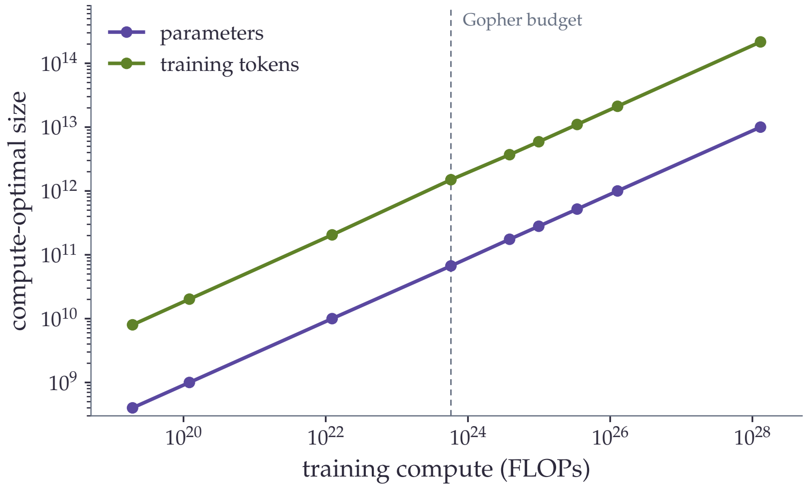

Figure 5.8: The compute-optimal model size and the compute-optimal number of training tokens rise together with the training budget, as the parallel lines show. Points are verbatim from Table 3 of Hoffmann et al. (2022); the paper's loss curves are not reproduced here.

Hoffmann et al. (2022) revisited this and found the trade-off was wrong. Fitting, on over 400 trained models, a loss law of the form

$$L(N, D) = E + \frac{A}{N^{\alpha}} + \frac{B}{D^{\beta}},$$

they estimated $E = 1.69$, $A = 406.4$, $B = 410.7$, $\alpha = 0.34$ and $\beta = 0.28$. Here E is the loss that would remain with unlimited data and parameters, a floor set by the language itself, while the two fractions measure how far a finite model and a finite dataset still sit above it. The cost of training, counted in arithmetic operations (FLOPs), grows with both, roughly as $6ND$, so for a fixed budget the law gives the best split between model size and data, the *compute-optimal* point. The answer is that the two should grow together: each time the model doubles, the data should double with it (Figure 5.8). By the authors' own projection this means about twenty training tokens per parameter, far more data than earlier models had used.

To prove the point they trained Chinchilla, a 70-billion-parameter model, on 1.4 trillion tokens, and it beat Gopher, a 280-billion model trained on the same compute, scoring 67.6% against 60.0% on MMLU, a multiple-choice exam spanning fifty-seven subjects. Most large models before it had simply been undertrained: built large, then fed too little data to fill them.

Chinchilla optimises the cost of *training* a model once. Touvron et al. (2023) pointed out that this ignores the cost of *using* it afterwards: a model served to many users is run countless times, and each run costs computation too, its *inference* cost.

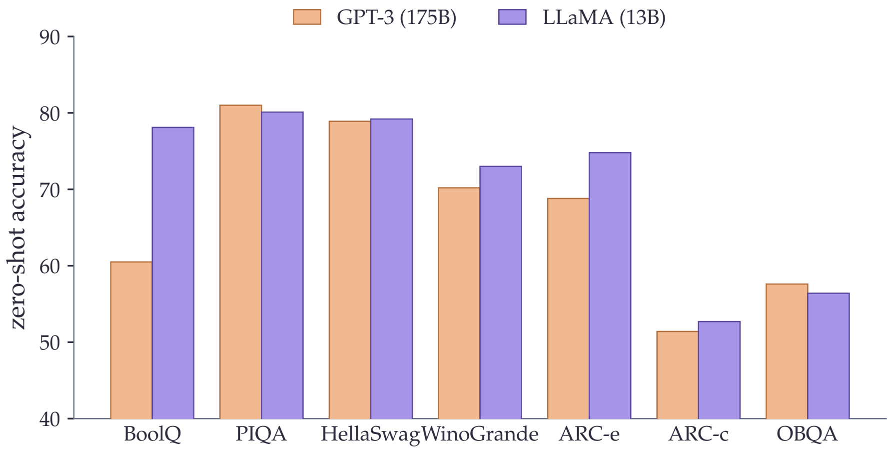

Figure 5.9: A 13-billion-parameter model trained on more tokens matches or beats GPT-3, which is more than ten times larger, on most of these zero-shot reasoning tasks. Numbers are verbatim from Table 4 of Touvron et al. (2023).

As they put it, "a smaller one trained longer will ultimately be cheaper at inference". They pushed small models well past the compute-optimal point, finding that a 7-billion model keeps improving even after a trillion tokens. The resulting LLaMA models, trained only on publicly available data, make the case plainly: the 13-billion model outperforms GPT-3 (175B) on most benchmarks "despite being 10× smaller" (Figure 5.9), and the 65-billion model is competitive with Chinchilla and the 540-billion PaLM. Because their weights were released, they also brought capable models within reach of ordinary research budgets.

5.5.2 Instruction Tuning

The previous section showed that a large base model can already perform a task from a few examples placed in its prompt. That ability is fragile. It leans on well-chosen demonstrations and on phrasing the task as a text to be continued, and it is weakest in the zero-shot case, with no examples at all. A base model is trained only to predict the next token, so asked a plain question it may answer, but it may equally continue with another question. *Instruction tuning* closes this gap. The model is fine-tuned on many existing datasets, each rewritten as a natural-language instruction paired with its answer, after which it follows instructions phrased in the same way, including instructions for tasks it never saw during tuning.

Wei et al. (2022a) introduced the idea and the name. Taking a 137-billion-parameter model and tuning it on more than sixty datasets phrased as instructions, they found that its zero-shot accuracy rose above that of the much larger GPT-3 on twenty of the twenty-five evaluation datasets they tried. The gain came not from more scale but from teaching the model to read instructions. Sanh et al. (2022) made the same case at a smaller size: an eleven-billion model tuned on a wide range of prompted tasks matched or beat GPT-3, about sixteen times larger, on nine of eleven held-out tasks. What matters most is the breadth of the tuning mixture. Increasing the number of distinct tasks improves generalisation to unseen ones, with most of the benefit reached after a few hundred tasks (Chung et al., 2024), and collections such as Super-NaturalInstructions were built to supply that breadth, gathering more than 1,600 tasks (Wang et al., 2022). Such instruction datasets are costly to assemble by hand, which opened a second route: Wang et al. (2023b) showed that a model can bootstrap its own instruction data, generating new instructions and answers from a small seed set and tuning on the filtered result, an approach that seeded much of the open instruction-tuning that followed.

5.5.3 Aligning Assistants with Human Feedback

Instruction tuning teaches a model to attempt the task, but it does not by itself make its answers the ones a person would prefer. A useful assistant should be, in an early statement of the goal, helpful, honest, and harmless (Askell et al., 2021). Reaching that goal from supervised data alone is hard: for an open-ended question there is no single correct target to imitate, only answers that are better or worse than others. The way around this is to learn from comparisons. Christiano et al. (2017) had shown, for control problems, that an agent can be trained from human judgments of which of two behaviours is better, with no hand-written reward, by fitting a *reward model* to those judgments and then optimising against it.

Ouyang et al. (2022) brought this recipe to language in three steps: fine-tune the model on human-written demonstrations, collect human rankings of model outputs and fit a reward model to them, then optimise the model against that reward with reinforcement learning. The result, *reinforcement learning from human feedback* (RLHF), was striking: human raters preferred the answers of a 1.3-billion-parameter aligned model over those of the hundred-times-larger 175-billion GPT-3. This is the recipe behind ChatGPT and the assistants that followed it, and it is what let a non-expert state a request in plain language and get a useful answer. Bai et al. (2022a) built a helpful and harmless assistant the same way and reported that alignment training improved performance across most benchmarks while remaining compatible with specialised skills such as coding. Because the expensive part is the human labelling, later work sought to reduce it: Constitutional AI replaces human judgments of harmful content with judgments the model itself makes against a short written list of principles, training from AI feedback rather than human feedback (Bai et al., 2022b).

5.5.4 Reasoning and Verifiable Rewards

The most recent shift concerns problems that have a checkable answer. Wei et al. (2022b) observed that prompting a model to produce a *chain of thought*, a series of intermediate steps before its final answer, sharply improves its accuracy on multi-step problems, with no change to its weights, and sampling several such chains and taking the majority answer improves it further (Wang et al., 2023a). This made reasoning something one could first elicit and then train for. On tasks such as mathematics or programming the answer can be checked automatically, by matching the solution or running the code, which yields a reward computed by a rule rather than predicted by a learned model. Training against such a signal has been named *reinforcement learning with verifiable rewards* (Lambert et al., 2024); because the reward is a correctness check, it cannot be gamed the way a learned reward model can.

The DeepSeekMath work introduced an efficient optimiser for this setting, Group Relative Policy Optimization, which drops the separate value network of standard policy optimisation and instead judges an answer against a group of sampled answers to the same question (Shao et al., 2024). DeepSeek-R1 then pushed the idea to its limit: a model trained by reinforcement learning alone, with no supervised fine-tuning first, developed reasoning behaviours such as self-checking and changing strategy from the reward signal, and this reasoning could afterwards be distilled into smaller models (Guo et al., 2025). The strength of the approach is also its boundary. Verifiable rewards exist for mathematics and code, where correctness is a computable rule, but not for open-ended writing, where there is no automatic check.

Two cautions close this picture. As models are tuned to follow instructions and increasingly trained on data shaped or generated by other models, the line between training and evaluation blurs, which raises the problem of *benchmark contamination*: when test items, or text closely resembling them, leak into the training data, reported scores stop measuring genuine generalisation. And all of these advances, from instruction tuning to verifiable-reward reasoning, concern general-purpose models trained on broad web text. Turning such a model into one that performs in a narrow, low-resource domain is a separate problem, and the most direct lever for it is to keep pretraining on text from that domain.

5.6 Continual Pretraining

The pivot just reached, specialising a model to a domain by continuing its pretraining on in-domain text, is the mechanism this thesis turns on, so we look at it closely. This section asks three questions in turn: what continued pretraining is, why it is delicate, and how recent work makes it reliable. The biomedical models that put it into practice come in the next section.

5.6.1 A Second Phase of Pretraining

The idea is simple. Between the generic pretraining of Section 5.4 and fine-tuning on a labelled task, one inserts a further phase of self-supervised training on text closer to the target. Gururangan et al. (2020) studied this across four domains, including biomedicine, and eight classification tasks, and named two variants. *Domain-adaptive pretraining* continues training on a large in-domain corpus; *task-adaptive pretraining* continues it on the unlabelled text of the task itself. They found that the two help in both high- and low-resource settings, that task-adaptive pretraining helps even after the domain phase, and that it pays off even when the task corpus is small (Figure 5.10).

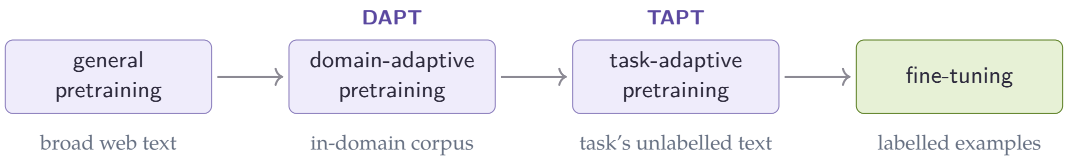

Figure 5.10: Continued pretraining inserts one or two self-supervised phases between general pretraining and fine-tuning: domain-adaptive pretraining on a large in-domain corpus, then task-adaptive pretraining on the task's own unlabelled text (Gururangan et al., 2020).

This second phase is one case of a broader topic. A recent survey organises continual learning for language models into continued pretraining, continual instruction tuning, and continual alignment, and groups the methods that protect earlier knowledge into replay, regularisation, and architecture-based families (Shi et al., 2025). The rest of this section keeps to the first case, continued pretraining, since it is the one this thesis uses.

5.6.2 Catastrophic Forgetting

Continuing to train a network has a cost. When a connectionist model is trained on new material, the weights that encoded the old material are overwritten, a failure named *catastrophic interference* when it was first described (McCloskey and Cohen, 1989). The standard remedy keeps the new training from straying too far: elastic weight consolidation adds a penalty that, in the authors' words, works by "selectively slowing down learning on the weights important for" the earlier tasks (Kirkpatrick et al., 2017). For a model adapted to a domain the same risk applies in reverse: pushing it toward biomedical text can erode the general competence it began with. The danger is uneven. Yıldız et al. (2024) report that smaller models are the most sensitive, showing the highest rates of both learning and forgetting, so a small domain encoder is more exposed than a large one.

Not every drop in a benchmark score is true forgetting, however. Zheng et al. (2025) argue that the early steps of a new training phase often disturb a model's *alignment*, its learned way of exposing what it knows, more than its knowledge, so that “performance drops often reflect a decline in task alignment rather than true knowledge loss” and much of the loss can be recovered. The lesson for what follows is to read a post-adaptation drop with care, and to design the adaptation so that it does not disturb more than it must.

5.6.3 Making Continual Pretraining Work

Recent work shows that, done carefully, continued pretraining need not forget and need not cost much. Ibrahim et al. (2024) reduce the recipe to three ingredients: re-warming the learning rate, which has decayed to almost nothing by the end of pretraining, then re-decaying it, and replaying a small fraction of the original data alongside the new domain. This combination, they report, is “sufficient to match the performance of fully re-training from scratch on all available data” while “using only a fraction of the compute”, and they verify it at both the 405-million and 10-billion parameter scales. Continued pretraining, in other words, can reach the same place as starting over, far more cheaply.

Two refinements sharpen this. First, less data can be better than more: choosing a small, relevant slice of the domain corpus rather than all of it can match or beat training on the whole. Xie et al. (2024) report data-selection strategies that “outperform vanilla continual pre-training’s performance with just 10% of corpus size and cost”, with no loss on general tasks. Second, the adaptation does not improve smoothly. Guo et al. (2024) document a *stability gap*, a temporary fall in in-domain performance at the start of continued pretraining followed by a recovery, and show that training for several passes over a high-quality subset and mixing in pretraining-like data smooths it out, raising the average medical-task accuracy of a three-billion model from 36.2% to 40.7% while using only 40% of the training budget. The common thread is that continued pretraining rewards care in the learning-rate schedule, the amount of replay, and the choice of data.

5.6.4 From Scratch or Continue?

The alternative to continuing is to pretrain a specialised model from scratch on in-domain text. The two poles have canonical biomedical examples. BioBERT continued the pretraining of BERT on PubMed abstracts and full-text articles (Lee et al., 2020), the continued-pretraining route; PubMedBERT was instead pretrained from scratch on biomedical text, with a vocabulary built from that text rather than inherited (Gu et al., 2022). The trade-off is clear in principle. Training from scratch can fit a vocabulary to the domain and cannot forget what it never learned, but it is expensive and throws away the general competence of an existing model; continuing is cheap and keeps that competence, but inherits a tokenizer and an initialisation built for general text.

Which wins is contested. Gu et al. (2022) found that pretraining from scratch on in-domain text, with an in-domain vocabulary, outperformed continued pretraining from a general model across a suite of biomedical tasks, and argued that the general-text vocabulary of a continued model holds it back. The general-purpose evidence above points the other way: with the right learning-rate schedule and a little replay, continued pretraining can match a from-scratch model at a fraction of the cost (Ibrahim et al., 2024). The disagreement is not settled, and it may depend on the architecture. Continued pretraining has often disappointed for generative decoder models: comparing biomedical large language models with their general-purpose counterparts, Dorfner et al. (2024) found the specialised models mostly the weaker of the two, sometimes by a wide margin, with one eight-billion biomedical model scoring 30% against 64.3% for the general model it was built from on New England Journal of Medicine case challenges. The bidirectional, masked objective of encoders may behave differently. Whether continued pretraining reliably specialises a model, and how the answer depends on whether the model is an encoder or a decoder, is the open question that the next section takes up through the biomedical models built to answer it.

5.7 Biomedical Language Models

The previous section left an open question: does continued pretraining reliably specialise a model, and does the answer depend on the architecture? The biomedical models built over the past few years are the testbed. We look at them in three groups, the English encoders, the French models, and the generative decoders, and the from-scratch-versus-continue split runs through all three.

5.7.1 English Biomedical Encoders

The first biomedical encoders took the continued-pretraining route. BioBERT continued the pretraining of BERT on PubMed abstracts and full-text articles, and reported clear gains over the general model: about half a point of F1 on biomedical named-entity recognition, nearly three on relation extraction, and twelve points of mean reciprocal rank on question answering (Lee et al., 2020). The gains were real and the recipe was cheap, since it reused the existing weights, and this established the mixed-domain paradigm: start from a general model, keep training on domain text.

A second line questioned that paradigm by training from scratch, with a vocabulary built from domain text rather than inherited. SciBERT measured what is at stake: between BERT's general vocabulary and one learned from scientific text, only 42% of frequent tokens overlap, so more than half of the common scientific words are split into pieces by a general tokenizer (Beltagy et al., 2019). Training from scratch on 1.14 million papers fixes this. PubMedBERT made the strongest version of the claim: pretraining from scratch on biomedical text, with an in-domain vocabulary, "substantially outperforms continual pretraining of generic language models", and they released the BLURB benchmark that became the standard yardstick for the field (Gu et al., 2022). BioLinkBERT added a twist, pretraining from scratch on PubMed with its citation links so that related documents are seen together, for a further gain on biomedical question answering (Yasunaga et al., 2022b). Clinical text adds a constraint of its own: ClinicalBERT was trained on the notes of the MIMIC-III intensive-care database, and while its weights are public the underlying notes are not, sitting behind credentialed access (Alsentzer et al., 2019). The current strongest encoder follows the continued route again, on a far larger and more varied corpus: BioClinical ModernBERT continues a modern long-context encoder on more than 53 billion tokens drawn from twenty datasets across institutions (Sounack et al., 2025), with the architectural changes that make it long-context deferred to Section 5.8.

5.7.2 French Language Models

French has its own lineage. CamemBERT adapted the RoBERTa recipe to French web text and made a point that matters for low-resource work: about four gigabytes of text gave results "as good as those obtained using larger datasets (130+GB)" (Martin et al., 2020). FlauBERT appeared in parallel, trained on a heterogeneous French corpus (Le et al., 2020). Later models refreshed the recipe: CamemBERTa adopted the DeBERTaV3 architecture and its replaced-token-detection objective for better data efficiency (Antoun et al., 2023), and CamemBERT 2.0 retrained the family on newer data to counter the drift of language over time (Antoun et al., 2024).

The two dedicated French biomedical encoders were both trained from scratch. DrBERT trained a RoBERTa model on NACHOS, a corpus of French biomedical and clinical text, and its experiments argued for the from-scratch route: models trained from scratch gave the best results, and continuing the pretraining of a French general model never reached the top on any of their tasks (Labrak et al., 2023). The picture is not one-sided, though, even in their own results: continuing the pretraining of an *English* biomedical model on French data did work, which they read as cross-lingual transfer. They also found that publicly available biomedical text could match or beat private hospital data, a point that bears directly on the use of public corpora for the clinic. AliBERT, the other French biomedical encoder, was likewise trained from scratch, with a regularised Unigram tokenizer built for the domain (Berhe et al., 2023). Whether continuing from a French general model can be made to match a from-scratch biomedical one is, on this evidence, still an open question.

5.7.3 Biomedical Decoders

Generative models were specialised in the same way, by continuing the pretraining of a general large language model on medical text, but at a much larger scale. BioMistral continued Mistral 7B on PubMed Central and reported that the adapted model beat the general instruction-tuned Mistral on eight of ten medical question-answering tasks; to keep the general ability that specialisation can erode, the authors also merged the domain and general models, combining their strengths (Labrak et al., 2024). Meditron continued Llama-2, at seven and seventy billion parameters, on a curated medical corpus of about 48 billion tokens that deliberately mixes in one percent of general data as replay, and its seventy-billion model outperformed GPT-3.5 and Med-PaLM on medical benchmarks (Chen et al., 2023). PMC-LLaMA continued LLaMA on biomedical papers and books (Wu et al., 2024).

Each of these reports gains on its own medical benchmarks, yet a broader picture is more sober. Evaluating biomedical models against the general models they were built from, on tasks chosen to fall outside the fine-tuning data, Dorfner et al. (2024) found the specialised models mostly the weaker of the two, especially away from pure medical knowledge. The contrast with the encoders is the lesson of this section. For encoders, continued pretraining on domain text is close to a default win on extraction tasks; for large decoders, specialising on a domain is a riskier trade, where in-domain gains can come at the cost of general ability and do not always survive an independent test. Why the two architectures should behave so differently is not settled, and it is one of the threads this thesis follows. These models, finally, are built on architectures that have themselves moved on from the original Transformer, which is where we turn next.

5.8 Architectures modernes

5.8.1 Modernisation des encoders

DeBERTa (He et al., ICLR 2021):

- Disentangled attention: separe contenu et position
- Premier a battre l'humain sur SuperGLUE
- -> influence sur ModernBERT, CamemBERT 2.0

MosaicBERT (Portes et al., NeurIPS 2023):

- 30% masking au lieu de 15% BERT
- FlashAttention + ALiBi + GLU + unpadding
- BERT-base en 1.13h sur 8x A100 (~$20)
- -> fondation directe de ModernBERT

5.8.2 Efficient Attention

FlashAttention (Dao et al., 2022):

- Reorganise le calcul pour la hierarchie memoire GPU (tiling, kernel fusion)
- Meme $O(L^2)$ mais 2-4x plus rapide en pratique
- Standard maintenant

Autres:

- GQA (Ainslie et al., 2023): partage de KV-heads -> Llama-2, Mistral
- RMSNorm (Zhang & Sennrich, 2019): LayerNorm simplifie

5.8.3 Long context

Le probleme: BERT = 512 tokens, comptes-rendus cliniques = souvent 2000+. Tronquer = perdre de l'info diagnostique.

- Longformer (Beltagy et al., 2020): 4096 tokens, sliding window + global attention
- BigBird (Zaheer et al., 2020): 4096, sparse attention
- Clinical Longformer (Li et al., 2022): 4096, Longformer continue pretrained sur MIMIC-III
- ModernBERT (Warner et al., 2024): 8192 tokens
  - FlashAttention v2/v3
  - RoPE
  - Alternance local/global attention
  - 30% masking (de MosaicBERT)
  - -> 16x le contexte de BERT

Adaptations cliniques 2025:

- Clinical ModernBERT (Lee et al., 2025): PubMed + MIMIC-IV + ontologies
- BioClinical ModernBERT (2025): 53.5B tokens, 20 datasets cliniques -> SOTA actuel

5.9 Tokenization

5.9.1 Methodes subword

- BPE (Sennrich et al., 2016): merge les paires les plus frequentes
- SentencePiece (Kudo & Richardson, 2018): language-independent
- Unigram (Kudo, 2018): selection probabiliste

5.9.2 Domain mismatch

SciBERT: vocab general vs scientifique = 42% intersection seulement.

Exemple biomed francais:

- Tokenizer general: "echocardiographie" -> ["echo", "##cardi", "##ographie"]
- Tokenizer domaine: "echocardiographie" -> ["echocardiographi", "e"]

Tradeoff:

- From-scratch = vocab custom mais cher
- Continual PT = tokenizer sous-optimal mais efficient

5.10 Limites et transition

- Rarete des donnees: texte clinique prive (CNIL), modeles hospitaliers pas partageables -> besoin d'alternatives publiques -> Ch.3 Corpus Annotation
- Knowledge gap: LMs captent des patterns, pas la structure des ontologies. Codage medical = connaissances hierarchiques -> approches knowledge-enhanced -> Ch.4 Clinical IE
- Documents longs: comptes-rendus hospitaliers depassent les limites de BERT. Longformer et ModernBERT aident mais faut adapter au domaine

-> ce chapitre a couvert les fondations techniques du language modeling. Le chapitre suivant traite de comment construire et annoter des corpus pour entrainer ces modeles.

6 Pretraining Data: Corpora and Knowledge

6.1 Introduction

Le chapitre precedent couvrait les modeles. Maintenant faut les nourrir.

Les LMs modernes sont data-hungry:

- GPT-3 (Brown et al., 2020): 300B tokens
- LLaMA (Touvron et al., 2023): 1-2T tokens
- Chinchilla (Hoffmann et al., 2022): compute-optimal = plus de data, modele plus petit

La qualite et la composition du corpus de pretraining determinent largement les capacites du modele. "Data is the new bottleneck."

Deux types de donnees de pretraining:

- Texte brut: web crawls, articles scientifiques, livres
- Connaissances structurees: ontologies medicales, knowledge graphs

Pourquoi c'est central pour la these:

- Part 1 = construire un corpus biomed francais a partir de sources publiques
- Ch.6 = injecter des connaissances d'ontologies medicales pendant le pretraining
- Triple rarete: francais + biomedical + clinique

-> plan du chapitre: on part des corpus web generalistes, on zoome sur le filtrage qualite, puis sur les sources biomed specifiques, et enfin sur les ontologies medicales et comment les integrer dans le pretraining

6.2 Web-Scale Pretraining Corpora

6.2.1 Common Crawl

Source de base de presque tous les grands corpus de pretraining:

- Web crawl ouvert, 2.5B pages par snapshot mensuel, 250-400 TiB non compresse
- Snapshots mensuels depuis 2008, petaoctets cumules
- Contenu tres heterogene: articles, forums, spam, code, pubs, boilerplate...
- Pas utilisable directement -> necessite des pipelines de nettoyage lourdes

6.2.2 Premiers grands corpus

C4 (Raffel et al., 2020):

- Colossal Clean Crawled Corpus, 750GB d'anglais extrait d'un seul snapshot CC (avril 2019)
- Filtres: langdetect >= 99% anglais, suppression phrases incompletes, dedup
- Dodge et al. (2021) documentent le contenu: brevets, sites militaires US, texte machine-generated, exemples de benchmarks
- Blocklist filtering retire disproportionnellement du texte de/sur les minorites

The Pile (Gao et al., 2020):

- 800GB anglais, 22 sources diversifiees (PubMed, ArXiv, GitHub, StackExchange, Wikipedia, livres, etc.)
- Idee: diversite des sources > volume brut d'une seule source
- Open-source (EleutherAI)

OSCAR (Ortiz Suarez et al., 2019):

- Pipeline asynchrone pour classifier CC par langue
- Optimise pour infra moyenne/basse ressource (I/O-bound)
- Produit le corpus de CamemBERT: 138GB de texte francais
- Shuffled au niveau ligne pour eviter les problemes de copyright (ironiquement)

6.2.3 Corpus modernes (2024-2025)

FineWeb (Penedo et al., 2024):

- 15T tokens, 96 snapshots CC
- Documentation detaillee des strategies de dedup et filtrage

- Pipeline: extraction texte, filtrage langue, heuristiques qualite, dedup URL + MinHash
- FineWeb-Edu: sous-ensemble 1.3T tokens filtre pour contenu educatif (cf section suivante)

RedPajama-V2 (Weber et al., 2024):

- Dataset ouvert pour entrainer des LLMs
- Inclut subset francais (RPv2-Fr)
- Metriques de qualite pre-calculees pour faciliter le filtrage

TxT360 (Tang et al., 2024):

- ”Perfect blend” de sources
- Dedup globale sur 99 snapshots CC: 20TB -> 4.83T tokens (reduction 80%)
- Pipeline similaire a FineWeb + near-dedup globale additionnelle

Autres:

- Dolma (Soldaini et al., 2024): 3T tokens, corpus ouvert pour recherche (OLMo)
- ROOTS (Laurencon et al., 2023): 1.6TB, multilingue composite (BigScience/BLOOM)

-> on a des milliards de tokens de texte web, mais la qualite varie enormement -> comment selectionner les bons documents?

6.3 Data Quality and Curation

6.3.1 Filtrage heuristique

Approches classiques, devenues standard depuis Gopher (Rae et al., 2022):

- Regles: longueur min/max du document, ratio mots/symboles, proportion de majuscules, lignes dupliquees
- Filtrage par perplexite: KenLM entraine sur Wikipedia, documents a perplexite trop haute = bruit
- Language ID: fastText pour filtrer par langue
- Ces heuristiques sont reprises par presque tous les pipelines modernes (FineWeb, C4, Dolma)

Limites: les heuristiques sont necessaires mais insuffisantes. Elles retirent le bruit evident mais ne distinguent pas la qualite du contenu.

6.3.2 Classifieurs de qualite

Paradigme recent: LLM-as-annotator -> distillation vers petit classifieur -> filtrage a grande echelle.

FineWeb-Edu (Penedo et al., 2024):

- Llama-3-70B-Instruct annote 460K pages web sur "educational value" (echelle 0-5)
- Echelle additive: le LLM evalue chaque critere et construit le score pas a pas
- Prompt cible le niveau primaire/college pour eviter de favoriser les papiers trop techniques
- Distillation vers classifieur de regression (Snowflake-arctic-embed, F1 = 82% en binaire avec seuil >= 3)
- Classification de 15T tokens: 6K heures H100
- Retire 92% des donnees, bat quand meme le corpus complet sur MMLU, ARC, OpenBookQA

DCLM / DataComp-LM (Li et al., 2024):

- Classifieur FastText entraine pour separer instructions synthetiques (Open Hermes 2.5) du web general
- Favorise les structures Q&A resolues: question courte + reponse directe + raisonnement bref
- Utilise dans le mix de pretraining d'OLMo2

Nemotron-CC (NVIDIA, 2024):

- Classifieur de qualite multi-criteres (accuracy, clarity, coherence, etc.)
- Approche plus granulaire que le score educatif unique de FineWeb-Edu

6.3.3 Organisation par domaine

WebOrganizer (Wettig et al., 2025):

- Organise le web en taxonomies par topic et par format
- Annotations LLM distillees vers classifieurs efficaces
- Montre que FineWeb-Edu a un biais implicite vers Science & Technology et Academic Writing
- L'efficacite de FineWeb-Edu vient en partie de preferences de domaine alignees avec les benchmarks (MMLU, HellaSwag)
- Combiner organisation par domaine + filtrage qualite > qualite seule

6.3.4 Deduplication

- URL-level: retirer les duplicats exacts d'URL
- Document-level: MinHash + LSH pour near-duplicates
- TxT360: dedup globale sur 99 snapshots CC, reduction de 80%
- Crucial: la duplication biaise le training vers les textes les plus copies (boilerplate, templates)

6.3.5 Contamination

Le filtrage agressif pose un probleme de contamination des benchmarks:

- Deng et al. (2024): ChatGPT et GPT-4 devinent 52% et 57% des options manquantes de MMLU -> les QA pairs sont sur le web
- Xu et al. (2024): survey des techniques de detection de contamination dans les LLMs
- InfiniGram (Liu et al., 2024): outil pour identifier des exact matches dans les training sets

Tension fondamentale: filtrer plus agressivement = meilleurs scores benchmarks, mais potentiellement via contamination implicite plutot que vraie qualite.

-> ok le filtrage marche pour les corpus web generalistes, mais en biomedical c'est different: les corpus sont plus petits, le domaine est tres specifique, les criteres de qualite ne sont pas les memes -> quelles sources biomed existent?

6.4 Biomedical and Clinical Corpora

6.4.1 Grandes sources anglophones

PubMed:

- Base de donnees de la NLM (National Library of Medicine)
- 39M citations et abstracts d'articles biomedicaux
- Texte structure, vocabulaire technique, qualite editoriale
- Abstracts seulement (pas le full-text)
- Utilise par BioBERT (Lee et al., 2019), PubMedBERT (Gu et al., 2022)

PMC Open Access:

- 4.5M articles full-text en acces libre
- 98% anglais (Labrak et al., 2024)
- Contenu heterogene: articles de recherche, reviews, case reports, editoriaux, guidelines
- Utilise par BioMistral (3B tokens), Meditron (46B tokens), PMC-LLaMA (75B tokens)
- Probleme: granularite article. Un article de recherche peut contenir a la fois des cas cliniques pertinents et des sections methodologiques non pertinentes

MIMIC (Johnson et al., 2016/2023):

- MIMIC-III, MIMIC-IV: notes cliniques du Beth Israel Deaconess Medical Center
- Acces restreint (data use agreement, formation ethique)
- Utilise par ClinicalBERT (Alsentzer et al., 2019), Clinical Longformer (Li et al., 2022), BioClinical ModernBERT (Sounack et al., 2025)
- Gold standard pour le texte clinique anglais, mais un seul hopital americain -> pas generalizable

6.4.2 Curation article-level pour le biomed

Meditron (Chen et al., 2023):

- Score 0-1 par article base sur: MeSH tags, type de publication, reputation du journal, recence, nombre de citations
- Upsampling des articles a haut score, filtrage des bas scores
- 46B tokens au total, 128x A100, 332h pour le 70B
- Limite: granularite article. Un article peut melanger contenu pertinent et non-pertinent

BioClinical ModernBERT (Sounack et al., 2025):

- 169B tokens: PubMed + PMC + 20 datasets cliniques (MIMIC-III, MIMIC-IV, CheXpert Plus, etc.)
- SOTA actuel sur les benchmarks biomed anglais
- Mais: la plupart des sources cliniques necessitent des data use agreements -> pas vraiment "public"

6.4.3 Sources francophones

Ressources publiques existantes:

- ISTEX: base de 27M references scientifiques, permet d'extraire des documents francais bio/med
- CLEAR (Grabar & Cardon, 2018): notices medicamenteuses, articles d'encyclopedie medicale en francais technique et simplifie
- E3C (Magnini et al., 2021): corpus multilingue de cas cliniques, notices, resumes de these medicale
- HAL / Halvest (Kulumba et al., 2024): articles scientifiques francais et anglais extraits du depot HAL (Hyper Articles en Ligne)
- Wikipedia biomedical: articles des portails medecine, pharmacie, biologie. Libre et multilingue

Modeles entraines from scratch sur corpus francais:

- DrBERT (Labrak et al., 2023): corpus de 7GB, from scratch sur 128x V100
  - Pretend que le continual PT ne marche pas pour le francais biomed
  - Mais: evaluation potentiellement biaisee (cf Ch.4 de la these)
- AliBERT (Berhe et al., 2023): ScienceDirect + theses Sudoc, tokenizer Unigram regularise, 48x A100
  - Modele non disponible publiquement

Tentatives de continual pretraining en francais biomed:

- Copara et al. (2020): 31K articles seulement -> aucune amelioration sur CamemBERT-large
- Le Clercq de Lannoy et al. (2022): PubMed + Cochrane + ISTEX + Wikipedia = 136M mots -> +2 F1 sur EMEA, rien sur MEDLINE
- Dura et al. (2022): 21M documents cliniques APHP -> +3% sur APMed (prive), mais donnees non partageables

Resultats décevants: DrBERT dit que continual PT marche pas, Copara et Le Clercq confirment. Mais le probleme est peut-etre le corpus (trop petit, pas assez cible), pas la methode.

Triple rarete: francais + biomedical + clinique. Les textes cliniques sont prives (CNIL), les modeles hospitaliers ne sont pas partageables entre etablissements. Les sources publiques francophones sont fragmentees et petites comparees a PubMed/PMC.

-> le texte biomedical aide pour le pretraining, mais les LMs captent des patterns statistiques, pas la structure des ontologies. Pour le codage medical (CIM-10, CCAM) il faut des connaissances hierarchiques explicites -> peut-on utiliser directement les ontologies comme donnees de pretraining?

6.5 Medical Ontologies and Knowledge Graphs

6.5.1 Principales ontologies medicales

UMLS (Lindberg et al., 1993):

- Unified Medical Language System, meta-ontologie biomedical de la NLM
- Integre 189 vocabulaires sources (SNOMED-CT, MeSH, ICD, ATC, etc.)
- 3.4M concepts, 16.7M noms de concepts uniques
- Standard international pour la recherche en NLP biomed
- Contient: synonymes, relations hierarchiques, relations associatives, definitions

SNOMED-CT:

- 371K concepts cliniques structures
- Relations hierarchiques (is-a) + associatives (finding site, causative agent, etc.)
- Standard pour les systemes d'information clinique internationaux
- Couverture: diagnostics, procedures, anatomie, substances, organismes

CIM-10 / ICD-10 (OMS, 2019):

- Classification internationale des maladies
- Hierarchie: chapitres -> blocs -> categories -> sous-categories
- Version francaise enrichie par l'ATIH avec: exclusions, inclusions, notes, libelles courts/longs
- Utilise pour le codage PMSI (Programme de Medicalisation des Systemes d'Information) en France
- 19K codes dans la version ATIH

CCAM (ATIH):

- Classification Commune des Actes Medicaux, 8K codes

- Specifique a la France, structure hierarchique
- Relations: actes associes, incompatibilites, modificateurs

ATC (OMS):

- Anatomical Therapeutic Chemical, 6K codes au 5eme niveau
- 5 niveaux: anatomique -> therapeutique -> pharmacologique -> chimique -> substance
- Standard pour la classification des medicaments

6.5.2 Representations formelles

- RDF (Resource Description Framework): triplets sujet-predicat-objet, standard W3C
- OWL (Web Ontology Language): logique descriptive, raisonnement automatique, classes, proprietes
- SKOS (Simple Knowledge Organization System): prefLabel, altLabel (synonymes), broader/narrower (hierarchie), related, exactMatch

Les ontologies codent explicitement ce que les LMs doivent apprendre implicitement:

- Hierarchie: E11 (diabete type 2) $\subset$ E10-E14 (diabete sucre)
- Exclusions: E11 $\neq$ E10 (diabete type 1) – a ne pas confondre
- Synonymes: un meme concept a plusieurs libelles (prefLabel + altLabel)
- Relations associatives: complications, traitements, liens entre ontologies

-> les ontologies sont riches en connaissances structurees, mais sous forme de graphes: pas directement exploitables par les LMs qui attendent du texte en language natu -> comment les integrer dans le pretraining?

6.6 Knowledge-Enhanced Pretraining

6.6.1 Embeddings statiques de graphes

Approches qui produisent des vecteurs fixes (pas contextuels) a partir de graphes:

- Node2Vec (Grover & Leskovec, 2016): random walks sur graphe -> skip-gram, apprentissage de representations de noeuds

- RDF2Vec (Ristoski & Paulheim, 2016): adapte le paradigme walk + skip-gram aux graphes RDF. Fondation des approches de walks sur ontologies
- Snomed2Vec (Agarwal et al., 2019): Node2Vec applique a SNOMED-CT, 5-6x amelioration sur concept similarity
- OWL2Vec* (Chen et al., 2021): walks sur ontologies OWL (teste sur Gene Ontology)

Limite: embeddings statiques, pas integres dans un transformer. Un concept = un vecteur fixe quel que soit le contexte. Pas de contextualisation.

-> on peut faire mieux en integrant les KG directement dans les transformers

6.6.2 Injection de connaissances dans les transformers

Premiere generation: injecter des entites dans l'architecture:

- ERNIE (Zhang et al., 2019): aligne entity embeddings (TransE) avec representations textuelles pendant le pretraining. Entity linking implicite
- KnowBERT (Peters et al., 2019): integre des entity linkers comme couches d'attention supplementaires dans BERT
- K-BERT (Liu et al., 2020): injecte des triples de KG comme soft-position embeddings dans l'input. Modifie la sequence d'entree directement

Deuxieme generation: multi-task pretraining language + KG:

- KEPLER (Wang et al., 2021): MLM + TransE knowledge embedding sur Wikidata simultanement. Les deux objectifs partagent le meme encoder
- CoLAKE (Sun et al., COLING 2020): construit des word-knowledge graphs qui unifient contexte textuel et contexte KG, puis pretraine avec MLM modifie sur ces graphes
- DRAGON (Yasunaga et al., NeurIPS 2022): joint pretraining bidirectionnel MLM + link prediction sur UMLS
  - PubMed pour le texte, UMLS pour le KG
  - +3% accuracy sur MedQA, +10% sur raisonnement multi-step
  - Reference pour le multi-task MLM + KG

6.6.3 Approches contrastives medicales

- SapBERT (Liu et al., NAACL 2021): self-alignment pretraining sur synonymes UMLS
  - Contrastive learning: synonymes UMLS = positifs, non-synonymes = negatifs
  - Metric learning scalable sur 4M+ concepts UMLS
  - SOTA entity linking biomed sur 6 benchmarks
  - Un seul modele pour tous les types d'entites ("one-model-for-all")
- CODER (Yuan et al., 2022): contrastive pretraining sur 87M triplets de relations UMLS
  - Relations + synonymes dans l'objectif contrastif
  - Medical term normalization

6.6.4 Graph-to-text pour le pretraining

Approche recente: transformer les graphes en texte lisible par les LMs.

"Textbooks Are All You Need" (Gunasekar et al., 2023):

- Donnees synthetiques de qualite textbook generees par GPT -> petits modeles tres performants
- Montre que la qualite et la structure des donnees importent plus que le volume brut
- Paradigme: generer des donnees synthetiques structurees plutot que crawler du texte brut

AntGLM-Med (Zhou et al., 2023):

- Reecrit des sous-graphes de KG medical en prose naturelle
- Utilise pour le continual pretraining de LLMs medicaux
- Plus proche precedent de la generation de texte a partir d'ontologies pour pretraining

EntiGraph (Yang et al., ICLR 2025):

- Algorithme d'augmentation entity-centric: decompose un corpus en entites puis genere du texte sur les relations entre entites
- 1.3M tokens reels -> 600M tokens synthetiques via GPT-4-turbo
- Scaling log-lineaire de l'accuracy avec le nombre de tokens synthetiques

- Cible les decoders (Llama 3 8B), pas encore teste sur encoders

Limites actuelles du knowledge-enhanced pretraining:

- La plupart des travaux sont en anglais et sur UMLS
- Graph-to-text pour le pretraining est recent (2023-2025)
- Pas encore explore pour les encoders bidirectionnels (DRAGON fait du link prediction, pas du graph-to-text)
- Les ontologies nationales (CIM-10 ATIH, CCAM) n'ont jamais ete utilisees pour du pretraining
- SapBERT et CODER exploitent UMLS mais uniquement via contrastive learning, pas via generation de texte

-> methodes prometteuses mais gap important: langues non-anglaises, ontologies nationales, pretraining d'encoders via graph-to-text

6.7 Limites et transition

- Rarete biomed non-anglais: PMC = 98% anglais, les corpus francais biomed sont petits et fragmentes. Les pipelines de filtrage (FineWeb-Edu, DCLM) sont concus pour l'anglais -> besoin de strategies de collection et de filtrage adaptees au domaine et a la langue
- Granularite de filtrage: les approches existantes (Meditron, FineWeb-Edu) filtrent au niveau document ou article. Mais un article biomed peut contenir a la fois des cas cliniques pertinents et des sections non pertinentes -> besoin de filtrage a granularite plus fine (paragraphe)
- Criteres de qualite non adaptes: les criteres "educatifs" de FineWeb-Edu ne correspondent pas au texte biomedical (cas cliniques, notices medicamenteuses, protocoles). Les biais vers Science & Technology identifies par WebOrganizer peuvent etre differents en biomed -> besoin de criteres domaine-specifiques
- Ontologies nationales sous-exploitees: UMLS domine la recherche, mais les systemes hospitaliers francais utilisent CIM-10/ATIH, CCAM, ATC. Aucun travail n'a integre ces ontologies dans le pretraining -> gap pour le codage medical en francais
- Graph-to-text naissant: AntGLM-Med et EntiGraph montrent la voie mais sont limites aux decoders en anglais. Pas encore explore pour les encoders bidirectionnels ni pour les ontologies non-anglophones -> possibilite de generer du texte "textbook" a partir des ontologies ATIH

-> ce chapitre a couvert les donnees de pretraining: sources textuelles web et biomed, methodes de curation et filtrage, ontologies medicales et leurs integrations dans le pretraining. Le chapitre suivant traite de la tache finale: l'extraction d'information clinique, et de comment les modeles pretrained sont utilises pour cette tache.

7 Clinical Information Extraction

7.1 Introduction

Clinical Information Extraction (IE) = extraire de l'information structuree a partir de texte clinique non structure.

Pourquoi c'est important:

- Les entrepots de donnees cliniques (CDW) se developpent dans les hopitaux
- Raghavan et al. (2014): jusqu'a 80% des entites cliniques sont absentes des donnees structurees -> le texte libre (comptes-rendus, courriers, prescriptions) est la source principale
- En France: PMSI (Programme de Medicalisation des Systemes d'Information), chaque sejour hospitalier doit etre code avec des codes CIM-10 et CCAM. 30M sejours/an (ATIH)
- Le codage est fait manuellement par des codeurs professionnels -> automatisation = enjeu majeur

Types de taches:

- Named Entity Recognition (NER): identifier les mentions d'entites dans le texte
- Relation Extraction (RE): extraire les relations entre entites
- Medical coding: assigner des codes d'ontologie (CIM-10, CCAM, ATC) a un document
- Entity linking / normalization: associer une mention a un concept dans une base de connaissances (UMLS, SNOMED-CT)

Pourquoi c'est central pour la these:

- Les taches de clinical IE sont le terrain d'evaluation final des modeles pretrained (Part 1-2)
- Les defis identifies ici (rarete des donnees, label space enorme, evaluation non standardisee) motivent les contributions de Part 3

-> plan du chapitre: on part des fondamentaux du NER, on zoome sur les benchmarks biomed, puis sur le codage medical, l'entity linking, les approches zero-shot, les donnees synthetiques, et enfin l'evaluation

7.2 Named Entity Recognition

7.2.1 Sequence Labeling

NER = identifier et classifier les mentions d'entites dans un texte.

Formulation classique: sequence labeling avec schema BIO (Beginning, Inside, Outside):

- Chaque token recoit un label: B-TYPE, I-TYPE, ou O
- Variantes: BILOU (Beginning, Inside, Last, Outside, Unit) pour plus de precision

CRF (Lafferty et al., 2001):

- Conditional Random Fields: modele les dependances entre labels consecutifs
- Evite les sequences invalides (ex: I-MALADIE apres B-TRAITEMENT)
- Standard pre-deep learning pour le NER

BiLSTM-CRF (Lample et al., 2016):

- Combine BiLSTM pour les representations contextuelles + CRF pour le decodage
- SOTA pre-BERT sur la plupart des benchmarks NER
- Character-level embeddings pour capter la morphologie

7.2.2 BERT-based NER

Devlin et al. (2019): BERT + linear layer sur chaque token -> NER.

- Simple: pretrained encoder + classification head, fine-tuning end-to-end
- Pas besoin de features manuelles ni de CRF (debat: CRF aide parfois marginalement)
- Representations contextuelles profondes: "cancer" dans "cancer du poumon" vs "cancer du sein" -> vecteurs differents
- Transfer learning: le modele pretrained apporte des connaissances linguistiques generales

7.2.3 Au-dela du token-level

Span-based NER:

- Au lieu de labelliser chaque token, on enumere les spans candidats et on classifie chaque span
- Avantage: gere mieux les entites imbriquees (nested entities)
- Exemple biomed: "cancer du poumon droit" = [MALADIE [ANATOMIE]]
- Les entites imbriquees sont frequentes en biomed: symptomes dans des maladies, anatomie dans des procedures

-> ok le NER marche bien en general, mais le domaine biomedical a ses specificites: vocabulaire technique, entites imbriquees, negation, style telegraphique

7.3 Biomedical and Clinical NER

7.3.1 Benchmarks anglais

NER sur texte scientifique (PubMed):

- BC5CDR (Li et al., 2016): chemical-disease relations, 1500 articles PubMed, 2 types d'entites (chemical, disease)
- NCBI Disease (Dogan et al., 2014): mentions de maladies dans 793 abstracts PubMed
- JNLPBA (Collier & Kim, 2004): genes/proteines, corpus GENIA, 5 types d'entites
- AnatEM (Pyysalo & Ananiadou, 2014): entites anatomiques, 1212 documents
- ChemProt (Krallinger et al., 2017): interactions chemical-protein, relation extraction

NER sur texte clinique:

- i2b2 shared tasks (Uzuner et al., 2010, 2011): NER clinique sur notes MIMIC, plusieurs editions (medication extraction, coreference, relations temporelles)
- n2c2 (successeur de i2b2): n2c2 2018 (Henry et al., 2020) = adverse drug events + criteres d'eligibilite aux essais cliniques. n2c2 2022 = contextualization of clinical NLP
- Taches specifiques sur notes cliniques: de-identification (DEID), social determinants (SDoH), phenotyping

Classification:

- GAD (Bravo et al., 2015): gene-disease associations, classification binaire
- Hallmarks of Cancer (Baker et al., 2016): classification multi-label de hallmarks tumoraux

BLURB (Gu et al., 2022): Biomedical Language Understanding and Reasoning Benchmark, unifie 13 datasets biomed en un seul benchmark. PubMedBERT = premier modele evalue.

7.3.2 Benchmarks francais

QUAERO (Neveol et al., 2014):

- EMEA: notices medicamenteuses en francais
- MEDLINE: titres d'articles scientifiques
- 10 types d'entites suivant les semantic groups UMLS (Anatomy, Chemical, Device, Disorder, Gene, Living Being, Object, Phenomena, Physiology, Procedure)
- Entites imbriquees presentes
- Utilise dans CLEF eHealth 2016 (Neveol et al., 2016)

CAS / DEFT 2020 (Cardon et al., 2020):

- Cas cliniques issus d'articles scientifiques francais
- Tache 3 de DEFT 2020 = extraction d'information clinique
- CAS1: 2 classes (pathologie, signes/symptomes)
- CAS2: 8 classes (anatomie, dose, examen, mode, moment, substance, traitement, valeur)
- Devenu benchmark standard pour le NER clinique francais

E3C (Magnini et al., 2021):

- Corpus multilingue (EN, FR, ES, IT, EU)
- Cas cliniques, notices medicamenteuses, resumes de these
- Layer 1 = annote manuellement, layer 2 = semi-annote, layer 3 = non annote
- 1 seule classe: entite clinique (pas de sous-types)

CLEF eHealth 2016 (Neveol et al., 2016):

- Task 2: NER + normalisation sur corpus QUAERO
- Codage CIM-10 sur certificats de deces (corpus CepiDC)
- BRATeval = outil d'evaluation officiel (exact match sur character offsets)
- 7 equipes participantes

DrBenchmark (Labrak et al., 2024, LREC-COLING):

- Premier benchmark unifie pour le francais biomedical (equivalent de BLURB pour l'anglais)
- 20 taches sur 12 datasets: NER (9 taches), classification (8), QA (1), similarite semantique (2)
- Regroupe QUAERO, E3C, CAS/DEFT-2020, DEFT-2021, Mantra-GSC, FrenchMedMCQA, PxCorpus, MorFITT, CLISTER, DiaMed
- Permet enfin une comparaison equitable entre modeles sur le francais biomedical
- Resultat cle: les modeles specialises biomed surpassent les modeles generaux

7.3.3 Specificites du texte clinique

Le texte clinique est tres different du texte scientifique (PubMed) et du texte general (web):

- Style telegraphique: phrases incompletes, listes de symptomes, abreviations non standard
- Abreviations: "IDM" (infarctus du myocarde), "HTA" (hypertension arterielle), "NFS" (numeration formule sanguine)
- Negation: "pas de signe de pneumonie" -> "pneumonie" est mentionnee mais absente. NegEx (Chapman et al., 2001): algorithme a base de regles pour detecter la negation, toujours utilise comme baseline
- Temporalite: antecedents vs diagnostics actuels vs hypotheses
- Incertitude: "probable", "a surveiller", "a eliminer"
- Fautes d'orthographe frequentes (saisie rapide par les cliniciens)

Consequence: les modeles entraines sur du texte scientifique (PubMed) ou general (web) sont sous-optimaux pour le texte clinique -> d'ou les modeles pretrained sur le domaine (BioBERT, PubMedBERT, ClinicalBERT, GatorTron, DrBERT, cf Section 5.7). Ces modeles specialises ameliorent les performances sur les taches biomed, mais restent supervises: il faut annoter des exemples pour chaque classe cible.

Alternative architecturale: Clinical Mamba (Yang et al., 2024) adapte les State Space Models (SSM) au domaine clinique. Complexite lineaire en la longueur de sequence (vs quadratique pour les transformers) -> pertinent pour les documents cliniques longs. Premiers resultats competitifs avec les transformers.

-> on a les benchmarks et les modeles pour le NER biomed, mais le codage medical c'est pas juste du NER: c'est assigner des codes d'une ontologie a un document entier, avec des milliers de labels possibles

7.4 Medical Coding

7.4.1 Le probleme

Medical coding = assigner des codes standardises (CIM-10, CCAM, ATC) a un document clinique.

En France:

- PMSI: chaque sejour hospitalier doit etre code
- DP (Diagnostic Principal), DAS (Diagnostics Associes), actes CCAM
- 30M sejours/an (ATIH), code manuellement par des codeurs professionnels
- Erreurs de codage = impact financier et epidemiologique

Defis:

- Label space enorme: 17K codes CIM-10 (ATIH), 8K codes CCAM, 6K codes ATC
- Documents longs: comptes-rendus de sortie = souvent 1000-3000 tokens, depassent la limite de BERT (512)
- Distribution tres desequilibree: quelques codes frequents (I10 Hypertension, E11 Diabete), longue traine de codes rares
- Multi-label: un sejour peut avoir 1 DP + plusieurs DAS

7.4.2 Approches

CAML (Mullenbach et al., 2018, NAACL):

- Convolutional Attention for Multi-Label classification
- CNN + label-wise attention: chaque code a son propre mecanisme d'attention sur le document
- Premiere approche explainable pour ICD coding: on peut visualiser quels passages du texte ont motive chaque code
- DR-CAML: regularisation par les descriptions textuelles des codes

HyperCore (Cao et al., 2020, ACL):

- Exploite la hierarchie ICD via embeddings hyperboliques (Poincare ball)
- L'espace hyperbolique capture naturellement les structures arborescentes (distance exponentielle avec la profondeur)
- + graphe de co-occurrence entre codes (GCN)

PLM-ICD (Huang et al., 2022, ClinicalNLP@NAACL):

- Premier a utiliser des pretrained language models pour ICD coding
- Label attention sur les representations du PLM
- Segment pooling pour gerer les documents longs (decoupe en segments, pool les representations)
- Resultat cle: meilleur encoder pretrained = meilleur codage -> la qualite du pretraining impacte directement le codage

MSMN (Yuan et al., 2022, ACL):

- Multiple Synonyms Matching Network
- Exploite les synonymes UMLS pour chaque code ICD
- Le texte clinique utilise des expressions variees pour un meme concept: "infarctus du myocarde", "IDM", "crise cardiaque" -> tous I21
- Multi-synonym matching: representations de codes plus riches que la seule description officielle

Et les LLMs? Soroush et al. (2024, NEJM AI):

- Evaluation de GPT-4 pour le codage ICD-10: exact match = 33.9% seulement

- Les LLMs generatifs peinent sur le codage: label space trop grand, codes trop specifiques
- Les approches specialisees (PLM-ICD, MSMN) restent superieures
- -> renforce le besoin d'encoders specialises plutot que de LLMs generiques

7.4.3 Codage en francais

Ressources specifiques:

- FRACCO (Pignat et al., 2024): corpus d'oncologie francais, 1301 cas cliniques annotes avec codes CIM-O-3.1, classification multi-label sur 100 codes
- FRASIMED (Zaghir et al., 2023): adaptation francaise de Cantemist (codage tumeurs, Miranda et al. 2020) et Distemist (codage maladies, Miranda et al. 2022), 2051 documents cliniques traduits de l'espagnol
- CepiDC / CLEF eHealth 2016: codage CIM-10 sur certificats de deces francais

-> le codage marche sur MIMIC (anglais), mais: 17K labels = impossible d'annoter des exemples pour chacun. En francais, les ressources sont encore plus rares. -> besoin d'approches qui generalisent a des labels non vus = zero-shot. Mais d'abord: comment associer mentions et concepts?

7.5 Entity Linking and Normalization

7.5.1 Le probleme

Entity linking = associer une mention textuelle a une entite dans une base de connaissances (UMLS, SNOMED-CT, CIM-10).

Normalization = mapper des variantes terminologiques au concept canonique:

- "infarctus du myocarde", "IDM", "crise cardiaque" -> I21.9 (CIM-10)
- "diabete de type 2", "DNID", "diabete non insulinodependant" -> E11 (CIM-10)

Difference avec le codage medical:

- Entity linking = mention-level: associer chaque mention a un concept
- Medical coding = document-level: assigner des codes a un document entier
- Entity linking peut etre vu comme une brique du codage: d'abord extraire les mentions (NER), puis les normaliser (linking), puis agreger au niveau document

7.5.2 Approches

SapBERT (Liu et al., 2021, NAACL):

- Self-Alignment Pretraining for BERT
- Contrastive learning sur synonymes UMLS: 4M+ concepts
- Synonymes = positifs, non-synonymes = negatifs
- SOTA entity linking biomed sur 6 benchmarks
- "One-model-for-all": un seul modele pour tous types d'entites (maladies, medicaments, procedures)
- Base: PubMedBERT fine-tune avec metric learning

CODER (Yuan et al., 2022):

- Contrastive pretraining sur 87M triplets de relations UMLS
- Ajoute les relations (is-a, part-of, causes) en plus des synonymes dans l'objectif contrastif
- Medical term normalization

BioSyn (Sung et al., 2020, ACL):

- Bi-encoder + cross-encoder pour entity normalization
- Marginal score aggregation: combine scores de plusieurs synonymes
- Montre que le re-ranking avec cross-encoder ameliore significativement

7.5.3 Limites

- La plupart des travaux sont en anglais et sur UMLS
- UMLS couvre bien l'anglais mais pas le francais clinique (abreviations, synonymes informels)
- SapBERT et CODER ne captent pas la hierarchie complete de l'ontologie (parents, exclusions, complications)
- Pas d'exploitation des relations RDF/SKOS (broader, narrower, related, exactMatch)

-> on sait normaliser des mentions isolees, mais comment extraire ET normaliser sans supervision, pour des milliers de labels jamais vus a l'entrainement?

7.6 Zero-shot and Open NER

7.6.1 Pourquoi zero-shot

- Les labels medicaux evoluent: nouveaux codes ICD, nouveaux criteres d'essais cliniques, nouvelles maladies (COVID-19)
- Impossible d'annoter des exemples pour chaque label (17K codes CIM-10)
- Les modeles NER classiques (BERT + linear head) ont un nombre fixe de classes de sortie
- -> besoin de modeles qui generalisent a des types d'entites non vus a l'entrainement

7.6.2 GLiNER

GLiNER (Zaratiana et al., 2024, NAACL):

- Formulation: NER = matching entre spans de texte et descriptions de types d'entites
- Architecture: un seul transformer bidirectionnel encode conjointement le texte et les types d'entites
- Les types d'entites sont donnes en langage naturel (pas des labels fixes)
- Token [ENT] devant chaque type -> representation du type
- Span representation layer -> representation de chaque span candidat
- Matching par produit scalaire + sigmoid dans un espace partage
- Bat ChatGPT sur zero-shot NER tout en etant beaucoup plus petit (200M params)

GLiNER2 (Zaratiana et al., 2025, EMNLP):

- Extension de GLiNER a un framework multi-taches unifie: NER, relation extraction, classification, extraction structuree
- Meme architecture de base mais avec des tokens de relation en plus des tokens d'entite
- -> un seul modele pour plusieurs taches IE, toujours en zero-shot

GLiNER-BioMed (Zaratiana et al., 2025):

- Adaptation de GLiNER au domaine biomedical

- Fine-tuning sur des datasets NER biomed (BC5CDR, NCBI, JNLPBA, etc.)
- +5.96% F1 par rapport au GLiNER general sur les benchmarks biomed
- Montre que l'adaptation au domaine est necessaire meme pour les modeles zero-shot

-> GLiNER et ses extensions montrent que le matching span-label est une direction prometteuse. Mais ou trouver les donnees d'entrainement?

Pre-entrainement sur Pile-NER:

- UniversalNER (Zhou et al., 2024, ICLR): distillation de ChatGPT
- ChatGPT annote des passages du Pile -> 45K paires, 13K types d'entites distincts
- Instruction-tuning de LLaMA sur ces donnees
- Bat ChatGPT de 7-9 F1 points en moyenne sur NER
- GLiNER pre-entraine sur ce dataset = forte capacite zero-shot

7.6.3 Autres approches

GoLLIE (Sainz et al., 2024, ICLR):

- Guideline-following Large Language model for IE
- Approche generative: le LLM genere des annotations structurees a partir de guidelines en langage naturel
- Zero-shot: les guidelines decrivent les types d'entites -> pas besoin d'exemples annotes
- Avantage: peut suivre des definitions complexes (criteres d'inclusion, codes CIM avec exclusions)
- Mais: modele genératif = plus lent et plus gros qu'un encoder

NuNER (Bogdanov et al., 2024, EMNLP):

- Foundation model NER compact (RoBERTa-base, 125M params)
- Pre-entraine sur annotations GPT-3.5 de C4: 24.4M mots, 4.38M annotations, 200K types d'entites
- Contrastive learning pour specialiser l'encoder
- Resultat cle: taille ET diversite du dataset d'entrainement = facteurs critiques

LLMs pour clinical NER:

- Agrawal et al. (2022): LLMs pour clinical IE, performances prometteuses
- Singhal et al. (2022): Med-PaLM, QA medicale a l'echelle
- Mais: Lehman et al. (2023), "We May Have Overestimated LLMs in Clinical NLP"
  - BERT-type models restent competitifs pour les taches extractives cliniques
  - LLMs trop gros pour deploiement hospitalier (contraintes infra)
  - Problemes de confidentialite avec APIs proprietaires (GPT-4, Claude)
  - -> les encoders restent pertinents pour les taches extractives cliniques

7.6.4 Limites du zero-shot actuel

- GLiNER pre-entraine sur donnees generales (Pile-NER), pas cliniques
- Les types d'entites sont des descriptions textuelles courtes: ne capte pas la structure hierarchique des ontologies
- Pas d'exploitation des relations entre codes (parent, exclusion, synonyme)
- Performances degradees sur les codes rares (longue traine)

-> zero-shot NER prometteur, mais: GLiNER ignore la structure ontologique, et les donnees d'entrainement sont generales. Les donnees cliniques sont privees -> comment entrainer sur du domaine clinique sans donnees reelles?

7.7 Synthetic Data for Clinical NLP

7.7.1 Le probleme de la rarete

- Donnees cliniques = privees (CNIL en France, HIPAA aux US)
- Modeles entraines sur donnees hospitaliers = non partageables entre etablissements
- Peu de datasets annotes publics en francais clinique: QUAERO, CAS, E3C = petits et principalement scientifiques, pas cliniques
- Les datasets cliniques accessibles (MIMIC) = anglais, un seul hopital americain

7.7.2 LLM-as-annotator

Paradigme recent: utiliser un LLM pour annoter des donnees, puis distiller vers un petit modele:

- FineWeb-Edu (Penedo et al., 2024): Llama-3-70B annote 500K docs, distille vers classifieur BERT-like (cf Ch.2)
- UniversalNER (Zhou et al., 2024): ChatGPT annote le Pile pour NER, distille vers LLaMA
- NuNER (Bogdanov et al., 2024): GPT-3.5 annote C4 pour NER, distille vers RoBERTa
- Avantage: le LLM capture des connaissances medicales, le petit modele est deployable en hopital

7.7.3 Generation de notes cliniques synthetiques

Premiers travaux: generer directement des notes cliniques synthetiques avec des modeles de langue.

Ive et al. (2020, npj Digital Medicine):

- Generation de notes cliniques synthetiques en psychiatrie (discharge summaries)
- Objectif: contourner la confidentialite des donnees
- Resultat: les modeles NLP entraines sur donnees synthetiques atteignent des performances comparables a ceux entraines sur donnees reelles
- Premier a valider experimentalement que le synthetique peut remplacer le reel pour les taches NLP cliniques

Chintagunta et al. (2021, NLPMC@ACL):

- GPT-3 pour augmenter des donnees de dialogue medical: 210 exemples annotes -> performances equivalentes a 6400 exemples (30x)
- Low-shot + ensemble: le LLM genere des variantes, un ensemble filtre les generations de mauvaise qualite
- Montre que le LLM-as-augmenter fonctionne specifiquement en domaine medical

Asclepius (Kweon et al., 2024, ACL Findings):

- LLM clinique entierement entraine sur notes synthetiques

- Pipeline: case reports publics (PMC) -> GPT-3.5 genere des notes cliniques realistes a partir des cas
- Aucune donnee clinique privee utilisee
- Performances comparables aux modeles entraines sur MIMIC (donnees reelles)
- -> preuve que des notes synthetiques de qualite suffisent pour entrainer des modeles cliniques

Approche texte + annotations simultanees:

- Le LLM genere le texte ET les annotations en meme temps
- Avantage: alignement texte-annotation garanti (pas de desynchronisation)
- Inputs structurees possibles: informations patient, codes CIM-10, guidelines therapeutiques

Evaluation de la qualite:

- Evaluation aveugle par medecins: notes synthetiques vs reelles
- Criteres: coherence medicale, completude, qualite linguistique
- Observation recurrente: les notes synthetiques sont souvent plus completes mais moins coherentes medicalement que les notes reelles

Risques:

- Style different des notes reelles (trop propre, pas assez telegraphique)
- Biais du LLM: peut generer des cas stereotypes
- Les annotations LLM ne sont pas gold-standard: heuristiques neurales, pas expertises medicales

-> les donnees synthetiques sont une solution a la rarete, mais comment evaluer correctement les modeles entraines dessus? Et plus generalement, comment comparer equitablement des modeles avec des tokenizers differents?

7.8 Evaluation

7.8.1 Metriques NER

seqeval (strict entity-level):

- Evaluation au niveau entite: une entite est correcte si et seulement si le type ET les frontieres sont exacts

- Schema IOB2: B-TYPE marque le debut, I-TYPE la continuation, O l'absence d'entite
- Micro-average: aggrege toutes les entites (favorise les classes frequentes)
- Macro-average: moyenne des scores par classe (traite toutes les classes egalement)
- Standard de facto pour le NER en NLP

BRATeval (CLEF eHealth 2016):

- Exact match sur character offsets dans le texte original
- Avantage: completement independant du preprocessing et de la tokenisation
- Utilise comme outil officiel pour les campagnes CLEF eHealth
- Mais: moins repandu que seqeval dans la communaute NLP

Token-level classification (weighted F1):

- Chaque token recoit un label, evaluation token par token
- Probleme majeur: le token "O" (non-entite) est largement majoritaire
- Le weighted F1 est domine par la classe O -> scores artificiellement eleves
- Utilise par DrBERT (Labrak et al., 2023): rend la comparaison avec d'autres modeles difficile

7.8.2 Impact de la tokenisation

- Tokenizers differents decoupent le texte differemment
- SciBERT (Beltagy et al., 2019): 42% intersection avec le vocabulaire BERT
- Un tokenizer qui sur-segmente les termes techniques = plus de tokens a predire = plus de chances d'erreur
- Consequence: les metriques token-level sont biaisees par le choix du tokenizer
- Exemple: "echocardiographie" = 3 tokens (CamemBERT) vs 2 tokens (tokenizer specialise) -> nombre de predictions different -> score different

7.8.3 Besoin d'un benchmark unifie

Problemes actuels:

- Chaque papier utilise des datasets, metriques, splits, et hyperparametres differents
- Pas de protocole standard pour le francais biomedical
- CRF vs linear head, gel des couches vs fine-tuning complet, micro vs macro F1
- Comparaison entre modeles = difficile et souvent biaisee

Comparaison equitable entre architectures:

- BERT-type (encoders) vs LLMs (decoders) vs GLiNER (bi-encoder): architectures tres differentes
- Tokenizers differents, procedures de fine-tuning differentes
- -> besoin d'une evaluation au niveau caractere, independante du tokenizer, qui permette une comparaison equitable

7.9 Limites et transition

- Rarete des donnees cliniques: les textes hospitaliers sont prives (CNIL), les modeles entraines dessus non partageables. Les corpus publics francais (QUAERO, CAS, E3C) sont petits et principalement scientifiques -> besoin d'alternatives aux donnees hospitalieres
- Label space enorme: 17K codes CIM-10, 8K CCAM, impossible d'annoter des exemples pour chaque code. Les approches supervisees classiques (BERT + linear head) ne scalent pas -> besoin d'approches zero-shot capables de generaliser a des labels non vus
- Zero-shot NER existant = pre-entraine sur donnees generales: GLiNER et UniversalNER sont entraines sur Pile-NER (texte web general), pas sur du domaine clinique. Performances degradees sur les codes rares -> besoin d'adapter au domaine medical
- Ontologies sous-exploitees dans le NER: GLiNER utilise des descriptions textuelles comme types d'entites mais ignore la structure hierarchique, les exclusions, les synonymes encodes dans RDF/SKOS -> besoin d'integrer la structure ontologique dans les embeddings de labels

- Evaluation non standardisee: pas de benchmark unifie pour le francais biomed, metriques biaisees par le tokenizer, protocoles incomparables entre papiers -> besoin d'un framework d'evaluation tokenizer-agnostic

- Encoders vs decoders: malgre l'engouement pour les LLMs, les encoders restent competitifs pour les taches extractives cliniques (Lehman et al., 2023) et sont deployables sur les infrastructures hospitalieres

-> ce chapitre a couvert les taches de clinical IE: NER, codage medical, normalisation, et les defis specifiques du domaine clinique. Les limites identifiees (rarete, label space, zero-shot non adapte, ontologies sous-exploitees, evaluation non standardisee) definissent les axes de recherche de la partie suivante.

Part III

Building a Biomedical Corpus

8 Collecting Biomedical Text

8.1 Introduction

Hospitals produce millions of clinical documents every year. These can be used to train language models internally, but the resulting models inherit the legal constraints of their training data: they cannot be released publicly or exchanged between institutions. For a model that is truly open, the training data must come from somewhere else.

Yet hospital records remain the richest source of medical information. Clinical data warehouses have made them increasingly accessible for research purposes. It is estimated that up to 80% of entities are missing from structured modalities (Raghavan et al., 2014), making clinical reports the richest source of medical information. Yet extracting this information requires high-performing language models adapted to the biomedical domain.

BERT-based models (Devlin et al., 2019) consistently demonstrate state-of-the-art results for a wide range of natural language processing tasks. The adaptation of BERT to the French language, particularly with the CamemBERT model (Martin et al., 2020), has replicated these performances in French natural language processing. However, the results of this model on biomedical data are disappointing (Cardon et al., 2020), as it exhibits lower performance compared to heuristic models on certain evaluation datasets. These results are predictable because CamemBERT is trained on plain language, often sourced from web pages such as forums. Biomedical data, especially clinical data, are significantly different. They contain technical terms that are very rare or absent in everyday language, and they have a radically distinct style, often telegraphic, rarely consisting of complete sentences, with varying abbreviations.

In this chapter, we address this gap by introducing *biomed-fr*, a public French biomedical corpus of 413 million words assembled from three complementary sources. We describe the corpus construction, analyze the vocabulary mismatch between general and biomedical French tokenizers, and situate our effort among existing French biomedical corpora. The use of this corpus for continual pretraining of CamemBERT, yielding CamemBERT-bio, is presented in Chapter 11.

Authors	Source	Size	Public
Copara et al. (2020)	PubMed	31K docs	Yes
Le Clercq de Lannoy et al. (2022)	Misc.	136M words	Partial
Dura et al. (2022)	APHP CDW	21M docs	No
Labrak et al. (2023)	Misc.	7.4 GB	Partial

Table 8.1: French biomedical corpora used for language model adaptation. Size units vary across works as reported by their respective authors.

8.2 Existing French Biomedical Corpora

Several works have attempted to adapt CamemBERT to the biomedical domain, each relying on different training data. Table 8.1 summarizes these corpora.

Copara et al. (2020) explored a second phase of pre-training on 31,000 French biomedical scientific articles. However, they did not observe a significant improvement on a clinical named entity recognition task using the large version of CamemBERT. This could be explained by the combination of a relatively small corpus (31K documents compared to the 18 billion words in BioBERT) and the large version of the CamemBERT model.

Le Clercq de Lannoy et al. (2022) also adapted CamemBERT to the biomedical domain. They aggregated documents from various sources, including PubMed, Cochrane, ISTEX, and Wikipedia, forming a larger partially public corpus of approximately 136 million words. They observed a 2-point improvement in F-measure on a named entity recognition evaluation set composed of drug notices (EMEA), but no significant improvement on a set composed of scientific article titles (MEDLINE).

Dura et al. (2022) continued the pre-training of CamemBERT on 21 million clinical documents from the APHP (Assistance Publique – Hôpitaux de Paris) clinical data warehouse. They observed a significant 3% improvement on APMed, a private clinical named entity recognition dataset owned by APHP. They also achieved similar scores to CamemBERT on EMEA and MEDLINE. Their new model performs better on clinical data, yet it obtains scores similar to CamemBERT in other biomedical domains.

Labrak et al. (2023) introduced a public French biomedical model named DrBERT. Through their experiments, they explored both continual-pretraining and from-scratch training strategies. Their findings indicated a superior performance when training from scratch, suggesting that continual-pretraining with CamemBERT for French biomedical data may not be as effective. However, when applied to PubMedBERT, continual-pretraining yielded results nearly on par with the from-scratch approach.

Finally, Berhe et al. (2023) introduced AliBERT, which was trained on a French biomedical corpus primarily comprising articles from ScienceDirect and theses collected through Sudoc. The model leverages a new regularized Unigram-based

Source	Size (M words)	Content
ISTEX	276	Scientific articles
CLEAR	73	Drug leaflets
E3C	64	Clinical cases, leaflets
Total	**413**	

Table 8.2: Composition of the *biomed-fr* corpus (in millions of words).

tokenizer and underwent extensive training on 48 GPUs for a total duration of 20 hours. Their pretrained model outperforms notable French non-domain-specific models, such as CamemBERT and FlauBERT, in two biomedical downstream tasks. Unfortunately, the model is not currently available.

8.3 The biomed-fr Corpus

We built a French biomedical corpus composed exclusively of public documents to minimize the usage constraints mentioned earlier. The documents come from three different sources (see Table 8.2), the main one being ISTEX. This new corpus, named *biomed-fr*, consists of 413 million words, equivalent to 2.7 GB of data. Martin et al. (2020) have shown that with only 4 GB of data, it is possible to achieve performance almost comparable to the model trained with the 138 GB OSCAR dataset (Ortiz Suárez et al., 2019). For an adaptation of CamemBERT to the biomedical domain, this amount of data can be considered sufficient.

ISTEX. The ISTEX database contains references to 27 million scientific publications. We extracted 108,183 French documents published in a biology or medical journal since 1990. Articles published before this date often contain numerous typographical errors, as they are often scanned articles that require optical character recognition algorithms, resulting in a certain number of errors. Such errors are found to a lesser extent in articles published after 1990. Some documents, although in French, contain passages in English. Therefore, there is an indeterminate amount of English in this corpus. However, it is unlikely that this will significantly impact pre-training. Typographical errors and the presence of other languages are aspects that can be addressed in future versions of *biomed-fr*.

CLEAR. The CLEAR corpus (Grabar and Cardon, 2018) consists of encyclopedia articles, drug leaflets, and abstracts of scientific articles. Each document is available in two versions: one in technical language and the other in simplified language. We retrieved all of these documents in both versions. Regarding the drug leaflets, we removed redundant sentences at the beginning and end of each document, such as

Term	General	Specialized
échocardiographie	écho-cardi-ographie	échocardiographi-e
transthoracique	trans-thorac-ique	trans-thoracique
glimépiride	g-lim-épi-ride	gli-m-épi-ride
cardiopathie	cardio-pathie	cardiopathie
diastoliques	dia-s-tol-iques	diastolique-s

Table 8.3: Comparison of tokenization between a general-purpose tokenizer and a specialized tokenizer for some biomedical technical terms. Hyphens indicate subword boundaries.

the website navigation bar from which the documents were extracted or information about the company selling the documents.

E3C. This corpus (Magnini et al., 2020) is composed of three layers. The first two layers are annotated or semi-annotated and will be used for evaluation in Chapter 11. The last layer is not annotated, and that is the one we retrieved for *biomed-fr*. It consists of medical specialty admission competitions, drug leaflets, and medical thesis abstracts. There may be duplicates of some leaflets found in the CLEAR corpus.

biomed-fr-small. By randomly selecting 10% of content from *biomed-fr*, we created a smaller corpus called *biomed-fr-small*. This subset allows us to study the impact of corpus size on downstream performance. Results of this comparison are presented in Chapter 11.

8.4 Vocabulary Analysis

CamemBERT-bio is a biomedical-adapted model based on CamemBERT. Unlike a new model trained from scratch, it shares the same vocabulary. The vocabulary of CamemBERT was constructed using SentencePiece (Kudo and Richardson, 2018) on an OSCAR sample. Therefore, it is a general-purpose vocabulary designed for everyday language. We can hypothesize that the tokenization of CamemBERT may result in oversplitting of biomedical technical terms.

To investigate this possibility, we trained a specialized tokenizer on *biomed-fr-small* and calculated the intersection of the two vocabularies.

We find a 45% intersection between the two vocabularies (Table 8.3), which is quite close to the 42% intersection found by Beltagy et al. (2019) between the vocabulary of BERT and SciBERT. Therefore, there is a significant difference in the most frequent terms.

The relatively modest performance gain of PubMedBERT compared to BioBERT and the similar performance of DrBERT compared to CamemBERT-bio despite their specialized vocabulary, together with the over-segmentation of technical terms and the low intersection rate, demonstrate the value of investigating vocabulary adaptation. However, taking into account the comparable performance levels and the significant difference in computational cost (see Chapter 11), we advocate for the continual-pretraining method as the preferred adaptation approach.

8.5 Limitations

It is important to discuss some limitations of our corpus.

Firstly, *biomed-fr* is limited in its diversity. This is because we chose to only use publicly available materials, which tend to lean towards scientific content. As a result, models trained on this corpus may not fully represent performances on real private clinical data. The gap between scientific and clinical text is a recurring challenge that we address in Chapter 9.

Furthermore, we did not explore any potential biases in our data and some parts of the dataset would benefit from further cleaning. These aspects could impact the outputs of models trained on this corpus and should be investigated.

Third, all documents from the selected sources were included indiscriminately, without assessing whether individual passages were genuinely useful for pretraining. Whether such filtering would improve downstream performance is an open question that we explore in Chapters 9 and 10.

Conclusion

More than half of the most frequent biomedical subwords are absent from CamemBERT's general vocabulary. A mismatch this large raises a natural question: should we abandon the general tokenizer entirely and train a new model from scratch? The answer, perhaps surprisingly, is no, as we show in Chapter 11.

Before we get there, however, a more immediate question deserves attention. We included all 413 million words of *biomed-fr* without filtering. Not all of this text is equally useful. In the next chapter, we investigate whether selecting data by quality can improve upon including everything indiscriminately.

9 Detecting Content Types

9.1 Introduction

For web data, the question of what to keep and what to discard has a straightforward answer. FineWeb-Edu (Penedo et al., 2024) showed that filtering entire documents by educational quality, discarding 92% of the data, actually improves downstream performance. A low-quality web page is low-quality throughout: if the source is bad, every paragraph is bad. But a scientific article is not a web page.

Figure 9.1 illustrates the difference. A web page about astronomy scores low throughout and can be safely discarded. A PMC article on pulmonary hypertension, by contrast, contains a textbook-quality review paragraph, a boilerplate committee note, and a clinical case report with real diagnostic values. Document-level filtering would either keep all three or discard all three. Neither is the right answer.

In the previous chapter, we assembled *biomed-fr* by including all available documents from three public sources. This approach treats every paragraph equally. How much of this text actually helps, what types of content does it contain, and can we be smarter about what we include?

In this chapter, we introduce a paragraph-level annotation framework that distinguishes content types, domains, and educational quality within scientific articles. We show that selective pretraining on the most relevant paragraphs matches the performance of training on the full corpus using one-third of the tokens. We also extract 2 million clinical case paragraphs from PubMed Central, providing an openly available alternative to restricted hospital records. Finally, we investigate the risks of neural quality filtering: not all selection strategies are equally safe, and some can amplify benchmark contamination by a factor of 20.

9.2 Content Heterogeneity in Scientific Articles

Large language models for biomedicine are typically pretrained on PubMed Central (PMC) Open Access articles. While valuable, this corpus presents challenges including heterogeneous quality, predominance of English content (approximately 98%), uneven representation of clinical cases, and variable educational value. Existing filtering approaches operate at the article level: BioMistral (Labrak et al., 2024) upsampled non-English articles, while Meditron (Chen et al., 2023) scored articles

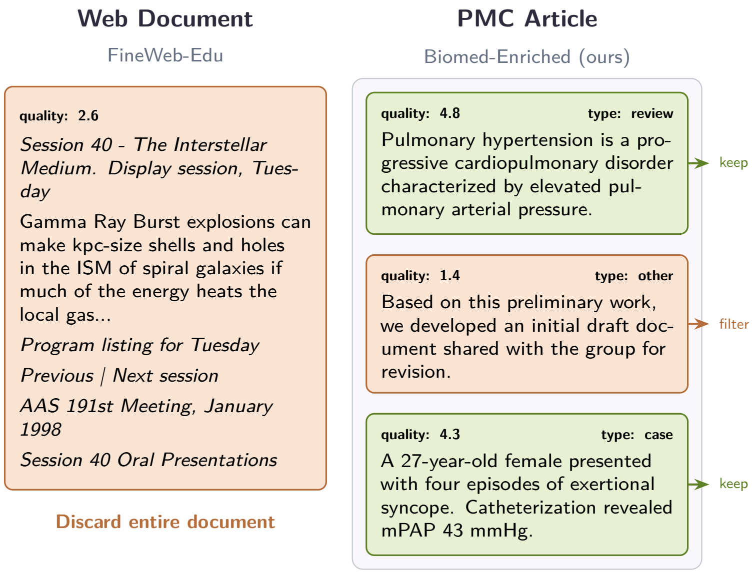

Figure 9.1: **Document-level vs. paragraph-level filtering.** A low-quality web page can be discarded entirely (left). A scientific article mixes high-value content (clinical cases, reviews) with boilerplate (right). Paragraph-level annotation allows keeping the valuable parts.

using MeSH tags, publication type, journal reputation, recency, and citation count. These methods assume that entire documents are either valuable or not.

This assumption does not hold for curated scientific corpora. Unlike web-scraped data where low-quality sources justify discarding entire documents, scientific articles are all potentially valuable. The real issue is variance *within* articles. To quantify this, we analyzed 10,000 PMC articles (378,000 paragraphs) and found that 91.6% of articles contain multiple document types (clinical case, study, review, other). Educational scores vary by 2.6 points on average within articles, spanning more than half of the 1–5 scale. Some paragraphs contain excellent educational or clinical content, while others add little domain knowledge but remain difficult to detect with simple heuristics.

9.3 Method

9.3.1 Data Collection and Preprocessing

We extracted text from the PubMed Central (PMC) Open Access Subset, containing approximately 4.5 million full-text scientific articles. Using a custom pipeline, we processed the raw XML files to extract article content, segment articles into 133 million individual paragraphs, filter out non-textual elements, and retain only paragraphs containing a minimum of 64 tokens.

	Single	Mixed
Document Type	8.4%	91.6%
Domain	44.7%	55.3%
Educational Score (1–5 scale)		
	Mean	**Median**
Intra-article std	0.68	0.70
Intra-article range	2.61	2.86

Table 9.1: Intra-document heterogeneity in 10K PMC articles (378K paragraphs). 91.6% of articles contain multiple document types. Educational scores vary by 2.6 points on average within articles.

Dimension	**Agreement**
Educational score	$r = 0.876$, MAE $= 0.326$
Domain	$F_1 = 0.954$
Document type	$F_1 = 0.925$, $\kappa = 0.821$

Table 9.2: Distillation quality: agreement between the XLM-RoBERTa student and Llama-3.1-70B teacher on held-out paragraphs.

9.3.2 Two-Stage Annotation Framework

We follow a two-stage annotation process on a single 8×A100 node: 160 GPU-hours for the first stage and 80 GPU-hours for the second.

Stage 1: LLM annotation. We used Llama-3.1-70B-Instruct (Grattafiori et al., 2024) to annotate a diverse subset of 400,000 paragraphs (approximately 0.33% of the corpus) across multiple dimensions: type classification (clinical case, study, review, other), domain categorization (clinical, biomedical, other), and educational quality (1–5 scale). Language was identified separately using langdetect.

Stage 2: Distillation. To scale annotation to the full corpus, we distilled the LLM annotations into a smaller XLM-RoBERTa-base model (Conneau et al., 2020) trained to jointly predict all annotation dimensions. On 39,800 held-out paragraphs, the student closely matches the Llama-3.1-70B labels (Table 9.2). It is important to note that these annotations are not gold-standard labels verified by medical experts, but neural heuristics for data curation. Similar to FineWeb-Edu (Penedo et al., 2024), they guide which content to prioritize during pretraining.

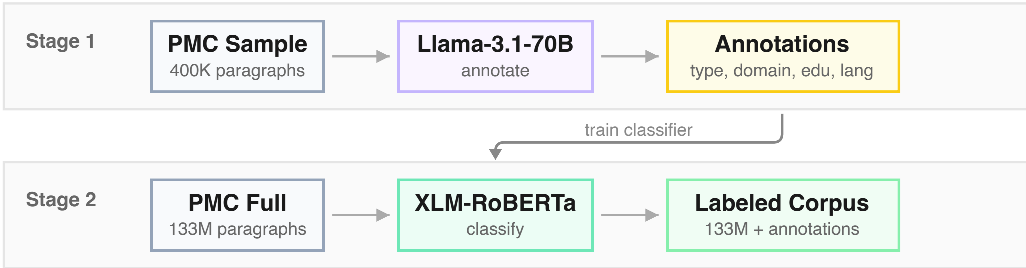

Figure 9.2: **Two-stage annotation pipeline.** Stage 1: Llama-3.1-70B annotates 0.33% of PMC paragraphs across four dimensions. Stage 2: a fine-tuned XLM-RoBERTa distills these annotations to label the full corpus, enabling flexible filtering strategies.

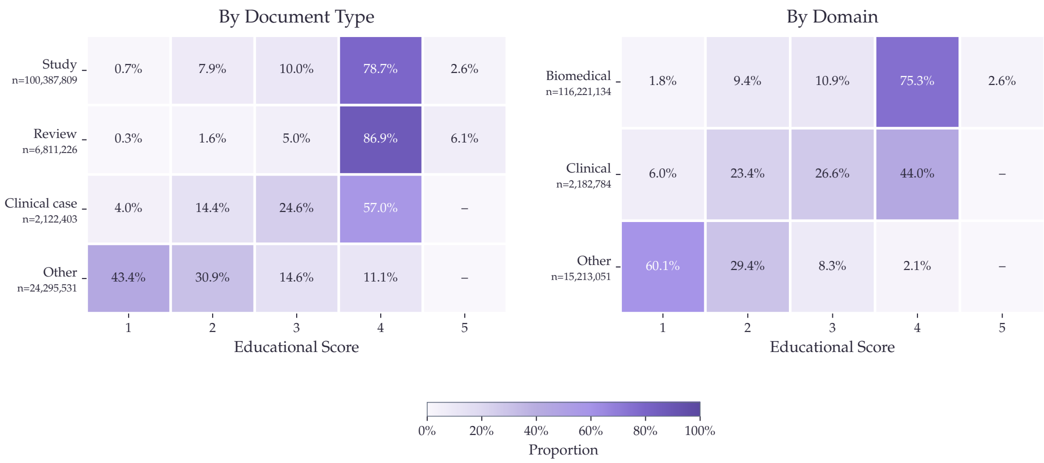

Figure 9.3: **Distribution of educational quality scores by document type and domain.** Reviews and studies show the highest proportion of high scores, while clinical texts display more variance. Content labeled “other” rarely reaches high scores.

9.3.3 Data Analysis

Most PMC paragraphs receive an educational score of 4, with a mean of 3.48 and median of 4.00. This distribution varies by content type: reviews and studies contain the highest proportion of educational content (86.9% and 78.7% scoring 4, respectively), while clinical cases show more variance (57.0% rated 4). Domain-wise, biomedical paragraphs score higher (75.3% at score 4) than clinical text (44.0%), and content labeled “other” rarely reaches high scores (2.1%). These patterns support the score threshold (≥ 3) used in BE-Educational and BE-All, and explain why combining domain and quality filters yields consistent gains across tasks.

Variant	Strategy
BE-Base	Complete unmodified PMC Open Access Subset (baseline).
BE-Educational	Removes paragraphs with educational quality scores below 3.
BE-Clinical	Upsamples 10× articles with predominantly clinical domain content.
BE-ClinicalCase	Upsamples 10× articles containing at least one clinical case paragraph.
BE-French	Upsamples 10× articles containing French text.
BE-All	Combines quality filtering (score ≥ 3), upsampling of clinical content, French text, and clinical cases, plus metadata prefixing.

Table 9.3: Biomed-Enriched dataset variants. All variants preserve the original article structure and use an 8K context window during pre-training.

9.3.4 Dataset Construction

Using the distilled model, we annotated the full PMC corpus and constructed several dataset variants through strategic filtering and upsampling, summarized in Table 9.3. Upsampling refers to duplicating articles in the training corpus, proportionally increasing their sampling probability.

For all variants, we preserved the original structure of the article and employed an 8K context window during pre-training. This ensures that models can process complete scientific articles, capturing dependencies where information presented in earlier paragraphs is essential for understanding later content.

9.4 Experimental Setup

Continual pre-training served as a method to evaluate the relevance and utility of our annotations. Our evaluation focuses on isolating the effects of data curation rather than pursuing state-of-the-art scores on benchmarks. A more powerful foundation model would likely yield higher absolute scores, but would probably obscure the precise impact of our dataset.

9.4.1 Decoder Setup

We selected OLMo2-7B-stage1 (Team et al., 2025) as our foundation model for continual pre-training, strategically choosing this intermediate checkpoint to more

Parameter	Value
Base model	OLMo2-7B-stage1
Peak LR	6.15e-5
Min LR	6.15e-6
LR Decay	Linear
Batch size	1024
Weight decay	0.1
Context	8,192 tokens
Hardware	128 MI250X
Tokens	33.6B / variant

(a) Decoder

Parameter	Value
Base model	ModernBERT-base
Peak LR	5e-4
Min LR	5e-7
LR Decay	one_minus_sqrt
Batch size	576
Weight decay	1e-5
Context	8,192 tokens
Hardware	4× H100
Tokens	70B

(b) Encoder

Table 9.4: Hyperparameters for continual pre-training.

clearly attribute performance changes to our data curation. Although stage 1 has already developed strong language modeling capabilities, it precedes the knowledge-intensive tuning of stage 2, providing an ideal balance of baseline capabilities without the risk of catastrophic forgetting of instruction-following abilities during domain adaptation. Notably, the data mix used in stage 1 includes DCLM (Li et al., 2024), which is a dataset obtained by filtering web-data using a classifier trained on instruct-data. Hence, OLMo2-7B already has relatively strong question-answering capabilities after stage 1.

Each Biomed-Enriched variant was trained with exactly 33.6 billion tokens, using identical hyperparameters. We follow the annealing strategy of OLMo2 used in the mid-training phase. By maintaining strict parameter parity across experiments, we created a controlled environment focused solely on measuring the effectiveness of our different data curation strategies.

9.4.2 Encoder Setup

We also conducted continual-pretraining experiments on encoder models. Starting from ModernBERT-base (Warner et al., 2024) after its context extension phase, we followed exactly the same hyperparameters as BioClinical-ModernBERT phase 1. Our baseline reproduces their standard continual pretraining setup on public PubMed/PMC. Our clinical-upsample recipe additionally applies educational filtering (score ≥ 3) and upsamples clinical cases 100× and clinical domain content 10×.

Table 9.4 summarizes the hyperparameters for both setups.

9.4.3 Evaluation

For decoders, our evaluation consists of zero-shot testing on MMLU medical subcategories, MedQA (Jin et al., 2021), MedMCQA (Pal et al., 2022), and PubMedQA (Jin et al., 2019). For the French adaptation assessment, we use 5-shot evaluation on

FrenchMedMCQA (Labrak et al., 2022). We compare our models against three baselines: OLMo2-7B-stage1, Llama-3.1-8B, and Meditron-70B.

For the encoder models, we evaluate on 11 tasks spanning clinical and biomedical domains. Clinical tasks include ChemProt, Phenotype, COS, SocialHistory, and DEID. Biomedical tasks include AnatEM, BC5CDR, JNLPBA, NCBI Disease, GAD, and Hallmarks of Cancer. Following the evaluation recipe of BioClinical-ModernBERT, we finetune for 10 epochs on clinical tasks and 20 epochs on biomedical tasks, with early stopping patience of 3 epochs for all. We compare against BioClinical-ModernBERT (Sounack et al., 2025), which was trained on 169B tokens from PubMed, PMC, and 20 clinical datasets including MIMIC-III, MIMIC-IV, CheXpert Plus radiology reports, and various clinical notes from multiple institutions, most requiring data use agreements.

9.5 Results

9.5.1 Decoder Results

Tables 9.5 and 9.6 present decoder results on medical QA and MMLU benchmarks. The combined strategy BE-All achieves the best average performance (61.08%), with the strongest gains on MedQA (+2.4 pts) and MMLU Clinical Knowledge (+1.5 pts). Different enrichment strategies show complementary strengths: clinical upsampling yields a +4 point improvement on MMLU Professional Medicine, while educational filtering improves Medical Genetics by +2 points and MedMCQA by +1.2 points. BE-All reaches target performance using roughly one-third of the training tokens compared to BE-Base, with stable improvements visible from early checkpoints. BE-French also demonstrates successful adaptation to French medical QA (40.5% vs 38.3% baseline, Figure 9.4), showing that language-specific upsampling generalizes beyond English.

9.5.2 Encoder Results

Tables 9.7 and 9.8 present encoder results across 11 clinical and biomedical benchmarks. Our BE-Clinical recipe achieves 77.08% average F_1, matching BioClinical-ModernBERT (77.05%) while using 2.5× fewer tokens and only public PubMed/PMC data. Adding a decay phase with high-quality educational content brings slight additional gains: the Biomedical decay variant (edu ≥ 4) achieves 77.32%.

Different decay targets show task-specific strengths: Study decay yields the best results on AnatEM (79.7%) and NCBI Disease (81.2%), Review decay excels on GAD (80.2%), and Clinical Case decay achieves the highest SocialHistory score (57.0%). These results suggest that public PMC data has more potential for clinical NLP than commonly exploited, and that targeted curation can close the gap with models trained on restricted datasets.

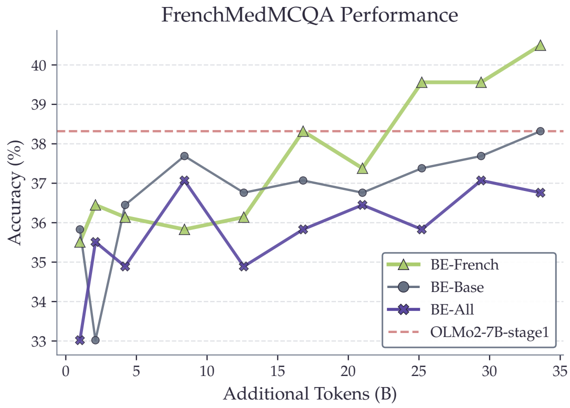

Figure 9.4: **FrenchMedMCQA performance.** BE-French outperforms other variants on French medical QA, demonstrating effective language-specific improvement through targeted upsampling.

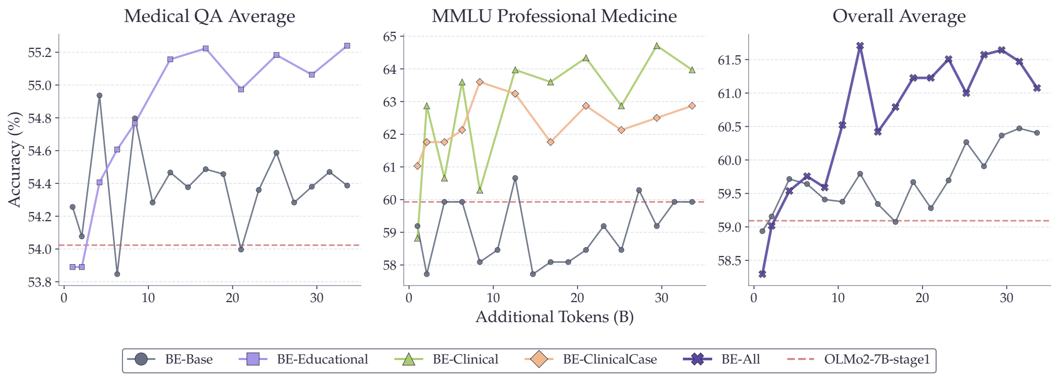

Figure 9.5: **Decoder training progression.** BE-All reaches target performance with approximately one-third of the training tokens required by BE-Base.

9.6 Discussion

The core advantage of paragraph-level annotation lies in its ability to identify valuable content that would be missed by document-level approaches. Clinical case descriptions, for instance, often appear in isolated sections of broader scientific articles. By operating at the paragraph level, we extracted 2 million clinical case paragraphs from PMC Open Access. This content is otherwise difficult to obtain at scale due to privacy restrictions on hospital records.

For decoders, our results reveal clear task-specific benefits from different enrichment strategies. Educational filtering improves knowledge-intensive QA tasks, clinical upsampling enhances clinical reasoning benchmarks, and the combination of both yields the best overall performance with faster convergence. The successful French adaptation through targeted upsampling further demonstrates that this framework

Model	MedQA	MedMCQA	PubMedQA	Avg
Reference Models				
Llama-3-8B	59.70	57.47	74.80	63.99
Meditron-70B	57.10	46.80	76.60	60.17
Our Variants				
OLMo2-7B-stage1	45.33	41.14	75.60	54.02
+ BE-Base	44.85	41.91	76.40	54.39
+ BE-Clinical	41.95	39.35	76.60	52.63
+ BE-ClinicalCase	42.11	39.52	76.60	52.74
+ BE-Educational	45.64	**43.08**	77.00	55.24
+ BE-All (33.6B)	**47.21**	42.79	76.60	55.53
+ BE-All (12.6B)	**47.21**	42.94	76.40	**55.52**

Bold indicates best result per column. Underline: second-best.

Table 9.5: **Decoder results: Medical QA benchmarks.** BE-All achieves the best QA average with both 33.6B and 12.6B tokens.

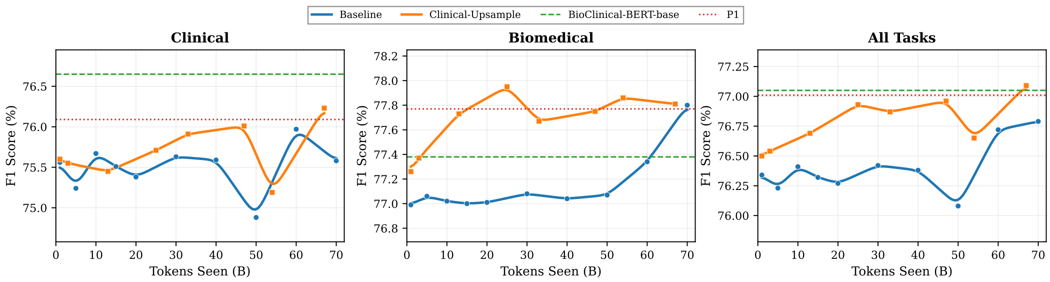

Figure 9.6: **Encoder training curves** on clinical, biomedical, and combined benchmarks. Our clinical-upsample recipe (orange) reaches higher performance faster on clinical tasks. At 67B tokens, we match BioClinical-ModernBERT (green dashed) which required 169B tokens and 20 clinical datasets.

generalizes beyond English. However, aggressive clinical upsampling can reduce performance on general biomedical tasks like College Biology, suggesting a trade-off between specialization and breadth.

For encoders, the decay phase experiments reveal that different paragraph types benefit different downstream tasks: Study paragraphs improve NER performance, Clinical Case paragraphs improve social history extraction, and Review paragraphs excel on relation extraction. The distinct task-specific strengths of each decay variant suggest a promising direction: model merging or checkpoint averaging could combine the benefits of specialized training runs without requiring additional compute. Rather than training general-purpose models on massive undifferentiated corpora, these findings support a more modular approach where data composition is tailored to downstream requirements.

Model	Anat	Clin	Bio	Med	Gen	Prof	Avg
Reference Models							
Llama-3-8B	68.89	74.72	78.47	61.85	83.00	70.22	72.86
Meditron-70B	53.30	66.70	76.30	63.00	69.00	71.60	66.65
Our Variants							
OLMo2-7B-stage1	54.81	63.40	<u>69.44</u>	53.18	<u>69.00</u>	59.93	61.63
+ BE-Base	<u>57.04</u>	64.15	**70.83**	**59.54**	<u>69.00</u>	59.93	63.42
+ BE-Clinical	53.33	63.40	65.28	58.38	66.00	**63.97**	61.73
+ BE-ClinicalCase	57.04	64.91	66.67	**59.54**	<u>69.00</u>	<u>62.87</u>	63.34
+ BE-Educational	57.04	<u>65.28</u>	68.06	56.65	**71.00**	58.82	62.81
+ BE-All (33.6B)	<u>60.00</u>	**65.66**	68.06	<u>58.96</u>	<u>69.00</u>	61.40	<u>63.85</u>
+ BE-All (12.6B)	**64.44**	64.53	68.75	57.23	**71.00**	<u>62.87</u>	**64.80**

Anat=Anatomy, Clin=Clinical Knowledge, Bio=College Biology, Med=College Medicine, Gen=Medical Genetics, Prof=Professional Medicine.

Table 9.6: **Decoder results: MMLU Medical subcategories.** BE-All (12.6B) achieves the best MMLU average despite using one-third of the tokens.

9.7 The Risks of Neural Quality Filtering

[1]

The results above show that selecting content by type and educational quality can improve pretraining efficiency. A natural next step is to apply neural quality classifiers, as done by FineWeb-Edu (Penedo et al., 2024) and DCLM (Li et al., 2024), to filter pretraining data more aggressively. However, this raises a question: do these classifiers treat all content equally, or do they have blind spots?

To investigate this, we designed Benchmark-In-A-HayStack (BiaHS), an experiment to test how different text-quality classifiers rank benchmark-like content within a large web corpus. We sampled 35 benchmark instances from three datasets: 15 samples from MMLU (3 samples each from anatomy, computer security, high school geography, moral scenarios, and college physics), 10 from GSM8K (Cobbe et al., 2021), and 10 from GPQA (Rein et al., 2023). Each benchmark sample was formatted as a standalone document containing the question, answer choices, and reference solution. These 35 benchmark documents were inserted as independent entries into a corpus of 99,965 documents sampled from FineWeb, yielding a final evaluation set of 100,000 documents where benchmarks constituted 0.035% of the total.

We scored all documents using four quality classifiers: DCLM (FastText-based, used to filter OLMo 2 training data), Textbook (FastText-based, used to filter OLMo 1 training data), FineWeb-Edu (transformer-based), and the GAPeron classifier (transformer-based). Each classifier produced a full ranking of all documents based on their predicted quality score.

[1]This section is based on our contribution to Godey et al. (2026).

Model	ChemPr	Pheno	COS	Social	DEID	Avg
Baselines						
ModernBERT-base	89.5	48.4	94.0	53.1	78.3	72.66
BioClinical-MdnBERT†	90.0	**60.7**	94.8	56.0	**81.8**	**76.66**
Our Models (Public Data Only)						
Baseline	89.8	59.6	94.5	55.5	78.5	75.58
BE-Clinical	89.9	59.9	94.8	56.8	79.7	76.22
With Decay Phase						
+ Review	**90.3**	59.7	94.6	55.4	79.6	75.92
+ Study	**90.3**	**60.3**	94.7	55.6	78.6	75.90
+ Clinical Case	89.5	59.4	94.5	**57.0**	79.3	75.94
+ Biomedical	90.1	59.9	**94.9**	56.4	79.6	76.18

† Trained on 169B tokens including 20 clinical datasets (MIMIC-III/IV, CheXpert, etc.). Our models use only public PubMed/PMC.
ChemPr=ChemProt, Pheno=Phenotype, Social=SocialHistory.

Table 9.7: **Encoder results: Clinical tasks.** Our BE-Clinical recipe matches BioClinical-ModernBERT on clinical tasks using only public data.

The results reveal a striking pattern. The DCLM classifier ranks all MMLU and GSM8K samples in the top-5 percentiles. If only the top 5% of documents are selected through filtering, the probability of encountering a benchmark sample in a training batch increases by a factor of approximately 20. The FineWeb-Edu classifier also ranks benchmark samples consistently higher, likely because its annotation prompt explicitly asks for documents with high educational value, which naturally favors exam-style items and step-by-step solutions.

The DCLM classifier pushes benchmark samples even higher and more consistently. Its FastText model is trained to separate synthetically generated instructions from high-scoring Reddit posts, from general web text. This training objective naturally favors solved Q&A structures with a short question, a direct answer, and brief reasoning, which aligns with the format of common benchmark datasets.

In contrast, the GAPeron classifier does not significantly push benchmark samples. It was trained on annotations focused on general content quality across multiple dimensions: accuracy, clarity, coherence, grammar, depth of information, and overall usefulness. This broader framing does not specifically reward the structured question-answer format typical of benchmark samples.

Implications. This experiment suggests that the way quality classifiers are trained can strongly influence contamination risks. As neural quality filtering becomes a standard step in data curation, these design choices deserve closer examination. For the Biomed-Enriched work presented earlier in this chapter, we rely on content-type

Model	AnatEM	BC5CDR	JNLPBA	NCBI	GAD	HoC	Avg
Baselines							
ModernBERT-base	77.2	87.9	74.3	77.7	76.8	66.6	76.75
BioClinical-MdnBERT[†]	**79.2**	88.7	74.8	78.7	75.8	67.0	77.37
Our Models (Public Data Only)							
Baseline	78.8	88.7	74.8	**81.3**	76.2	67.0	77.80
BE-Clinical	78.5	88.1	74.6	79.4	77.7	68.5	77.80
With Decay Phase							
+ Review	78.5	88.5	74.7	79.6	**80.2**	68.8	<u>78.38</u>
+ Study	**79.7**	88.4	**74.8**	81.2	78.7	67.7	**78.42**
+ Clinical Case	78.7	88.3	74.7	80.0	77.9	68.1	77.95
+ Biomedical	79.0	**88.6**	74.7	79.9	78.4	**69.0**	78.27

[†] Same model as Table 9.7. Trained on 169B tokens with 20 clinical datasets.

Table 9.8: **Encoder results: Biomedical tasks.** Study decay achieves 78.42%, outperforming BioClinical-ModernBERT (77.37%) with only public data.

classification and educational quality scoring rather than general quality filtering, which avoids this particular risk.

9.8 Limitations

Our decoder experiments used 7B parameter models. Larger models may respond differently to data curation choices. The two-stage annotation pipeline relies on XLM-RoBERTa-base, which may have limited capacity for nuanced paragraph classification. Our clinical case paragraphs come from published case reports, which may differ stylistically from actual hospital records. The educational quality scores are neural heuristics rather than gold-standard pedagogical assessments.

For the BiaHS experiment, the evaluation was conducted on a relatively small number of benchmark samples (35) inserted into a single web corpus (FineWeb). While the results are directionally clear, a larger-scale study across multiple corpora would strengthen the conclusions.

Conclusion

Two findings emerge from this chapter. First, paragraph-level annotation reveals that scientific articles are far more heterogeneous than their metadata suggests: 91.6% mix content types, and educational quality varies by 2.6 points within a single article. Selecting the right paragraphs allows matching full-corpus performance with a third of the tokens, and extracting 2 million clinical case paragraphs provides public clinical text without data use agreements.

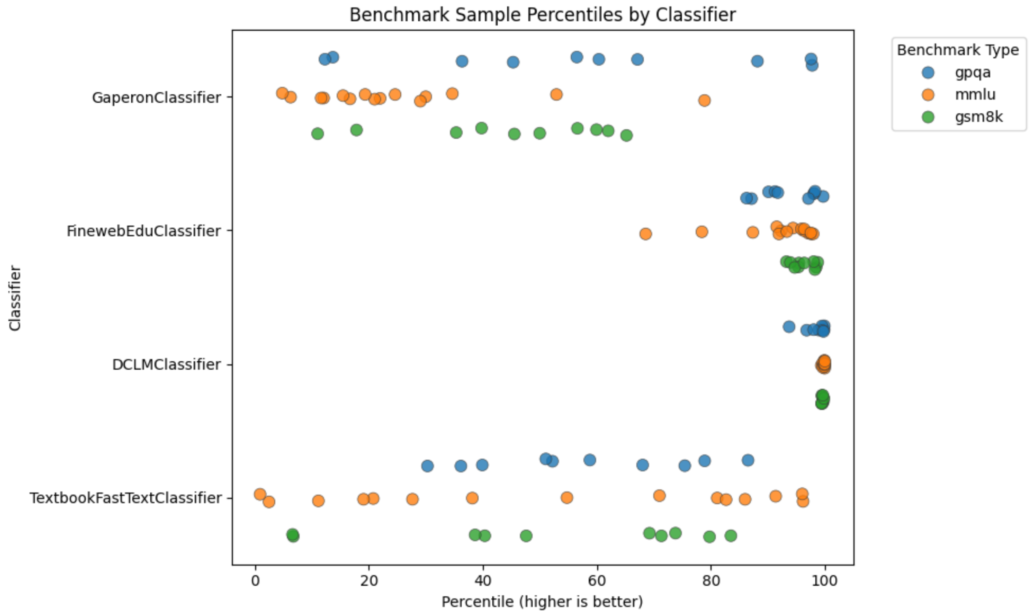

Figure 9.7: **BiaHS: Benchmark-In-A-HayStack results.** Percentile rank of 35 benchmark samples (MMLU, GSM8K, GPQA) within 100K FineWeb documents, as scored by four quality classifiers. DCLM ranks all benchmark samples in the top-5 percentiles.

Second, not all filtering is created equal. Neural quality classifiers trained on instructional or educational content can amplify benchmark contamination by a factor of 20, effectively turning a data curation step into a memorization accelerator. Content-type classification, by contrast, selects for relevance rather than format, and does not carry this risk.

We now have both a general biomedical corpus (*biomed-fr*, Chapter 8) and a method for selecting the most relevant paragraphs within it. The next chapter addresses how to combine these resources into a curated training mix, and how quality signals can be used at scale to build a better corpus for pretraining ModernCamemBERT-bio.

10 Which Quality Signals Matter?

10.1 Introduction

Here is one passage as it came out of an ISTEX PDF:

```
… vise ~ remonter l&apos;organe prolab6 (correction de la pt6se) et
h le soutenir dans sa position id6ale plus qu&apos;/~ le fixer (fixation).
Ce sout~nement est assur6 par le renforcement du p6rin6e grace aux
tissus naturels que l&apos;on r6pare (rapphie) ou que l&apos;on consolide
avec du mat6riel local (plastie, ligamentopexie)…
```

And here is what it should read:

> … la chirurgie vise à remonter l'organe prolapsé (correction de la ptose) et à le soutenir dans sa position idéale plus qu'à le fixer (fixation). Ce soutien est assuré par le renforcement du périnée grâce aux tissus naturels que l'on répare (raphie) ou que l'on consolide avec du matériel local (plastie, ligamentopexie)…

OCR had read every *é* as a *6*, apostrophes came as `l'`, and tildes had drifted into the middle of words. Multiply this by the hundreds of thousands of similar PDFs in HAL and ISTEX, and assembling a 10-billion-token biomedical training mixture stops being a collection problem and becomes a repair problem.

Chapter 8 introduced *biomed-fr*, a 413-million-word public French biomedical corpus. Chapter 9 showed that paragraph-level annotation by a large language model can identify content types and educational quality more precisely than document-level filters, and that neural quality classifiers carry contamination risks. This chapter applies these lessons at scale. We assemble *biomed-fr-v2*, a 10-billion-token mixture from six French sources, and ask which quality signals actually improve downstream performance once the dust settles. The corpus is the training data for ModernCamemBERT-bio, presented in Chapter 12.

The contributions of this chapter are: (1) a multi-source French biomedical corpus 25 times larger than *biomed-fr*, including a curation pipeline that handles the OCR artifacts, boilerplate, and structural noise of HAL and ISTEX; (2) an LLM-based annotation of 2.16 million paragraphs along four quality signals and nine content types; (3) ablation studies that quantify the effect of each signal and each type on downstream performance, with one notable surprise: writing-quality and terminological-precision filters *degrade* performance.

10.2 Sources

We assemble *biomed-fr-v2* from six complementary French-language sources. HAL, ISTEX, and FineWiki-bio cover scientific biomedical literature. EMEA contributes drug package inserts. E3C contributes clinical case reports. Synthetic textbooks generated from medical ontologies complete the mixture with the vocabulary of French administrative nomenclatures. The final composition is summarized in Table 10.4.

- **HAL.** Medical theses, research articles, and reports deposited on the HAL open archive, filtered to the biomedicine section.
- **ISTEX.** French scientific articles from the ISTEX corpus, filtered to biology and medicine.
- **FineWiki-bio.** Biomedical subset of FineWiki, the Wikipedia counterpart to FineWeb (Penedo et al., 2024), restricted to French and filtered for the biomedical domain.
- **EMEA** (Névéol et al., 2014). Drug package inserts published in French by the European Medicines Agency.
- **E3C** (Magnini et al., 2020). Multilingual clinical corpus (layer 3, French partition), composed of clinical cases extracted from medical journals and theses.
- **Synthetic ontology textbooks.** Educational text generated from three French medical ontologies published by the Agence du Numérique en Santé in RDF/OWL format: CIM-10 (diagnoses, 19,161 codes), CCAM (procedures, 38,191 codes), and ATC (medications, 6,950 codes). Random walks through the ontology graph (1.3 million walks total) traverse hierarchical relations and, for CIM-10, causal and exclusion relations. A large language model (Qwen3-235B-A22B-Instruct) reformulates each walk into a paragraph of medical prose while preserving the original codes and relations. The full pipeline is described in Chapter 13.

HAL and ISTEX are distributed in heterogeneous formats: some documents come with structured XML, others only as PDFs, sometimes scans of printed material. For documents without clean XML, we apply the extraction and cleaning pipeline described in Section 10.3. FineWiki-bio, EMEA, E3C, and the synthetic textbooks are already of high quality and are integrated directly into the final mixture, bypassing this pipeline.

10.3 Curation Pipeline

A portion of HAL and ISTEX documents come with clean XML and proceed directly to annotation. The other portion is available only as PDFs, sometimes scans of printed material (particularly for articles published before the 1990s). Once converted to text, these PDFs contain OCR artifacts (incorrect ligatures, character substitutions, fragmented words), bibliographic references, repeated headers, and irrelevant content. We therefore apply an LLM-based extraction and cleaning pipeline, described below. The other sources (FineWiki-bio, EMEA, E3C, synthetic textbooks) are already clean and bypass this pipeline entirely.

Extraction and cleaning. Each document is split into segments of approximately 5,000 tokens at paragraph boundaries. A large language model (Qwen3-Next-80B-A3B-Instruct) receives each segment and produces structured JSON output containing semantically complete passages, with references, metadata, headers, and captions removed, and OCR errors corrected (ligatures, encoding). Extracted passages must contain between 30 and 600 words and at least 60% alphabetic characters. We then apply Jaccard-similarity deduplication (threshold 0.80, sliding window of 5 passages), followed by the FineWeb-2 filter chain for French: encoding correction, repetition filters, and Gopher and FineWeb quality filters.

Fidelity verification. To ensure the LLM does not introduce content absent from the source, we measure character-level overlap between each extracted passage and its source document on 300 ISTEX passages (after typographic normalization). 94.4% of passages have $\geq$ 90% overlap and 84.1% have $\geq$ 99% overlap; the median rate of words absent from the source is zero (mean: 3.4%). Manual inspection of edge cases shows that divergences come from OCR corrections (such as the surgical passage shown at the start of this chapter) or from converting tables into prose. The opposite case is also common: a clean 1,143-character passage on cancer-related depression is extracted with 100% character overlap and no additional words introduced. These two cases bracket the typical behavior of the pipeline: it cleans aggressively when the source is corrupted by OCR, and copies verbatim when the source is already clean.

LLM quality annotation. The same LLM annotates each passage along four quality signals scored from 1 to 10: educational value (`edu`, mean 6.5), content richness (`cont`, 6.8), writing quality (`writ`, 7.6), and terminological precision (`term`, 7.6). The codes in parentheses are the abbreviations used in ablation tables. A content type is also assigned among nine categories: `research_findings` (526k passages), `medical_knowledge` (388k), `clinical_guidance` (218k), `research_methodology`,

Type removed	Avg F_1	Δ
Removal helps		
`drug_information`	64.35	+0.67pp
`research_findings`	64.24	+0.56pp
`policy_administrative`	64.07	+0.39pp
Controls		
Random (same volume)	63.72	+0.04pp
Full corpus (reference)	63.68	ref.
Removal hurts		
`medical_knowledge`	63.55	−0.13pp
`research_methodology`	63.51	−0.17pp

Table 10.1: **Content-type ablation.** We remove one type and measure Δ relative to the full corpus.

`drug_information`, `background_review`, `policy_administrative`, `patient_case` (36k), and `other`. In total, 2.16 million passages are annotated.

10.4 Content-Type Ablation

We measure the effect of each content type by ablation: we remove one type and compare against the full corpus on three evaluation tasks (DiaMed, FrACCO-30, and EMEA, 9 seeds). Table 10.1 shows that removing `drug_information` *improves* performance by +0.67pp, while removing `medical_knowledge` degrades it by −0.13pp. A random control (removing the same amount of text at random) gives +0.04pp, confirming that the effect is specific to the content type.

We exclude `drug_information` from the final corpus (redundant with the EMEA source already included separately) and `other`. Paragraphs shorter than 50 tokens are also excluded.

10.5 Quality Filtering and Upsampling

We then test the effect of thresholding on the quality signals (Table 10.2). Joint filtering with `edu`≥ 7 and `cont`≥ 7 produces a +0.48pp gain; a stricter threshold (`edu`≥ 8 and `cont`≥ 8) reaches +0.55pp. By contrast, the writing-quality and terminological-precision signals *degrade* performance when used as filters (−0.26 and −0.33pp). This degradation likely comes from the fact that these filters eliminate informal but informative clinical documents, such as case notes and lightly edited reports.

Rather than applying a strict filter, we adopt a quality-weighted upsampling inspired by FineWeb (Penedo et al., 2024). Each article receives a quality ratio, defined as the proportion of its paragraphs satisfying `edu`≥ 7 AND `cont`≥ 7. Articles

Filter	Avg F_1	Δ
Helpful filters		
`edu`≥ 8 AND `cont`≥ 8	**59.37**	+0.55pp
`edu`≥ 7 AND `cont`≥ 7	<u>59.30</u>	**+0.48pp**
`edu`≥ 6 AND `cont`≥ 6	58.95	+0.13pp
No filter (reference)	58.82	ref.
Harmful filters		
`term`≥ 7	58.57	−0.26pp
`writ`≥ 7	58.49	−0.33pp

Table 10.2: **Quality-signal ablation.** Educational value and content richness improve performance; writing quality and terminological precision degrade it.

Ratio of `edu`≥ 7 AND `cont`≥ 7 paragraphs	Coefficient
0%	excluded
1–25%	4.3×
26–50%	8.5×
51–75%	12.8×
76–99%	21.3×
100%	34.1×

Table 10.3: **Upsampling coefficients** per article-quality bucket.

are then upsampled in proportion to this ratio (Table 10.3). Articles with 0% high-quality paragraphs are excluded from the final corpus.

10.6 Final Composition

The final training mixture, after cleaning and quality-weighted upsampling, contains 10 billion tokens (Table 10.4). These 10 billion include the repetitions induced by upsampling (coefficients ranging from 4.3× to 34.1× depending on quality, see Section 10.5); the underlying unique content is more limited. HAL and ISTEX articles (passed through the pipeline of Section 10.3) and FineWiki-bio (integrated as-is) are grouped under "curated biomedical articles." The three other sources (EMEA, E3C, synthetic textbooks) are also integrated as-is.

Conclusion

Two findings emerge from the ablations. First, not all content types contribute equally: removing drug information actually improves downstream performance, while removing medical knowledge degrades it. Second, not all quality signals

Source	Tokens	%	Content
Curated biomedical articles	7B	70%	HAL + ISTEX + FineWiki-bio
Synthetic ontology textbooks	2B	20%	CIM-10, CCAM, ATC
EMEA (drug inserts)	600M	6%	Pharmacology
E3C (clinical sentences)	400M	4%	Patient cases
Total	**10B**	**100%**	

Table 10.4: **Composition of the *biomed-fr-v2* training mixture** after quality-weighted upsampling.

work as filters: educational value and content richness help, but writing-quality and terminological-precision filters *hurt* performance, likely because they remove the informal but informative clinical documents that ultimately matter most. The lesson, perhaps, is that curation should be guided by what is being said, not by how well it is written.

The corpus is now ready. The next two chapters use it to train French biomedical encoders. Chapter 11 starts with the simplest recipe (continual pretraining with MLM on *biomed-fr*) and shows that even a modest public corpus produces a competitive model. Chapter 12 returns to *biomed-fr-v2* with a more ambitious recipe: a temporary detour through causal language modeling that, somewhat unexpectedly, leaves a permanent imprint on the encoder.

Part IV

Pretraining Language Models

11 Encoder Models for French Biomedicine

11.1 Introduction

In Chapter 8, we found that 55% of biomedical subwords are absent from CamemBERT's vocabulary. PubMedBERT (Gu et al., 2022) turned this kind of observation into a principle: for domains with enough unlabeled text, training a language model from scratch with a specialized vocabulary outperforms continual pretraining. The gain over BioBERT (Lee et al., 2020) was modest, but the conclusion was adopted widely.

For French, Labrak et al. (2023) went further. They reported that continual pretraining from CamemBERT led to F_1 losses of up to 20 points, and released DrBERT, trained from scratch on 128 V100 GPUs for 20 hours. The claim was strong, the compute was expensive, and the implication was clear: continual pretraining does not work for French biomedical models. Yet the size of the reported loss is difficult to reconcile with what we know about continual pretraining. A model that has already seen 138 GB of French text should not lose 20 points from seeing additional biomedical text. Something in the evaluation deserves closer examination.

In this chapter, we show that continual pretraining on *biomed-fr* (Chapter 8) yields an average improvement of +2.54 F_1 across five biomedical NER benchmarks, using 2 V100 GPUs for 39 hours. We also identify the methodological differences that led to the contradictory conclusion.

11.2 Pre-training

For the adaptation of CamemBERT to the biomedical domain, we conducted a second phase of pre-training on both versions of the *biomed-fr* corpus, starting from the weights and configuration of the camembert-base model. We applied the Masked Language Modeling (MLM) task with whole-word masking, following the method of Martin et al. (2020). We used the Adam optimizer (Kingma and Ba, 2015) with $\beta_1 = 0.9$ and $\beta_2 = 0.98$, and a learning rate of 5×10^{-5}. We performed 50,000 steps over 39 hours using two Tesla V100 GPUs. A batch size of 8 per GPU and gradient accumulation over 16 steps were used to achieve an effective batch size of 256.

11.3 Fine-tuning and Evaluation

Regarding model evaluation, we collected three named entity recognition evaluation datasets. These datasets cover various styles, allowing us to assess the model's versatility across different subdomains of biomedicine.

QUAERO. The QUAERO corpus (Névéol et al., 2014) consists of two evaluation sets: EMEA, containing drug leaflets, and MEDLINE, containing scientific article titles. The entities are manually annotated following 10 semantic groups from the UMLS (Lindberg et al., 1993). As some of these entities are nested, we kept only the entities with the coarsest granularity.

E3C. For evaluation, unlike the *biomed-fr* corpus, we use layers 1 and 2. These layers contain documents of different types, including clinical cases extracted from scientific articles. Layer 2 is semi-annotated, and it is used as the training set for fine-tuning, with 10% dedicated to the validation set. We evaluate on layer 1, which is fully manually annotated. There is only one class, and the objective is to find clinical entities in the text, regardless of their type.

CAS. The CAS corpus (Grouin et al., 2019) also consists of clinical cases from scientific articles. We focus on task 3 of DEFT 2020 (Cardon et al., 2020), which is an information extraction task based on CAS. It includes two subtasks, and thus two sets of annotations. In the first subtask, two classes need to be identified: *pathology* and *signs or symptoms*. The second subtask concerns associated information, including *anatomy*, *dose*, *examination*, *mode*, *timing*, *substance*, *treatment*, and *value*. These two tasks will be referred to as CAS1 and CAS2, respectively.

Fine-tuning. For fine-tuning, we used Optuna (Akiba et al., 2019) for hyperparameter selection. We set the learning rate to 5×10^{-5}, the warmup ratio to 0.224, and the batch size to 16. We performed 2000 steps. Predictions were made using a simple linear layer on top of the model. None of the CamemBERT layers were frozen.

Evaluation. Scores are measured using the seqeval tool (Nakayama, 2018) in strict mode with micro-average and the IOB2 scheme. For each evaluation, the best fine-tuned model on the validation set is selected to measure the final score on the test set. We average the results over 10 evaluations with different seeds.

11.4 Results and Discussion

CamemBERT vs CamemBERT-bio. We observe a significant performance gain on all evaluation datasets with our new model (Table 11.1). On average, we achieve a 2.54-

Dataset	CamemBERT	CamemBERT-bio biomed-fr-small	CamemBERT-bio biomed-fr
Clinical			
CAS1	70.50	<u>72.94</u>	**73.03**
CAS2	79.02	<u>80.00</u>	**81.66**
E3C	67.63	<u>67.96</u>	**69.85**
Leaflets			
EMEA	74.14	<u>75.93</u>	**76.71**
Scientific			
MEDLINE	<u>65.73</u>	65.48	**68.47**
Average	71.40	<u>72.46</u>	**73.94**

F_1 scores (micro-average, seqeval strict, IOB2). Mean over 10 seeds. biomed-fr-s = biomed-fr-small (10% subset).

Table 11.1: **CamemBERT vs CamemBERT-bio** on five biomedical NER benchmarks. Continual pretraining on *biomed-fr* yields +2.54 F_1 on average.

point improvement in F-score. This gain is observed across all styles, demonstrating the model's versatility for both clinical and scientific domains.

BIOMED-FR-SMALL VS BIOMED-FR. We observe a decrease in performance with the biomed-fr-small dataset, but there is still a significant gain on certain datasets compared to CamemBERT. This confirms that the size of the corpus positively influences the performance, even in a specialized domain like biomedicine.

COMPARISON WITH THE STATE OF THE ART. We compared the performance of CamemBERT-bio with various previously mentioned approaches (Table 11.2). CamemBERT-bio achieves the best results for almost all evaluation datasets. Dura et al. (2022) did not observe improvement on EMEA and MEDLINE compared to CamemBERT because their pre-training corpus consists of documents from APHP, making it a less diverse corpus. However, they gain several points on their evaluation dataset, which is also based on APHP documents. van Mulligen et al. (2016) presents the highest score on MEDLINE and the best recall on EMEA. Their approach is based on a knowledge-based model, allowing them to achieve the best recall on both QUAERO evaluation datasets. Furthermore, their approach is the only one in this table capable of handling nested entities, giving them an advantage.

It is important to note that these different CamemBERT-based approaches have various experimental setups. The presence of CRF layers instead of a simple linear layer after CamemBERT, freezing CamemBERT layers, and variations in hyperpa-

Authors	CAS1	CAS2	EMEA	MEDLINE
seqeval				
Dura et al. (2022)	–	–	72.90	59.70
Ours	**73.03**	**81.66**	**76.71**	**68.47**
BRATeval				
Le Clercq de Lannoy et al. (2022)	–	–	67.40	55.30
van Mulligen et al. (2016)	–	–	74.90	**69.80**
Copara et al. (2020)	61.53	73.70	–	–
Ours	**84.97**	**83.25**	**77.80**	56.16

F_1 scores. BRATeval is the evaluation tool from the CLEF eHealth 2016 campaign (Névéol et al., 2016).

Table 11.2: **Comparison with existing approaches** on four NER tasks, using two evaluation tools.

rameters are examples of differences observed in addition to the pre-training corpus, which makes the comparison more challenging.

11.5 Evaluating Models: Methodology Matters

In a recent work, Labrak et al. (2023) introduced a new French biomedical language model named DrBERT. The authors contend that performing continual-pretraining on biomedical data from CamemBERT leads to reduced performance. They used a different methodology for the evaluation of named entity recognition, prompting us to examine the implications of each approach on the interpretation of the results.

Our evaluation approach is centered around micro F_1, which is measured using seqeval in strict mode with the IOB2 scheme. Their approach is based on weighted F_1 with independent token classification. Every token has a label and the model is evaluated for each of them, whereas with seqeval, the model is evaluated for each entity. Notably, the "O" token, representing non-entity, is the most frequent token and thus holds significant weight in the evaluation process. As all tokens are labeled, the number of predictions depends on the tokenizer, thereby influencing the final score.

In order to explore the impact of token labeling variations on the evaluation, we conducted an additional experiment on EMEA and MEDLINE where we reproduced the DrBERT methodology that we named *token-with-O*, and another one where we excluded the "O" token and only labeled the first token of each entity, referred to as *entity-without-O*.

We observe significant differences in scores between the two methodologies (Table 11.3). The methodology *token-with-O* consistently reports higher scores across all tasks compared to the alternative method, *entity-without-O*. Notably, the best-

Method	Model	EMEA w-F_1	EMEA m-F_1	MEDLINE w-F_1	MEDLINE m-F_1
token-with-O					
	DrBERT-7GB	87.45	34.95	75.52	**15.07**
	CamemBERT-bio	**90.37**	36.27	**77.89**	14.82
	CamemBERT	88.33	**47.45**	76.20	11.92
entity-without-O					
	DrBERT-7GB	66.72	**24.72**	60.70	**10.80**
	CamemBERT-bio	**73.53**	24.15	**62.04**	8.70
	CamemBERT	71.85	22.71	60.95	9.41

w-F_1 = weighted F_1, m-F_1 = macro F_1. Averaged over 10 runs.

Table 11.3: **Impact of evaluation methodology.** The best model varies depending on the metric and the treatment of the "O" token.

Model	Time	Hardware	GPU-h	kg CO_2
DrBERT	20h	128× V100	2,560	26.11
AliBERT	20h	48× A100	960	8.16
CamemBERT-bio	39h	2× V100	78	**0.80**

Table 11.4: **Carbon emissions during training.** CamemBERT-bio emits 32× less CO_2 than DrBERT. Estimation based on 34g CO_2eq. per kWh (France, 2022–2023).

performing model varies depending on the chosen methodology for one metric. CamemBERT achieves the highest macro F_1 score on the EMEA dataset, whereas with *entity-without-O*, DrBERT-7GB emerges as the top model. This observation underscores the influential role of the "O" class in determining the best-performing model.

In terms of macro F_1, DrBERT consistently outperforms CamemBERT-bio when evaluated using the *entity-without-O* methodology. On the other hand, CamemBERT-bio exhibits better performance across all other metrics. This indicates that DrBERT demonstrates a more balanced performance across different classes.

Furthermore, Labrak et al. (2023) suggested that continual-pretraining from a French model on French biomedical data is not effective, as it results in an F_1-score loss of up to 20 points. However, considering our success with continual-pretraining and the points we raised about the methodology, we suggest a reassessment of these findings.

11.6 Environmental Impact

Considering estimated carbon emissions during training, AliBERT emits 10 times the amount of CamemBERT-bio, while DrBERT releases 32 times more (Table 11.4). While a direct comparison with AliBERT was not possible, the minor performance variance with DrBERT, set against the pronounced disparities in computational and environmental costs, leads us to advocate for continual-pretraining as the preferred adaptation method for biomedical language models.

11.7 Limitations

Our evaluation focused on one task, named entity recognition. This might mean our understanding of how well our model performs is restricted. Future studies should aim to assess our model across a wider range of tasks and datasets to get a clearer picture of its strengths and weaknesses.

Additionally, the corpus limitations discussed in Chapter 8 apply here as well: *biomed-fr* is dominated by scientific literature and does not contain genuine clinical notes. Models pretrained on this corpus may still underperform on real hospital data compared to models trained on private clinical corpora (Dura et al., 2022).

Conclusion

The vocabulary mismatch identified in Chapter 8 is real, but it does not prevent effective adaptation: +2.54 F_1 at 32× less CO_2 than the from-scratch alternative. The 20-point loss reported by Labrak et al. (2023) turns out to be an artifact of the evaluation protocol, not a property of continual pretraining itself.

Continual pretraining works for encoders. The bidirectional nature of MLM may make these models particularly robust to domain shifts. But the field has moved toward autoregressive decoders, where the dynamics of continual pretraining are different, and the results, as the next chapter discusses, are less encouraging.

12 Beyond Masked Language Modeling

12.1 Introduction

Masked language modeling sees only 15% of its input. At every training step, 85% of the tokens receive no gradient signal. Causal language modeling, by contrast, predicts every token from its predecessors: every position contributes, every position learns. The price is that it can only look backward. The question is whether an encoder can benefit from this dense training signal without permanently losing its ability to look in both directions.

In this chapter, we show that it can. We introduce a two-phase pretraining strategy for ModernCamemBERT-bio: first, continual pretraining with a causal language modeling objective, then a short "decay" phase that converts the model back to masked language modeling. The resulting encoder outperforms the standard MLM-only approach by +2.9 F_1 on average across 8 French biomedical tasks (Base) and +1.5 F_1 (Large), winning on all 8. The effect transfers to English at three scales, with gains of +0.3 to +0.8 F_1 depending on model size.

The CLM phase leaves a lasting imprint on low transformer layers. CKA analysis shows that layers 0–7 diverge 5–44× more between CLM and MLM models than seed noise alone, and this divergence persists through extended MLM decay. Freeze interventions establish a double dissociation: freezing low layers during CLM eliminates the downstream benefit; freezing mid layers preserves it. The advantage cannot be localized post hoc: transplanting layer groups from CLM to MLM recovers at most 35% of the gap. The effect is emergent, requiring the coordinated interaction between all layers.

12.2 Motivation

12.2.1 When Decoder Continue-Pretraining Stops Working

12.2.2 Architecture and Objective

CamemBERT-bio (Chapter 11) demonstrated that continual pretraining works for biomedical encoders. However, it used the original CamemBERT architecture: 512-token context, absolute position embeddings, no Flash Attention. Clinical documents

Figure 12.1: **The CLM detour pipeline.** Starting from a pretrained MLM encoder, we continue pretraining with a causal objective (Phase 1), then decay back to MLM (Phase 2). The model does not return to an MLM-like state.

are often longer than 512 tokens, and the architecture has not kept pace with recent advances.

ModernBERT (Warner et al., 2024) addresses these limitations with Flash Attention, RoPE position embeddings, alternating local/global attention, and an 8,192-token context window. For biomedical adaptation, we start from ModernCamemBERT (French) and ModernBERT (English), in both Base (150M) and Large (350M) sizes.

The standard approach would be to continue MLM pretraining on biomedical data, as we did for CamemBERT-bio. But MLM has a structural limitation: only 15% of tokens are masked and receive gradient signal at each step. Causal language modeling does not have this limitation: every token predicts the next, and every position contributes to the loss. The idea is to use CLM for the compute-intensive pretraining phase, and switch back to MLM only at the end, to recover bidirectional attention. The key question is whether this switch erases the benefits of CLM or preserves them.

Related approaches. A handful of recent works explore objective switching for encoder pretraining. Gisserot-Boukhlef et al. (2025) pretrain encoders *from scratch* (210M–1B parameters, 100B tokens) and find that a biphasic CLM-then-MLM schedule outperforms pure MLM under fixed compute. Weller et al. (2025) train matched encoder/decoder pairs (up to 1B parameters, 1.7T tokens) and show that continued pretraining on the reverse objective does not bridge the encoder-decoder performance gap. AntLM (Team, 2024) alternates between CLM and MLM epochs at small scale (10M words). Our approach differs in three ways: we study the round-trip MLM→CLM→MLM with continued pretraining (not from scratch); we focus on the lasting effect of the CLM phase after returning to MLM; and we provide causal evidence via freeze interventions of where the benefit comes from. As Gisserot-Boukhlef et al. (2025) note, understanding how training objectives transfer to downstream tasks "remains largely unexplored."

12.3 Method

12.3.1 The CLM Detour

Our approach consists of two phases, illustrated in Figure 12.1.

Phase 1: CLM pretraining. Starting from a pretrained ModernBERT encoder, we replace the bidirectional attention mask with a causal mask and train with next-token prediction. The model architecture is unchanged: only the attention mask and loss computation differ. This forces the model to compress information into each token representation, since it cannot look ahead.

Phase 2: MLM decay. After CLM pretraining, we restore bidirectional attention and train with MLM at 15% masking for approximately 10% of the Phase 1 budget. The optimizer state is kept between phases; only the learning rate scheduler resets. Phase 2 decays the learning rate from peak to 10% of peak.

Controlled comparison. The MLM baseline follows the same two-phase structure with 30% masking in Phase 1 and 15% in Phase 2, identical schedule and optimizer. Both trajectories use identical architecture, data, and total compute:

$$\theta_A = T_{\text{MLM}}(T_{\text{CLM}}(\theta_0, n), 0.1n) \qquad \text{(CLM detour)}$$
$$\theta_B = T_{\text{MLM}}(\theta_0, 1.1n) \qquad \text{(MLM baseline)}$$

The only difference is the Phase 1 objective. Any performance gap is attributable to the training objective alone.

12.3.2 Freeze Interventions

To test which layers carry the CLM benefit, we run three freeze experiments on the 22-layer French Base model, where the CLM-MLM gap is largest (+2.9pp):

- **Experiment 1** (low freeze, CLM phase): Layers 0–7 frozen during CLM, then normal decay. Tests whether low-layer modification during CLM is necessary.
- **Experiment 2** (low freeze, decay phase): Normal CLM, then layers 0–7 frozen during decay. Tests whether low-layer CLM changes persist without further updates.
- **Experiment 3** (mid freeze, CLM phase): Layers 8–14 frozen during CLM, then normal decay. Together with Experiment 1, tests selectivity: if freezing low layers eliminates the benefit while freezing mid layers preserves it, this constitutes a double dissociation.

12.3.3 Data

French. We compile 10B tokens (Base) or 25B tokens (Large) from four sources. The main source (7B tokens) is French biomedical literature (scientific articles, clinical guidelines, medical theses), where each paragraph is scored for educational value using Qwen3-235B and articles are upsampled based on their proportion of high-scoring paragraphs, following the Biomed-Enriched approach (Chapter 9). The remaining sources are synthetic medical QA from French coding systems (2B), clinical cases from the European Clinical Case Corpus (E3C, 400M), and drug package inserts from the European Medicines Agency (600M).

English. For the 50B scale, we mix three components: 60% biomedical literature from Biomed-Enriched (PMC Open Access articles filtered by educational value ≥ 3.0), 20% medical instruction-following datasets (MedS-Ins, ReasonMed, MediFlow, UltraMedical, EHR-Ins-Reasoning), and 20% MIMIC-III discharge summaries and clinical notes. The smaller 10B variant uses Biomed-Enriched with clinical case paragraphs upsampled 100× and other clinical documents 10× (80%), plus MediFlow medical instructions (20%), without MIMIC.

12.3.4 Training Details

All runs use decoupled AdamW with peak lr 2×10^{-4}, $\beta_1 = 0.9$, $\beta_2 = 0.98$, weight decay 10^{-5}, and a global batch size of 384 sequences (~3.1M tokens). Phase 1 uses linear warmup over 100M tokens then constant learning rate. Documents are packed into 8,192-token sequences with end-of-sequence tokens between documents. Training uses bf16 mixed precision on 4×H100 GPUs.

12.3.5 Fine-tuning Protocol

French tasks. Classification (DiaMED) and multilabel tasks (MedDialog, FrACCO, CANTEMIST, DisTEMIST) use a linear head on the CLS token, trained for 15 epochs with $\text{lr} = 2 \times 10^{-5}$, batch size 4, and 10% warmup. Multilabel tasks use max sequence length 4096 with weighted BCEWithLogitsLoss; classification uses 2048. NER tasks (EMEA, Medline) use a span-based architecture with BiLSTM, CRF, and biaffine scorer, trained for 4000 steps with $\text{lr} = 5 \times 10^{-5}$ for the encoder and 10^{-3} for the head, batch size 16, max length 1024. All tasks use AdamW with weight decay 0.01 and select the best checkpoint by validation F_1.

English tasks. We follow the BioClinical-ModernBERT evaluation protocol (Sounack et al., 2025): $\text{lr} = 5 \times 10^{-5}$, weight decay 0.01, batch size 16, 10 epochs for most tasks (20 for NER). All models are fine-tuned with the same hyperparameters per task.

12.3.6 Evaluation

We evaluate on 8 French and 11 English biomedical tasks (Table 12.1), using 9 seeds for French and 5 for English. French baselines include ModernCamemBERT, DrBERT (Labrak et al., 2023), CamemBERT-bio, and CamemBERT (Martin et al.,

Table 12.1: **Evaluation tasks.** Max sequence length reflects the fine-tuning configuration; longer values indicate tasks with longer input documents.

Task	Type	Lang	Max len.
French			
DiaMED	Classification	FR	2048
MedDialog-FR	Multilabel	FR	4096
FrACCO (30/100)	Multilabel	FR	4096
CANTEMIST	Multilabel	FR	4096
DisTEMIST	Multilabel	FR	4096
QUAERO (EMEA/Medline)	NER	FR	1024
English			
ChemProt	Relation Extr.	EN	512
Phenotyping	Classification	EN	8192
COS	NER	EN	512
Social History	NER	EN	512
De-identification	NER	EN	8192
AnatEM	NER	EN	512
BC5CDR	NER	EN	512
JNLPBA	NER	EN	512
NCBI Disease	NER	EN	512
GAD	Classification	EN	512
HoC	Classification	EN	512

2020). English baselines include ModernBERT, PubMedBERT (Gu et al., 2022), BioBERT (Lee et al., 2020), and BioClinical-ModernBERT (Sounack et al., 2025).

12.4 Results

12.4.1 French Biomedical Tasks

Table 12.2 presents the French results. For Base, the CLM detour achieves 60.8% average F_1, outperforming the MLM baseline on all 8 tasks (+2.9pp). For Large, CLM reaches 63.3% versus 61.8% for MLM (+1.5pp). CamemBERT-bio and CamemBERT average 36% and 35% overall, limited by their 512-token context on long clinical documents. We release the CLM-detour models as ModernCamemBERT-bio (Base and Large).

12.4.2 English Biomedical Tasks

Table 12.3 shows the English results at three scales. CLM outperforms MLM on average at all scales, with the gap widening at Large scale (+0.8pp, 7/11 task wins). The smaller English effect is expected: ModernBERT was pretrained on web data that commonly includes biomedical corpora such as PubMed, so its low layers

Table 12.2: **French biomedical results** (macro F_1, 9 seeds). Bold: best per column; underline: second best.

		Multilabel Classification					CLS	NER		
	Ctx	FrACCO-30	FrACCO-100	CANTEMIST	DISTEMIST	MedDialog	DiaMED	EMEA	Medline	Avg
Baselines										
ModernCamemBERT	8192	70.1	55.3	63.3	20.2	60.6	56.4	63.4	55.3	55.6
DrBERT	512	53.0	35.6	37.9	21.4	63.6	57.0	**68.0**	**62.3**	49.9
CamemBERT-bio	512	41.9	20.1	12.8	9.6	38.6	47.7	61.6	56.6	36.1
CamemBERT	512	40.8	19.4	11.9	9.5	37.4	40.6	61.5	54.7	34.5
Our Models (Base, 150M)										
MLM baseline	8192	69.9	56.8	64.9	23.5	62.5	63.4	65.4	56.8	57.9
CLM detour	8192	74.8	60.1	71.0	25.5	63.6	**67.4**	65.9	58.2	60.8
Our Models (Large, 350M)										
MLM baseline	8192	79.4	63.3	72.6	29.1	**64.5**	61.2	66.1	58.0	61.8
CLM detour	8192	**80.7**	**65.4**	**74.4**	**30.4**	**64.5**	64.8	67.0	59.5	**63.3**

Table 12.3: **English biomedical results** across 11 benchmarks (5 seeds). Base at 10B and 50B scales; Large at 50B. Bold: best; underline: second best.

		Clinical Tasks					BigBIO Tasks						
	Ctx	ChemPr	Pheno	COS	Social	DEID	AnatEM	BC5CDR	JNLPBA	NCBI	GAD	HoC	Avg
Baselines													
ModernBERT-base	8192	89.5	48.4	94.0	53.1	78.3	77.2	87.9	74.3	77.7	76.8	66.6	74.9
PubMedBERT	512	90.2	52.0	95.0	48.7	80.4	**83.3**	89.7	74.9	**82.1**	79.3	**71.0**	77.0
BioClinical-MdnBERT	8192	90.0	60.7	94.8	56.0	81.8	79.2	88.7	74.8	78.7	75.8	67.0	77.0
Base 150M (10B tokens)													
MLM baseline	8192	90.4	61.3	94.4	55.5	79.3	78.9	88.3	74.9	79.2	78.6	68.9	77.3
CLM detour	8192	90.3	61.0	94.3	55.7	82.6	79.9	89.1	74.2	80.4	79.3	69.4	77.8
Base 150M (50B tokens)													
MLM baseline	8192	**90.5**	61.6	95.1	55.7	81.6	80.0	88.7	74.9	80.3	78.7	67.1	77.7
CLM detour	8192	90.1	**61.9**	**95.2**	54.2	83.2	81.0	89.1	74.5	80.1	78.8	70.0	78.0
Large 350M (50B tokens)													
MLM baseline	8192	**90.5**	61.0	94.9	55.0	82.3	82.0	89.4	**75.5**	81.8	76.4	67.8	77.9
CLM detour	8192	90.4	61.3	94.7	**56.5**	**84.2**	83.2	**89.8**	75.3	81.7	**79.7**	69.3	**78.7**

already partially encode biomedical features. ModernCamemBERT, by contrast, was pretrained on general French web text without biomedical sources, leaving a larger domain gap and correspondingly stronger CLM benefit (+2.9pp vs +0.3pp). We release the English models as ModernBERT-bio.

12.4.3 Optimal Decay Ratio

We sweep the decay length from 2.5% to 50% of the CLM phase at two scales, evaluating on 3 French tasks (DiaMED, FrACCO-30, FrACCO-100; Table 12.4). At both scales, 10% decay gives the best performance; shorter decay (2.5–4%) scores about 1.0pp below optimal, and longer decay (20–50%) provides no additional benefit. This ratio matches the cooldown lengths used in language model pretraining.

12.4.4 Needle-in-a-Haystack

Long-context encoding is critical in biomedical NLP, where clinical documents routinely span thousands of tokens and key information (diagnoses, coding labels) can

Table 12.4: **Decay ratio sweep** on 3 French tasks (DiaMED, FrACCO-30, FrACCO-100; 9 seeds each). 10% decay is optimal at both scales.

Decay tokens	Decay / Phase 1	CLM+decay F_1	MLM+decay F_1
10B Phase 1 (Base, 150M)			
0.25B	2.5%	71.6%	69.1%
1.0B	**10%**	**72.6%**	70.4%
2.0B	20%	71.9%	70.4%
3.0B	30%	72.1%	69.7%
5.0B	50%	71.7%	70.4%
25B Phase 1 (Large, 350M)			
1.0B	4%	73.2%	70.1%
2.5B	**10%**	**73.5%**	70.5%
5.0B	20%	72.7%	70.1%

appear anywhere in the document. We design a synthetic needle-in-a-haystack task in French to evaluate long-context retrieval without entity-type confounds.

Protocol. A medical fact (the "needle") is inserted into a clinical document (the "haystack") at a controlled position, and the model must determine whether a given query fact is present (binary classification). Negative examples use the same template with different slot values, so the task requires precise retrieval rather than shallow pattern matching. We define 8 French medical fact templates covering drugs, vital signs, lab results, diagnoses, procedures, allergies, treatment duration, and symptoms. Each template has 2–4 slots filled from small lexicons:

- *Le patient a reçu 200 mg de morphine par voie intraveineuse.*
  Distractor: *Le patient a reçu 500 mg de tramadol par voie orale.*
- *La tension artérielle mesurée était de 160/90 mmHg.*
- *Le diagnostic retenu est cirrhose hépatique de stade III.*
- *Le patient rapporte une dyspnée évoluant depuis une semaine.*

We generate 1,500 balanced positive/negative pairs across 5 lengths (512–8,192 tokens) and 3 positions (start, middle, end), split 70/15/15 for train/validation/test. We freeze each encoder and train a 2-layer MLP probe (Dropout → Linear → GELU → Dropout → Linear) on CLS representations for 3 epochs (lr = 2×10^{-5}, batch size 4, AdamW), selecting the best checkpoint by validation accuracy.

Results. CLM outperforms MLM at every context length and position (+10.7pp overall: 62.2% vs 51.6%). MLM accuracy degrades monotonically with document length (57% at 512 tokens to 43% at 8,192), while CLM degrades more slowly and remains above MLM throughout. The CLM advantage is largest at mid-document positions, where the inserted fact is farthest from both sequence boundaries. This is consistent with the downstream pattern: the French tasks with the largest CLM gains (FrACCO, CANTEMIST) use sequence lengths of 4,096 tokens during fine-tuning.

12.5 Analysis: Why Does the CLM Detour Work?

12.5.1 CKA Methodology

We measure representational similarity with linear Centered Kernel Alignment (Kornblith et al., 2019). Given two representation matrices $X \in \mathbb{R}^{n \times p}$ and $Y \in \mathbb{R}^{n \times q}$ (one per model, n samples, mean-pooled over tokens), we center each column, compute Gram matrices $K = XX^\top$ and $L = YY^\top$, and define:

$$\mathrm{CKA}(X, Y) = \frac{\mathrm{tr}(KL)}{\sqrt{\mathrm{tr}(K^2)\,\mathrm{tr}(L^2)}}$$

We report divergence $d = 1 - \mathrm{CKA}$, so higher values indicate greater representational difference. CKA equals 1 when representations are identical up to invertible linear transformation, 0 when there is no linear relationship. All computations use float64 arithmetic. For French we use 500 held-out texts from DiaMED and FrACCO; for English, PubMed abstracts. Results are averaged over 3 seeds.

12.5.2 The CLM Imprint Persists in Low Layers

Comparing layer-by-layer CKA between CLM-detour and MLM-only models, the CLM phase modifies low-layer representations in a way that subsequent MLM training does not undo, even when the decay budget matches the CLM phase. CKA divergence reaches approximately 56.5% after 1.5B tokens of MLM decay and remains stable, with only 0.6pp variation over 8B additional tokens.

To isolate where CLM differs from MLM *specifically* rather than just from training noise, we normalize each layer's CLM-MLM divergence by a seed-noise control (two MLM models trained with different random seeds on identical data):

$$r_\ell = \frac{1 - \mathrm{CKA}(\mathrm{CLM}_\ell,\ \mathrm{MLM}_\ell^{s_1})}{1 - \mathrm{CKA}(\mathrm{MLM}_\ell^{s_2},\ \mathrm{MLM}_\ell^{s_1})}$$

A ratio of 1 means no CLM-specific effect. Low layers (0–7) show ratios of 5–44×: CLM changes them far more than random seed variation does. Mid and deep layers are near 1×: both objectives modify them similarly, so CKA cannot distinguish CLM from MLM there.

Table 12.5: **Freeze intervention results** (French Base, 8 tasks, 9 seeds). Freezing low layers during CLM eliminates the benefit ($p = 0.36$ vs MLM baseline). Freezing mid layers preserves it. Freezing low layers during decay has negligible effect.

Condition	Avg F_1	Δ vs MLM
References		
MLM baseline	57.9	—
CLM detour	**60.8**	+2.9
Freeze interventions		
Freeze L0–7 (CLM phase)	58.4	+0.5
Freeze L8–14 (CLM phase)	60.3	+2.4
Freeze L0–7 (decay phase)	60.3	+2.4

Table 12.6: **Layer transplantation** on DiaMED (3 seeds, linear probing). No transplant recovers more than 35% of the 7.1pp gap.

Transplant	F_1 (%)	Δ	Recovery
References			
MLM baseline	32.8	—	0%
CLM detour	39.9	+7.1	100%
Layer group transplants			
Low (0–7)	31.9	−0.9	−12%
Mid (8–14)	35.3	+2.5	+35%
Late (15–21)	32.4	−0.4	−6%

12.5.3 Causal Evidence: Freeze Interventions

Do these low-layer changes actually drive the downstream improvement? We answer with the freeze interventions described in §12.3.2, summarized in Table 12.5.

Freezing low layers (0–7) during CLM drops F_1 from 60.8% to 58.4%. We cannot reject the null hypothesis that the resulting model matches the MLM baseline ($p = 0.36$, paired bootstrap across 8 tasks × 9 seeds). Freezing mid layers (8–14) during CLM preserves the full benefit (60.3%). Together, these two experiments establish a double dissociation: low layers are necessary and mid layers are not. Freezing low layers during decay has negligible effect (−0.5pp, 60.3%), confirming that MLM decay preserves the CLM imprint in low layers.

12.5.4 Non-Localizability

The freeze experiments show that low layers are necessary *during training*. We now ask whether the resulting benefit can be localized *after training*, by copying full parameter blocks from the CLM model into the MLM model.

Table 12.7: **Component-level transplants** on DiaMED (3 seeds, linear probing). Neither attention nor MLP modules alone recover the full gap.

Transplant	F_1 (%)	Δ	Recovery
All attention	33.9	+1.1	+15%
All MLP	35.3	+2.5	+35%

Table 12.8: **CKA divergence scales with model size and is asymmetric.** The encoder-decoder comparison uses different model families (see text).

Model	Params	Div.	CI
Encoders (MLM→CLM→MLM)			
ModernCamemBERT Base	150M	56.5%	±0.7%
ModernCamemBERT Large	350M	67.2%	±0.3%
Decoder (CLM→MLM→CLM)			
Gemma-3	270M	26.3%	±1.0%

No transplant recovers more than 35% of the gap (Table 12.6). Low and late layer transplants hurt performance. Inserting CLM low layers into an MLM model creates a mismatch with the mid and deep layers that co-adapted under MLM, yielding representations worse than either pure model.

Component-level transplants (Table 12.7) confirm the same pattern. Replacing all attention modules from CLM into MLM recovers only 15% of the gap; replacing all MLP blocks recovers 35%. No single component type captures the full benefit.

Unlike freeze interventions, which act during training, transplants replace already-trained weights and cannot reconstruct the cross-layer co-adaptation that developed during CLM. The freeze experiments show low layers are necessary during training; the transplant experiments show the resulting benefit cannot be isolated post hoc.

12.5.5 Scaling and Architectural Asymmetry

The CLM imprint increases with model capacity (Table 12.8): Large (350M) reaches 67.2% ± 0.3% divergence versus 56.5% ± 0.7% for Base (150M). Testing the reverse on Gemma-3 (270M decoder), MLM-then-CLM retains only 26.3% ± 1.0% divergence versus 67.2% for an encoder of similar scale. One explanation is the density of the decay signal: MLM decay at 15% masking provides weak erasure, while CLM decay at 100% provides strong erasure. This comparison is exploratory: it is confounded by model family and language, and a controlled comparison using encoder and decoder variants of the same family would be needed to confirm the asymmetry.

12.6 Limitations

Our freeze interventions use French Base models only. The CKA patterns that motivate them (low-layer divergence well above seed noise) are consistent across French Base, French Large, and English Base, but replication of the freeze interventions on English and Large models would strengthen the causal claim. The encoder-decoder asymmetry comparison is confounded by model family. The position trade-off at 2048+ tokens means that the approach may not be suitable for tasks requiring very long context, such as full discharge summary coding.

Conclusion

Training an encoder like a decoder, then converting it back, produces stronger representations than training with MLM alone. The CLM detour reshapes low transformer layers in ways that subsequent MLM decay does not undo: a lasting imprint that scales with model capacity and produces consistent downstream gains across 8 French and 11 English biomedical tasks.

The practical implication is straightforward: for biomedical encoder pretraining, the CLM detour is a free lunch. Same data, same compute, better results. MLM decay of roughly 10% of the detour length suffices, and the effect is strongest when the domain gap between pretraining and target data is large.

Parts 1 and 2 of this thesis have focused on data and pretraining: how to collect biomedical text, how to filter it, and how to train encoder models on it. But a pretrained model is only useful if it can be deployed on real tasks. In the next part, we turn to adaptation: how to extract clinical information when labeled data is scarce, labels number in the thousands, and the only available text is public.

13 Knowledge-Enriched Pretraining

13.1 Introduction

Medical coding is the task of assigning standardized codes from medical ontologies to clinical documents. It is central to hospital billing, epidemiological surveillance, and clinical research. In France, over 30 million hospital stays per year require accurate coding using the CIM-10 classification (the French adaptation of ICD-10), CCAM procedure codes, and ATC medication codes. This task requires understanding not only medical vocabulary but also the relational structure of medical ontologies: hierarchical links between codes, causal associations, differential diagnoses, and exclusion rules. These relationships are explicitly encoded in ontology graphs but are absent from clinical text corpora.

Pretrained biomedical language models such as BioBERT (Lee et al., 2020), PubMedBERT (Gu et al., 2022), and CamemBERT-bio (Touchent et al., 2023) learn medical vocabulary from large text corpora. However, they do not capture the relational structure of medical ontologies, since this structure is not expressed in running text. Knowledge graph approaches such as RDF2Vec (Ristoski and Paulheim, 2016) and Snomed2Vec (Agarwal et al., 2019) encode ontology structure through random walks, but produce static, non-contextualized embeddings that cannot benefit from transformer pretraining. Knowledge-enhanced transformers such as DRAGON (Yasunaga et al., 2022a) and KEPLER (Wang et al., 2021) integrate knowledge graph structure into language model pretraining, but they rely on existing text-graph alignment rather than generating new training data from ontology structure alone.

We propose OntoBook, a pretraining method that converts medical ontology structure into training signal for encoder language models. Our approach proceeds in three stages. First, we generate random walks through ontology graphs to produce sequences that capture hierarchical, causal, and differential relationships between medical codes. Second, we reformulate these walks into fluent medical prose using a large language model, creating synthetic textbooks grounded in ontology structure. Third, we train an encoder with two objectives on the same textbook data: masked language modeling and relation prediction between code pairs. We refer to this co-training on shared data as *alignment*, and our experiments show that it is a necessary condition for the method to work.

We evaluate OntoBook on three French medical coding benchmarks and present ablation studies on the contribution of each component. We release 1.3 million LLM-

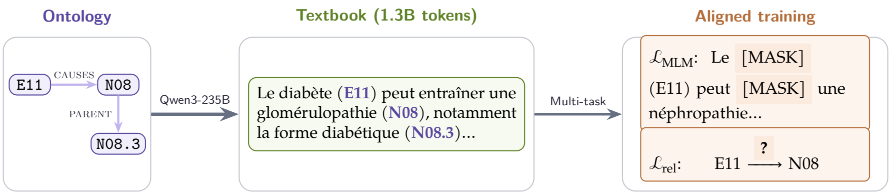

Figure 13.1: **OntoBook pipeline.** An ontology subgraph is converted into a random walk, then reformulated into textbook prose by Qwen3-235B. The encoder trains on this text with two aligned objectives: masked language modeling and relation prediction between code pairs.

reformulated medical textbooks across three French ontologies (CIM-10, CCAM, ATC) and all pretrained model checkpoints.

13.2 Related Work

This section reviews three research directions: biomedical language model pretraining, knowledge-enhanced language models, and automatic medical coding.

13.2.1 Biomedical Pretrained Language Models

Domain-specific pretraining has proven essential for biomedical NLP. BioBERT (Lee et al., 2020) further pretrained BERT on PubMed abstracts and PMC full-text articles, improving named entity recognition and relation extraction. PubMedBERT (Gu et al., 2022) demonstrated that pretraining from scratch on in-domain text outperforms mixed-domain initialization. ClinicalBERT (Huang et al., 2019) targeted clinical notes from MIMIC-III. For French, CamemBERT-bio (Touchent et al., 2023) adapted CamemBERT (Martin et al., 2020) to biomedical text, while DrBERT (Labrak et al., 2023) was trained on French biomedical and clinical text from the NACHOS corpus. These models learn from flat text corpora and do not capture the hierarchical structure of medical ontologies.

13.2.2 Knowledge-Enhanced Language Models

Random walks on knowledge graphs generate sequences capturing relational structure. RDF2Vec (Ristoski and Paulheim, 2016) adapted the walk-then-embed paradigm to RDF graphs. In the biomedical domain, Snomed2Vec (Agarwal et al., 2019) applied random walk and Poincaré embeddings to SNOMED-CT for clinical prediction tasks. OWL2Vec* (Chen et al., 2021) combined random walk sequences with lexical and logical features from OWL ontologies. These methods produce static embeddings (Word2Vec-style) and cannot be used for transformer pretraining.

Several approaches inject knowledge graph structure into transformers. ERNIE (Zhang et al., 2019) aligns entity embeddings with text representations. K-BERT (Liu et al., 2020) injects knowledge triples into the input sequence using soft-position indices and a visibility mask. For multi-task pretraining, KEPLER (Wang et al., 2021) combines MLM with TransE-style knowledge embedding on Wikidata. DRAGON (Yasunaga et al., 2022a) jointly trains MLM and knowledge graph link prediction, with a biomedical variant on UMLS achieving +3% on MedQA. CODER (Yuan et al., 2022) uses UMLS relation triplets for contrastive pretraining, and SAPBERT (Liu et al., 2021) uses contrastive learning on UMLS synonyms for biomedical entity linking. Closest to the present work, BioOntoBERT (Shashikumar et al., 2023) generates sentences from biomedical ontologies using Onto2Sen templates and pretrains BERT with MLM on the resulting corpus. However, the template-based generation produces formulaic sentences (e.g., "X is a type of Y") rather than fluent prose, and the method uses only MLM without multi-task relation prediction.

Recent work has explored synthetic data generation from knowledge sources. The Phi series (Gunasekar et al., 2023) demonstrated that LLM-generated textbook-quality data enables strong small-model performance. EntiGraph (Yang et al., 2025b) generates entity-centric synthetic data from knowledge sources but targets decoder language models rather than encoder pretraining.

13.2.3 Automatic Medical Coding

Medical coding assigns ICD, procedure, or medication codes to clinical text. CAML (Mullenbach et al., 2018) introduced label-wise attention for explainable predictions. PLM-ICD (Huang et al., 2022) showed that fine-tuning pretrained encoders with label attention directly improves coding performance. Kirchler et al. (2026) recently demonstrated that LLM embeddings improve transferability of EHR-based predictions across countries and coding systems, highlighting the importance of encoding medical knowledge structure for cross-system generalization. These methods improve the classification head or the input representations but rely on generic encoders pretrained on flat text corpora.

Table 13.1 summarizes how OntoBook combines components not jointly present in prior work.

13.3 Method

OntoBook converts medical ontology structure into pretraining signal through three stages: (1) generating random walks through ontology graphs, (2) reformulating these walks into fluent textbook prose using a large language model, and (3) training an encoder with multi-task learning combining masked language modeling and

Method	Walks	LLM	MLM+Rel
Prior work			
Snomed2Vec	✓		
BioOntoBERT	✓		
DRAGON			✓
KEPLER			✓
Phi-1		✓	
Ours			
OntoBook	✓	✓	✓

Table 13.1: **Comparison with related approaches.** OntoBook uniquely combines all three components for biomedical encoder pretraining.

relation prediction. We first describe the ontologies used, then detail each stage. Figure 13.1 illustrates the full pipeline.

13.3.1 Medical Ontologies

We use three French medical ontologies published by the Agence du Numérique en Santé (ANS) through its terminology server (SMT) in RDF/OWL format. CIM-10 FR PMSI is the French adaptation of the WHO's ICD-10, enriched by the Agence Technique de l'Information sur l'Hospitalisation (ATIH) for the French hospital information system (Programme de Médicalisation des Systèmes d'Information, PMSI). It contains 19,161 diagnostic codes organized in a 5-level hierarchy (chapters, blocks, categories, subcategories), with textual annotations including 23,282 synonyms, 8,171 inclusion notes, and 381 definitions. CIM-10 is the only ontology with semantic edges beyond the hierarchy: 1,317 causal edges (e.g., E11 type 2 diabetes *causes* N08.3 diabetic glomerulopathy), 5,739 exclusion edges, and 341 manifestation edges, yielding a mean node degree of 2.77 compared to 2.00 for the purely hierarchical ontologies.

CCAM (Classification Commune des Actes Médicaux) is the French procedure classification maintained by the Caisse Nationale d'Assurance Maladie (CNAM), containing 38,191 procedure codes in a 4-level hierarchy with 8,991 synonyms and 1,188 disjointness constraints between mutually exclusive procedures. ATC (Anatomical Therapeutic Chemical) is the WHO medication classification with 6,950 substance codes in a strict 5-level hierarchy from anatomical group to chemical substance, with no semantic edges, no synonyms, and no definitions beyond labels. This contrast in relational richness directly shapes the walk generation strategy: CIM-10 walks can follow causal and differential diagnosis paths, whereas CCAM and ATC walks are limited to hierarchical and sibling traversals. Table 13.2 summarizes the structural properties of the three ontologies.

	CIM-10	CCAM	ATC
Graph structure			
Codes	19,161	38,191	6,950
Hierarchy depth	5	4	5
Hierarchical edges	19,139	38,191	6,950
Causal edges	1,317	—	—
Manifestation edges	341	—	—
Exclusion edges	5,739	—	—
Disjoint edges	—	1,188	—
Mean degree	2.77	2.06	2.00
Textual attributes			
Synonyms	23,282	8,991	0
Inclusion notes	8,171	—	—
Definitions	381	151	0

Table 13.2: **Structural properties** of the three French medical ontologies used for walk generation. CIM-10 is the only ontology with semantic edges beyond the hierarchy.

13.3.2 Ontology Walk Generation

We generate random walks through ontology graphs to capture relational structure in a sequential format suitable for language model pretraining. We sample walks starting from each code in the ontology. At each step, the next node is selected via weighted sampling where edge type weights depend on the walk type. This biased sampling is necessary because semantic edges are rare (Table 13.2), and a uniform walk would rarely traverse them. Each walk records the traversed codes along with their labels and the relation types connecting them. Walk length is sampled uniformly between 8 and 20 steps. We track visited nodes to avoid cycles, with a fallback that allows revisiting nodes when all neighbors have been explored.

For CIM-10, we define five walk types, each emphasizing different aspects of medical knowledge: *Etiologie* (causal chains), *Diagnostic Différentiel* (codes to distinguish clinically), *Codage Double* (mandatory code pairs linking etiology to manifestation), *Syndrome* (multi-system manifestations), and *Cross-Chapter* (connections between different ICD chapters, e.g., diabetes E11 to its renal complication N08). We generate 402,328 walks for CIM-10 (average 4,830 characters), 762,958 for CCAM, and 138,980 for ATC, totaling 1,304,266 walks.

13.3.3 Textbook Generation

Raw walks contain structured markers (e.g., "`>> À distinguer de:`") that are useful for parsing but not natural for language model pretraining. We use a large language model to reformulate walks into fluent medical prose resembling textbook paragraphs.

Structured Walk

```
[F02.0] Démence de la maladie de Pick
>> Cette pathologie peut provoquer:
[G31.0] Atrophie cérébrale circonscrite
>> Parmi les causes possibles:
[F02.02] ...avec symptômes hallucinatoires
```

↓ *Qwen3-235B*

Medical Textbook

> Le code **F02.0**, désignant la démence de la maladie de Pick, s'inscrit dans un réseau de connexions inter-systèmes entre les troubles mentaux et les affections neurologiques. Ce code entretient une relation étroite avec **G31.0** (atrophie cérébrale circonscrite), établissant un lien étiologique réciproque...

Figure 13.2: **Walk-to-textbook transformation.** The structured walk is reformulated into fluent medical prose while preserving all codes and relations.

We prompt Qwen3-235B-A22B-Instruct (Yang et al., 2025a) with FP8 quantization to transform each walk into a coherent paragraph. The prompt instructs the model to: (1) preserve all medical codes and their relationships exactly as stated, (2) use natural medical language without lists or bullet points, (3) not add any information beyond what appears in the walk. This ensures the textbook content remains grounded in the ontology structure. We apply two filters: we discard reformulations shorter than 50 characters or that fail to mention the source codes. The reformulation runs on 4 NVIDIA H100 GPUs using vLLM (Kwon et al., 2023) with guided decoding to ensure valid JSON output, taking approximately 20 hours for all 1.3 million walks. The resulting textbooks contain approximately 1.3 billion tokens across all three ontologies. Figure 13.2 shows an example transformation.

Figure 13.3 shows the system prompt used for CIM-10 walks (translated from French for readability; prompts for CCAM and ATC follow the same structure with ontology-specific terminology).

13.3.4 OntoBook Training

We train a ModernCamemBERT-base encoder (Antoun et al., 2025), initialized from a French checkpoint trained on 1 trillion tokens.

The model optimizes two objectives jointly. The first, $\mathcal{L}_{\text{MLM}}$, follows standard masked language modeling (Devlin et al., 2019): we mask 15% of tokens from textbook paragraphs and train to predict them. The second, $\mathcal{L}_{\text{rel}}$, classifies the relationship between pairs of medical codes. Our main model uses CIM-10; we explore transfer from CCAM and ATC in Section 13.4.5. We extract 282,907 code pairs from the CIM-10 ontology across 6 relation types: Parent, Child, Sibling,

Reformulation Prompt (CIM-10, translated)

You are an expert medical writer specializing in high-quality educational material.

Reformulate the provided CIM-10 walk into fluent, textbook-quality medical prose.

Strict rules:
(1) **No invention**: never add information not in the source.
(2) **Total fidelity**: preserve all definitions, codes, clinical notes, exclusions.
(3) **Completeness**: do not omit any source information.
(4) **Ignore metadata**: never mention walk types or structural headers.

Style: Continuous narrative prose, no bullet points or markdown. Integrate CIM-10 codes naturally. Preserve exact medical terminology.

Goal: Transform the walk into text suitable for a medical nosology textbook chapter while remaining 100% faithful to the original content.

Figure 13.3: System prompt for LLM-based walk reformulation. Temperature is set to 0 for deterministic, faithful output.

Differential Diagnosis, Causes, and Caused_By. For each pair, we retrieve the corresponding textbook descriptions generated in Section 13.3.3. The input format is "`[CLS]` text$_A$ `[SEP]` text$_B$ `[SEP]`", where text$_A$ and text$_B$ are the textbook descriptions of the two codes, and a separate linear classification head on the `[CLS]` token predicts the relation type, trained with cross-entropy loss over the 6 classes.

The combined loss is:

$$\mathcal{L} = \mathcal{L}_{\text{MLM}} + \lambda \mathcal{L}_{\text{rel}} \tag{13.1}$$

where $\lambda = 1.0$. We train for 4 epochs with learning rate 2×10^{-5}, batch size 128, and AdamW optimizer with weight decay 0.01 and 1,000 warmup steps. Training takes approximately 4 hours per epoch on one NVIDIA H100 GPU. The key design choice is that both objectives operate on the *same textbook data*. We call this property *alignment*, and we show in Section 13.4.3 that it is essential for performance.

13.4 Experiments

13.4.1 Experimental Setup

Evaluation Benchmarks. We evaluate on the standard French medical coding evaluation suite:

- **FRACCO** (Pignat et al., 2025): a native French oncology corpus with 1,301 clinical cases annotated with ICD-O-3.1 codes, framed as multi-label classification over the top 100 codes.
- **Cantemist-FR** (Zaghir et al., 2024): the French adaptation of the Spanish tumor coding task (Miranda-Escalada et al., 2020), with 2,051 clinical documents.

Model	FRACCO	Cant.	Dist.	Avg.
External baselines				
CamemBERT-bio	20.2$\pm$0.2	12.1$\pm$0.4	9.0$\pm$0.2	13.8
DrBERT	36.3$\pm$0.7	37.7$\pm$1.0	22.5$\pm$0.7	32.1
ModernCamemBERT	56.4$\pm$1.0	63.5$\pm$1.3	23.4$\pm$1.7	47.7
Our models				
CodeInfill+MLM	56.5$\pm$0.2	66.4$\pm$1.9	20.0$\pm$0.9	47.6
MLM-only	55.8$\pm$0.4	66.0$\pm$1.1	24.2$\pm$5.8	48.7
OntoBook	**58.3**$\pm$0.3	**67.1**$\pm$1.1	**32.2**$\pm$1.1	**52.5**

Table 13.3: **Main results** on French medical coding (micro-F_1 %). External baselines averaged over 9 seeds; our models over 3 seeds. Best results in **bold**.

- **Distemist-FR** (Zaghir et al., 2024): the French adaptation of the disease mention coding task (Miranda-Escalada et al., 2022).

Cantemist-FR and Distemist-FR are cross-lingual projections from Spanish. FRACCO is the only native French benchmark. All results are micro-F1 scores averaged over 3 random seeds.

Baselines. We compare against French biomedical language models: CamemBERT-bio (Touchent et al., 2023), DrBERT (Labrak et al., 2023), and ModernCamemBERT (Antoun et al., 2025) (which serves as our initialization checkpoint). We also report results for three ablations of our method: MLM-only (trained with $\mathcal{L}_{\text{MLM}}$ only on our textbooks), Rel-only (trained with $\mathcal{L}_{\text{rel}}$ only), and CodeInfill+MLM, which replaces relation prediction with a code infilling objective that masks medical code tokens (e.g., "E11", "N08.3") in textbook text and trains the model to predict them. We do not compare against knowledge-enhanced models such as DRAGON (Yasunaga et al., 2022a) or SAPBERT (Liu et al., 2021) as they are English encoders that cannot be applied to French text.

Fine-tuning. For downstream evaluation, each pretrained encoder receives a clinical document as input and predicts a set of medical codes (multi-label classification over the top 100 codes per benchmark). We fine-tune with a linear classification head for 10 epochs, learning rate 2×10^{-5}, and batch size 16. External baselines (CamemBERT-bio, DrBERT, ModernCamemBERT) are averaged over 9 seeds; our models over 3 seeds due to the large number of configurations evaluated.

13.4.2 Main Results

Table 13.3 compares OntoBook against French biomedical baselines.

Configuration	FRACCO	Cant.	Dist.	Avg.	Δ
Loss ablations					
OntoBook (aligned)	**58.33**	**67.06**	**32.24**	**52.54**	—
− $\mathcal{L}_{\text{rel}}$ (MLM-only)	55.81	66.01	24.23	48.68	−3.86
− $\mathcal{L}_{\text{MLM}}$ (Rel-only)	48.24	51.15	20.18	39.86	−12.68
− Alignment (misaligned)	33.39	20.11	12.63	22.04	−30.50
Data ablation					
MLM-only (raw walks)	55.51	66.00	22.76	48.09	−4.45

Table 13.4: **Ablation study** (micro-F_1 %). Misalignment degrades all benchmarks, with the largest drop on Cantemist (−46.95). All variants use the same base model and training budget.

OntoBook outperforms all baselines on all three benchmarks. The improvement over MLM-only pretraining is significant on FRACCO (+2.5 micro-F1, $p < 0.001$) and Distemist (+8.0 micro-F1, $p < 0.001$), with consistent but not individually significant gains on Cantemist ($p = 0.11$). The gain is largest on Distemist, suggesting that ontology-aware pretraining particularly helps with fine-grained disease coding. OntoBook also reduces variance on Distemist from ± 5.8 (MLM-only) to ± 1.1.

13.4.3 Ablation Study

Table 13.4 ablates the key components of OntoBook.

Alignment is the most critical factor. Training $\mathcal{L}_{\text{MLM}}$ and $\mathcal{L}_{\text{rel}}$ on different data causes catastrophic degradation across all benchmarks, with Cantemist dropping from 67.06 to 20.11 (−46.95). Relation prediction alone also fails (−12.68 average F1), confirming that $\mathcal{L}_{\text{rel}}$ cannot serve as a standalone pretraining objective. The last row trains MLM-only on raw structured walks, removing both reformulation and relation prediction. The raw walk model (48.09) is close to the textbook MLM-only model (48.68), indicating that reformulation alone contributes little to MLM pretraining (+0.59 F1). However, the gap to full OntoBook is 4.45 points, reflecting the combined effect of adding both reformulation and relation prediction. This suggests that reformulation primarily benefits the relation prediction objective: on raw walks with structural markers (e.g., "`>> À distinguer de:`"), the classifier can exploit formatting shortcuts rather than learning medical semantics, whereas fluent textbook prose forces the model to learn from content rather than format.

13.4.4 Training Dynamics and Hyperparameter Sensitivity

Figure 13.4 shows training dynamics and hyperparameter sensitivity. Epoch 1 (51.98) and epoch 2 (52.54) both outperform MLM-only pretraining (dashed line), but epoch 3 (49.43) degrades below it, possibly due to overfitting on the relation prediction

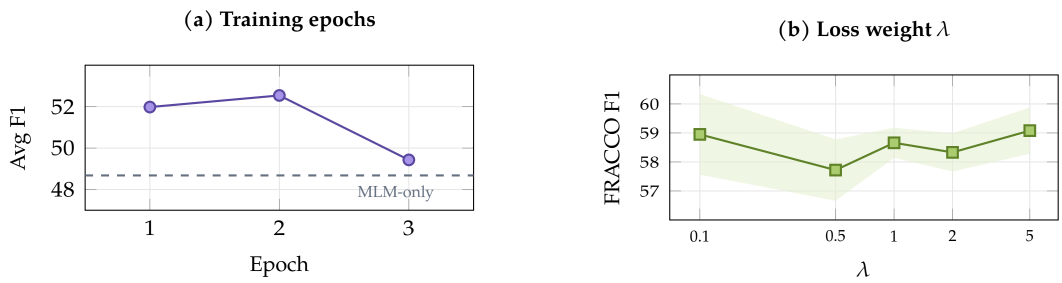

Figure 13.4: **Training dynamics.** (a) Average F_1 peaks at epoch 2 then degrades. (b) Performance is stable across λ values (shaded: ± 1 std).

Ontology source	FRACCO	Cant.	Dist.	Avg.
Main configuration				
CIM-10 (OntoBook)	58.33	67.06	32.24	52.54
Other ontology sources				
All (CIM-10+CCAM+ATC)	59.31	**67.64**	31.44	52.80
CCAM only	58.59	66.54	34.26	**53.13**
ATC only	**59.66**	63.03	**36.05**	52.91

Table 13.5: **Effect of ontology source** on downstream performance (micro-F_1 %). Each row trains a separate model using walks and relations from only that ontology. All models use the same training procedure and base checkpoint.

task. All results in this paper use the epoch 2 checkpoint. The loss weight λ is robust across $\{0.1, 0.5, 1.0, 2.0, 5.0\}$: all values achieve 57.7–59.1% F1 on FRACCO. We use $\lambda = 1.0$ as it achieves the lowest variance ($\pm 0.52\%$). This robustness suggests that the exact balance between objectives matters less than ensuring both are present and aligned.

13.4.5 Multi-Ontology Transfer

The main OntoBook model uses CIM-10 walks (Table 13.3). To test whether other ontologies also provide useful pretraining signal, we train separate models on walks from each ontology individually and on all three combined.

All three individual ontologies improve over the MLM-only baseline, indicating that the pipeline generalizes across different medical knowledge structures. ATC (medications) yields the best Distemist score (+3.81 over CIM-10), despite Distemist being a disease coding task, suggesting that drug-disease relationships transfer well to disease understanding. CCAM (procedures) achieves the best average micro-F1 (53.13). Combining all three ontologies does not outperform the best individual ontologies. We hypothesize that this is because the relation prediction head must accommodate heterogeneous edge types across ontologies.

Model	Chapter	Depth	Dist. ρ
Baselines			
Base	97.5	99.3	0.195
MLM-only	98.9	99.4	0.287
CodeInfill+MLM	**99.0**	**99.8**	**0.397**
Rel-only	98.0	98.2	0.203
Ours			
OntoBook	98.7	99.7	0.073

Table 13.6: **Probing analysis** on CIM-10 code embeddings (% accuracy for Chapter and Depth, Spearman ρ for Distance). CodeInfill+MLM achieves the best geometric organization but OntoBook achieves the best downstream performance (Table 13.3).

13.4.6 Probing Analysis

To understand how OntoBook organizes code representations, we evaluate with three probing tasks on CIM-10 code embeddings (mean-pooled over code descriptions): chapter classification (26 classes), hierarchy depth prediction (4 levels), and distance correlation (Spearman ρ between cosine distances and ontology tree distances). We compare against a code infilling variant (CodeInfill+MLM) that masks and predicts code tokens during pretraining.

CodeInfill+MLM achieves the best probing metrics ($\rho = 0.397$) but the worst downstream performance among our models (47.6 average micro-F1, Table 13.3), while OntoBook has lower distance correlation ($\rho = 0.073$) despite achieving the best downstream performance (52.5 micro-F1). This inverse relationship suggests that for encoder pretraining, representational flexibility is more valuable than strict geometric preservation of ontology structure.

13.5 Discussion

The 30.50-point gap between aligned and misaligned training highlights the central role of objective alignment. When $\mathcal{L}_{\text{MLM}}$ trains on textbooks while $\mathcal{L}_{\text{rel}}$ trains on unrelated data, the model receives conflicting gradient signals: the MLM objective pushes representations toward textbook language patterns, while the relation objective pushes them toward a different data distribution. Aligned training avoids this conflict by ensuring both objectives reinforce the same semantic structure. Relation prediction guides attention toward ontologically meaningful patterns in the textbook text, while MLM ensures the model builds coherent representations of that text. This explains why relation prediction alone fails (−12.68 F1): it is a discriminative task that can exploit surface-level shortcuts, as demonstrated by the small gap between raw-walk and textbook MLM-only models (+0.59 F1). Combined with aligned MLM, however, it provides a complementary signal that improves downstream coding performance.

The cross-ontology transfer results suggest that our approach generalizes beyond CIM-10: training on CCAM or ATC alone also improves over the baseline despite targeting different medical domains. However, combining all ontologies does not outperform individual ones, possibly because the relation prediction head must accommodate heterogeneous edge types.

The probing analysis reveals a tension between geometric organization and downstream performance. CodeInfill+MLM achieves the best probing metrics ($\rho = 0.397$) but the worst downstream scores among our models (47.6 micro-F1), while OntoBook achieves the opposite ($\rho = 0.073$, 52.5 micro-F1). This suggests that for downstream coding tasks, flexible representations adapt better than rigid geometric structure that must be partially unlearned during fine-tuning.

LLM reformulation plays a dual role. For MLM alone, the effect is modest (+0.59 F1), but reformulation is essential for relation prediction: on raw walks, structural markers provide formatting shortcuts that the classifier can exploit without learning medical semantics, whereas fluent prose forces the model to learn from content rather than format. The full gap between OntoBook and MLM-only on raw walks is 4.45 points.

Limitations. All experiments use French medical coding. Extending to English ontologies such as ICD-11 and SNOMED-CT is left for future work. The reformulation step requires significant compute for LLM inference, and sensitivity to the choice of LLM has not been evaluated. Cantemist-FR and Distemist-FR are cross-lingual projections from Spanish (Zaghir et al., 2024), which may introduce translation artifacts; FRACCO is the only native French benchmark. All experiments use ModernCamemBERT; generalization to other architectures remains untested.

Future Work. Future work will extend OntoBook to English medical coding using UMLS and SNOMED-CT ontologies. A promising direction is cross-ontology walks that traverse edges between different ontologies (e.g., from a CIM-10 diagnosis to its ATC treatment to the CCAM procedure), generating richer walks that capture inter-ontology relationships in a single sequence.

13.6 Conclusion

We presented OntoBook, a method that converts medical ontology structure into pretraining signal for encoder language models through random walks, LLM-based textbook reformulation, and multi-task learning combining masked language modeling and relation prediction. Our key finding is that alignment between objectives is essential: both tasks must train on the same data, as misalignment causes a 30-point degradation. On French medical coding benchmarks, OntoBook improves over MLM-only pretraining by +2.5 micro-F1 on FRACCO and +8.0 on Distemist. We release 1.3 million LLM-reformulated medical textbooks across three French ontologies (CIM-10, CCAM, ATC) and pretrained model checkpoints.

Part V

Adapting to Clinical Tasks

14 Limits of Direct Fine-Tuning

15 Architectures for Low-Resource Extraction

16 Synthetic Data for Task Adaptation

16.1 Introduction

Clinical text is scarce, but structured medical knowledge is not. France maintains medical ontologies that span tens of thousands of codes (CIM-10 for diagnoses, CCAM for procedures, ATC for medications), and PubMed Central contains millions of clinical case descriptions embedded in scientific articles. Both are publicly available, both are richly structured, and neither can be used directly to train a language model. The question is whether we can convert them into synthetic training data that is closer to what clinical applications actually need.

This chapter presents two instances of this idea. The first, OntoBook, generates synthetic medical textbooks from ontology graphs by reformulating random walks with a large language model. The second, currently in progress, reformulates the clinical case paragraphs extracted in Chapter 9 into synthetic clinical reports. Both share the same recipe: take a structured source, hand it to an LLM with a tight prompt, and get back fluent medical text that preserves the structure. They differ in input (graph vs. paragraph) and target (encoder pretraining vs. task adaptation). The full method and downstream results for OntoBook are presented in Chapter 13; here we focus on the data artifacts themselves.

16.2 Synthetic Textbooks from Medical Ontologies

16.2.1 Walk Generation

We generate random walks through ontology graphs to capture relational structure in a sequential format suitable for language model pretraining. We sample walks starting from each code in the ontology. At each step, the next node is selected via weighted sampling where edge type weights depend on the walk type. This biased sampling is necessary because semantic edges are rare (Table 13.2 in Chapter 13), and a uniform walk would rarely traverse them. Each walk records the traversed codes along with their labels and the relation types connecting them. Walk length is sampled uniformly between 8 and 20 steps. We track visited nodes to avoid cycles, with a fallback that allows revisiting nodes when all neighbors have been explored.

For CIM-10, we define five walk types, each emphasizing different aspects of medical knowledge:

- **Étiologie** (causal chains): traverses CAUSES and CAUSED_BY edges to follow disease etiology.
- **Diagnostic Différentiel**: traverses EXCLUSION edges to enumerate codes to distinguish clinically.
- **Codage Double**: follows mandatory code pairs linking etiology to manifestation, encoding the dual-coding rules of CIM-10.
- **Syndrome**: traverses MANIFESTATION edges to follow multi-system manifestations of a single underlying condition.
- **Cross-Chapter**: connects different ICD chapters, e.g., a metabolic diagnosis (E11, type 2 diabetes) and its renal complication (N08.3, diabetic glomerulopathy).

We generate 402,328 walks for CIM-10 (average 4,830 characters), 762,958 for CCAM, and 138,980 for ATC, totaling 1,304,266 walks across all three ontologies.

16.2.2 Walk-to-Textbook Reformulation

Raw walks contain structured markers (e.g., `>> À distinguer de:`) that are useful for parsing but not natural for language model pretraining. We use a large language model to reformulate walks into fluent medical prose resembling textbook paragraphs.

We prompt Qwen3-235B-A22B-Instruct (Yang et al., 2025a) with FP8 quantization to transform each walk into a coherent paragraph. The prompt instructs the model to: (1) preserve all medical codes and their relationships exactly as stated, (2) use natural medical language without lists or bullet points, (3) not add any information beyond what appears in the walk. This ensures the textbook content remains grounded in the ontology structure. We apply two filters: we discard reformulations shorter than 50 characters or that fail to mention the source codes. The reformulation runs on 4 NVIDIA H100 GPUs using vLLM (Kwon et al., 2023) with guided decoding to ensure valid JSON output, taking approximately 20 hours for all 1.3 million walks. The resulting textbooks contain approximately 1.3 billion tokens across all three ontologies.

Chapter 13 shows one walk-to-textbook transformation in detail (Figure 13.2). The same LLM prompt structure is used for CCAM and ATC, with ontology-specific terminology substituted for medical-coding domain terms.

16.2.3 The Role of Alignment

The key result of OntoBook, presented in Chapter 13, is that aligned multi-task training (MLM and relation prediction on the same textbook data) produces a +3.86 F_1 improvement on French medical coding, while misaligned training (different data per task) causes a −30.50 point degradation. The synthetic textbooks themselves are necessary but not sufficient: training MLM on raw structured walks (without LLM reformulation) yields nearly the same downstream performance as training on the reformulated textbooks (+0.59 F_1). The reformulation matters specifically for the relation prediction objective, where structural markers in raw walks would let the classifier exploit formatting shortcuts rather than medical semantics.

16.3 Synthetic Clinical Reports from Case Paragraphs

This section is a placeholder. The project that motivates it exists, but the paper is still in progress.

16.4 Two Instances, One Pattern

Conclusion

This is the last chapter of Part 3. The thesis conclusion revisits the research questions raised in the introduction and discusses what remains open.

17 Conclusion

Acronyms

PCA	Principal component analysis
SNF	Smith normal form
TDA	Topological data analysis

Glossary

LaTeX	A document preparation system
$\mathbb{R}$	The set of real numbers

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Appendix